

GH, KH and pH in Freshwater Aquariums

A Master Research Reference and Data Compilation

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- R5: Problem → Possible Explanation → Test Next
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- R7: Testing Method → Strengths → Weaknesses
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Bibliography

- Peer-Reviewed Scientific Literature
- Aquatic Veterinary Sources
- Aquaculture and Fisheries Sources
- Government and Water Treatment Authorities
- Established Aquarium Organizations and Reputable Publications
- Manufacturer Technical Documentation
- Hobbyist and Community Sources — Used for Identifying Recurring Questions and Practices, Not as Scientific Evidence
- Sources Deliberately Excluded

Document Notes

About This Document

Purpose and Scope

This document is a **research reference compilation**, not an article, not a script, and not a narrative. It exists to serve as a retrieval corpus: a dense, explicitly labeled, heavily cross-referenced body of source material about **pH, general hardness (GH), and carbonate hardness (KH / alkalinity)** in freshwater aquariums.

It is written for machine retrieval first and human reading second. That means:

- Terminology is repeated deliberately rather than replaced with pronouns or synonyms.
- Synonyms and alternate names are stated explicitly (“KH, carbonate hardness, alkalinity, acid-neutralizing capacity, buffering capacity”).
- Conclusions are stated close to the evidence supporting them.
- Redundancy across sections is intentional. A reader skipping to Section 17 should not need Section 5 to understand it.
- Uncertainty is labeled rather than smoothed over.

No editorial conclusion is drawn about what a podcast episode should say. Where evidence conflicts, both positions are documented.

Evidence Classification Key

Every substantive claim in this document is intended to fall into one of the following categories. Where a claim is not explicitly tagged, its category should be inferred from the surrounding language (“established,” “commonly reported,” “anecdotal,” etc.).

Tag	Meaning	Example of the kind of claim
[EST] Established scientific principle	Textbook chemistry or physiology; not seriously disputed	The pH scale is logarithmic
[STR] Strong evidence	Multiple peer-reviewed studies or authoritative agency consensus	Nitrification consumes alkalinity
[INF] Reasonable inference	Sound extrapolation from established principles to the aquarium case, but not directly tested in aquariums	Therefore a heavily stocked, rarely-changed aquarium will lose KH over time
[AQC] Aquaculture / fisheries practice	Standard commercial or agency practice, well documented, but usually developed for ponds/RAS rather than glass aquariums	Liming ponds to raise alkalinity above 20 mg/L
[VET] Aquatic veterinary source	Clinical guidance from veterinary references	Old Tank Syndrome parameter profile
[HOB] Common aquarium practice	Widely used by hobbyists, plausible mechanism, limited formal testing	Crushed coral in a filter bag
[ANC] Anecdotal observation	Reported experience without controlled comparison	“My shrimp colony crashed after a large water change”
[CON] Contested claim	Credible sources actively disagree	Whether a 1.0-unit pH change is inherently dangerous
[MYTH] Unsupported or misleading folklore	Repeated widely; mechanism absent, backwards, or overstated	“KH controls pH”

A Note on Source Reliability in This Subject Area

Two problems recur when researching aquarium water chemistry and should be flagged up front, because they affect how the rest of this document is written.

Problem 1: Circular hobby citation. A very large fraction of aquarium care-sheet content is copied from other aquarium care-sheet content. A parameter range published in the 1970s can propagate through forty years of websites without ever being retested. When this document reports a “commonly cited range,” that phrase means *commonly cited*, not *empirically validated*. See Section 30.

Problem 2: Fabricated or unverifiable citations on aquarium content sites. During research for this document, multiple recently published aquarium websites were found citing specific-sounding studies — with named institutions, years, and precise percentages — that could not be located in any indexed literature database. Examples encountered included a claimed 2020 study from a named university documenting a specific percentage increase in shrimp molting mortality below a specific dGH, and a claimed 2021 *Journal of Fish Biology* study reporting a specific percentage cortisol elevation in fish outside “optimal hardness.” **These claims are not reproduced as facts anywhere in this document**, because their underlying sources could not be verified. This appears to be a growing contamination problem in AI-assisted hobby content, and it is worth treating any precise-sounding percentage on a hobby site with suspicion unless a primary citation is given.

Consequence for this document: where a number is given, this document either (a) cites a traceable authoritative source, (b) shows the arithmetic derivation, or (c) explicitly labels the number as a hobby convention of unknown provenance.

How Sections Relate

- **Sections 1-3** define each parameter independently.
- **Section 4** is the integration section and is the single most important section for anyone confused about how the three interact.
- **Sections 5-7** cover buffering and the two dominant pieces of hobby advice about pH.
- **Sections 8-10** cover biology, source water, and RO.
- **Sections 11-18** are the intervention sections: how to change each parameter, and what each method actually does.
- **Sections 19-23** cover system dynamics over time: CO₂, nitrification, tank aging, water changes, evaporation.
- **Sections 24-27** cover measurement, units, and physiology.
- **Sections 28-31** cover invertebrates, plants, care sheets, and acclimation.
- **Sections 32-36** are the evaluation sections: myths, scenarios, decision frameworks, and gaps discovered during research.

Section 1: Fundamentals of pH

Search terms for this section: pH, hydrogen ion, hydronium, acidity, alkalinity, acidic, basic, alkaline, neutral pH, logarithmic scale, pH meter, pH test kit, pH strip, calibration, diurnal pH swing, degassing, CO₂ and pH, carbonic acid, organic acid, tap water pH, aquarium pH.

1.1 What pH Measures

[EST] pH is a measure of the **activity of hydrogen ions (H⁺)** in an aqueous solution, defined as the negative base-10 logarithm of hydrogen ion activity:

$$\text{pH} = -\log_{10}[\text{H}^+]$$

In practice, the free proton does not exist alone in water; it associates with water molecules to form the hydronium ion (H₃O⁺). For aquarium purposes the distinction is immaterial: pH is a measure of **how acidic or how basic the water is right now**.

The Southern Regional Aquaculture Center's foundational fact sheet on this topic defines pH exactly this way: the SRAC fact sheet frames pH as an index of the hydrogen ion concentration in water, formally the negative base-10 logarithm of the molar concentration of H⁺ (Wurts & Durborow, 1992, SRAC 464).

Plain-English version for retrieval: pH is a *snapshot* of how acidic the water is. It is not a measure of how much acid the water can absorb, and it is not a measure of how much mineral is dissolved in the water. Those are different measurements (KH and GH respectively). This distinction is the single most common source of confusion in the hobby.

1.2 The Scale, and Why the Logarithm Matters

[EST] The conventional pH scale runs 0–14 in dilute aqueous solution at 25 °C, though values outside that range are chemically possible in concentrated solutions.

Because the scale is logarithmic:

pH difference	Change in hydrogen ion activity
0.1 unit	≈ 1.26×
0.2 unit	≈ 1.58×
0.3 unit	≈ 2×
0.5 unit	≈ 3.16×
1.0 unit	10×
2.0 units	100×
3.0 units	1,000×

Why pH 6 is meaningfully different from pH 7: water at pH 6.0 contains ten times the hydrogen ion activity of water at pH 7.0. Water at pH 5.0 contains one hundred times as much. This is the arithmetic behind the hobby's anxiety about pH swings.

Important qualification [CON]: the tenfold arithmetic is real, but it does not by itself establish biological significance. A tenfold change in a quantity that is already extremely small may or may not matter to a given organism. At pH 7.0, [H⁺] ≈ 1×10⁻⁷ mol/L. At pH 6.0, ≈ 1×10⁻⁶ mol/L. Both are minuscule compared with the concentrations of sodium, chloride, calcium and bicarbonate the fish is simultaneously regulating. The biological question is not “how many times more H⁺” but “does this H⁺ concentration interfere with gill ion transport and acid-base regulation in this species.” See Sections 6 and 27. This is a place where the hobby's mathematical framing can mislead.

1.3 Neutral pH and Why Neutral Is Not Automatically Ideal

[EST] “Neutral” pH is defined as the point where [H⁺] = [OH⁻]. In pure water at 25 °C this occurs at pH 7.00, because the ion product of water (K_w) is 1.0×10⁻¹⁴ at that temperature.

[EST] Neutral pH is **temperature-dependent**, because K_w is temperature-dependent. Approximate neutral points:

Temperature	Neutral pH (pure water)
0 °C	≈ 7.47
10 °C	≈ 7.27
20 °C	≈ 7.08
25 °C	7.00
30 °C	≈ 6.92
40 °C	≈ 6.77

Implication for tropical aquariums [INF]: at 26–28 °C, chemically neutral water sits at roughly pH 6.94–6.96, not 7.00. This is a small effect and is not a practical management concern, but it does undercut the intuition that 7.0 is a magic biological midpoint. It is a chemical definition tied to a temperature, not a biological optimum.

[MYTH] “**Neutral pH is ideal for fish.**” There is no general basis for this. Vast numbers of freshwater fish evolved in water that is chronically acidic (Amazonian blackwater, pH 4–6) or chronically alkaline (East African rift lakes, pH 8–9). Neutrality is a property of water, not a preference of animals. See Sections 8 and 32.

1.4 How Aquarium pH Is Measured

1.4.1 Liquid colorimetric test kits

Mechanism: an indicator dye (commonly bromothymol blue for the 6.0–7.6 range; phenol red, cresol red, or thymol blue for high-range kits) changes colour across a pH range. The user compares the colour against a printed card.

Strengths: inexpensive, no calibration, adequate for tracking direction of change.

Weaknesses: - Resolution is typically 0.2–0.4 pH units at best. - Colour cards fade; lighting and background colour bias the read. - Standard “low range” kits are commonly limited to about 6.0 at the bottom and “high range” kits to about 8.8 at the top. **A tank reading “6.0” on a low-range kit may actually be 5.4 or 4.8.** This is a very common source of misdiagnosis in soft-water and aquasoil tanks. - Tannin-stained water tints the sample and corrupts colour matching. - Some kits are affected by high alkalinity, requiring the sample to be read promptly.

1.4.2 Test strips

Strengths: fastest; useful for a rough screen; many strips test GH, KH, pH, nitrite and nitrate simultaneously.

Weaknesses: generally the least precise option; pads are humidity-sensitive and degrade once the bottle has been opened; read-time sensitivity (reading late causes drift); resolution often ± 0.5 pH.

[HOB] Consensus hobby view, consistent with the general analytical chemistry position: strips are acceptable for *screening* and for *detecting large changes*, and poor for *precise values*.

1.4.3 Digital pH meters and probes

Mechanism: a glass electrode develops a potential across a pH-sensitive membrane, referenced against a stable internal reference electrode. The instrument converts millivolts to pH.

Strengths: resolution to 0.01 pH; continuous logging possible; not affected by water colour.

Weaknesses and requirements: - **Calibration is mandatory and recurring.** Two-point (pH 7.00 and pH 4.01) or three-point calibration with fresh certified buffers. An uncalibrated meter can be wrong by a full pH unit while displaying two decimal places, which is worse than a liquid kit because it projects false confidence. - Electrodes must be stored wet, in storage solution (typically KCl), never in distilled/RO water. Storing a probe dry or in pure water destroys the reference junction. - Junction fouling from proteins, biofilm and tannins causes slow drift. - Probes have finite life; typical service life is roughly 1–2 years of regular use. - Temperature compensation matters. ATC (automatic temperature compensation) corrects the electrode slope but does **not** correct for the fact that the sample's actual pH changes with temperature.

Accuracy vs. precision [EST]: *Precision* is repeatability (a meter giving 6.82, 6.82, 6.83). *Accuracy* is closeness to truth (whether 6.82 is actually 6.82 or actually 7.30). A digital meter is more precise than a liquid kit by default; it is only more accurate if calibrated. This distinction should be stated explicitly whenever meters are discussed.

1.4.4 Why different tests give different results on the same water

Sources of disagreement, roughly in order of practical importance:

1. **Calibration state of the meter** (largest single factor).
2. **Sample degassing between draw and read** — CO₂ escapes or is absorbed; see 1.5.
3. **Time of day the sample was taken** in a planted or algae-containing system; see 1.6.
4. **Kit range limits** — a value pegged at the end of a card's range is not a measurement, it is a floor or ceiling.
5. **Reagent age** — expired indicator solutions shift colour.
6. **Operator colour perception and lighting.**
7. **Sample location** — water near a CO₂ diffuser, near the surface, or in a filter chamber is not identical to bulk tank water.
8. **Temperature difference** between the sample and the reference condition.

1.5 Why Tap Water pH Changes After Sitting or Aeration

[EST] This is one of the most frequently asked questions in the hobby and it has a clean chemical answer.

Municipal and especially well water is often delivered **under pressure and supersaturated with dissolved carbon dioxide (CO₂)**. Dissolved CO₂ forms carbonic acid:



Excess dissolved CO₂ therefore depresses pH. When the water is drawn into an open bucket, the CO₂ partial pressure above the water is atmospheric (~0.04%), far below the water's dissolved load. **CO₂ outgasses to the air.** As it leaves, the equilibrium shifts left, H⁺ is consumed, and pH rises.

Typical observed pattern [HOB, mechanism EST]: tap water tests 7.0 fresh from the faucet and 7.8 after 24 hours of sitting, or after 1 hour of vigorous aeration. The *alkalinity (KH)* does not change during this process — only the CO₂ and therefore the pH.

Critical corollary, frequently reversed in hobby writing: surface agitation and aeration **drive CO₂ out of** water that is supersaturated with CO₂, and **draw CO₂ into** water that is undersaturated. In a normal aquarium at equilibrium with air, agitation moves the system *toward* atmospheric equilibrium (~2–3 ppm CO₂ in a non-injected tank). In a CO₂-injected planted tank, surface agitation **removes injected CO₂ and raises pH**. Agitation is not a way to add CO₂ to a planted aquarium.

Practical testing protocol [INF]: to characterise source water honestly, test the same tap sample twice: 1. Immediately on draw (“as-delivered pH”). 2. After 24 hours of standing, or after 1 hour of airstone aeration (“equilibrated pH”).

The **equilibrated value** is the better predictor of what the aquarium will settle toward. The as-delivered value predicts the transient pH shock during a water change. Both numbers are useful and they answer different questions. See Sections 9 and 22.

1.6 Why pH Changes Over the Course of a Day (Diurnal pH Swing)

[STR] In any system containing photosynthetic organisms — plants, algae, biofilm, phytoplankton — CO₂ is consumed during daylight and produced continuously by respiration. Because CO₂ is acidic in solution, pH follows an inverse curve to CO₂:

- **Daylight:** photosynthesis > respiration → CO₂ falls → **pH rises**, typically peaking in late afternoon.
- **Night:** photosynthesis stops, respiration continues → CO₂ accumulates → **pH falls**, typically bottoming near dawn.

This is thoroughly documented in aquaculture. the Merck Veterinary Manual describes CO₂ in plant- or algae-containing systems as cycling on a daily basis, with daytime photosynthesis pulling CO₂ down and pushing pH up, and overnight respiration reversing both (Merck Veterinary Manual, *Environmental Diseases of Aquatic Animals in Aquatic Systems*).

Magnitude depends on alkalinity [STR, AQC]. This is the key point. The size of the daily swing is governed by buffering capacity. The Southern Regional Aquaculture Center reports that ponds holding less than about 20 mg/L total alkalinity as CaCO₃ can swing across roughly pH 6 to pH 10 within a single day, and that swings of that size are associated with stress, depressed growth, and mortality in farmed aquatic animals (Wurts & Durborow, 1992; Wurts & Masser, 2013, SRAC 4100).

Aquarium translation [INF]: 20 mg/L as CaCO₃ ≈ **1.1 dKH**. Below roughly 1 dKH, the aquarium has effectively no carbonate buffer and pH will move with any biological or chemical input. Between roughly 2 and 4 dKH there is meaningful but finite resistance. Above roughly 5–6 dKH, daily biological CO₂ cycling produces small pH movement.

Important caveat: the pond literature describes systems with dense phytoplankton blooms and far higher photosynthetic biomass per volume than most home aquariums. A lightly planted 20-gallon aquarium at 3 dKH will not swing from 6 to 10. The *mechanism* transfers; the *magnitude* does not. This is a place where

aquaculture data is regularly over-applied in hobby writing.

Practical measurement note: an aquarist who tests at 8 p.m. on Monday and 7 a.m. on Saturday and reports “my pH is unstable” may simply be sampling opposite ends of a normal daily cycle. **Time-of-day consistency is required before a pH trend can be interpreted at all.**

1.7 Every Mechanism That Moves Aquarium pH

This subsection is a consolidated index; each item is treated in depth in the referenced section.

Processes that lower pH (add acid or remove base)

Mechanism	How it works	Speed	Section
Dissolved CO ₂ / carbonic acid	$\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{HCO}_3^-$	Minutes-hours; reversible	19
Nitrification	Ammonia oxidation produces H ⁺ and consumes alkalinity	Continuous, cumulative	20
Organic acid accumulation (humic, fulvic, tannic)	Weak acids from decomposing organics, botanicals, driftwood	Weeks-months	16
Active substrate (aquasoil) ion exchange	Cation exchange strips Ca ²⁺ /Mg ²⁺ ; acidic components released	Days-months; finite lifespan	18
Peat, botanicals, blackwater extracts	Humic/fulvic acid release; cation chelation	Days-weeks	16
Commercial “pH Down” (acid buffers, phosphate)	Direct acid addition or phosphate buffer imposition	Immediate; often rebounds	15
Very soft, low-KH source water	Removes resistance rather than adding acid	Enabling condition	5
Respiration at night	CO ₂ accumulation	Overnight cycle	1.6, 19

Processes that raise pH (add base or remove acid)

Mechanism	How it works	Speed	Section
CO ₂ outgassing / surface agitation	Removes carbonic acid	Minutes-hours	1.5, 19
Photosynthesis	Consumes CO ₂	Daylight hours	1.6, 19
Carbonate mineral dissolution (crushed coral, aragonite, limestone, shell)	$\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca}^{2+} + 2\text{HCO}_3^-$	Slow, self-limiting	17
Sodium or potassium bicarbonate	Direct alkalinity addition	Immediate	13
Sodium carbonate (soda ash), lime, calcium hydroxide	Strong base addition	Immediate; hazardous	13, 15
Water change with higher-pH, higher-KH source water	Dilution/replacement	Immediate	22
Commercial “pH Up”	Base or buffer addition	Immediate; often rebounds	15
Denitrification (anaerobic zones)	Recovers roughly half the alkalinity nitrification consumed	Slow, variable	20

Note on the two tables above: the mechanisms are not equivalent in kind. Some change pH *by changing alkalinity* (crushed coral, bicarbonate, aquasoil exhaustion of KH). Others change pH *without changing alkalinity* (CO₂ addition or removal). This distinction is central to Sections 4, 5 and 19 and is arguably the most under-explained idea in hobby water chemistry.

Section 2: Fundamentals of GH (General Hardness)

Search terms for this section: GH, general hardness, total hardness, dGH, degrees of general hardness, German degrees, °dH, calcium, magnesium, Ca²⁺, Mg²⁺, divalent cations, multivalent cations, ppm as CaCO₃, mg/L as CaCO₃, soft water, hard water, mineral content, osmoregulation.

2.1 What GH Measures

[EST] General hardness (GH), also called **total hardness** or simply **hardness**, is the summed concentration of **divalent (doubly charged) metal cations** dissolved in the water, expressed as an equivalent concentration of calcium carbonate (CaCO₃).

In virtually all natural and tap waters this is dominated by two ions:

- **Calcium (Ca²⁺)**
- **Magnesium (Mg²⁺)**

[EST] Other multivalent cations technically contribute to hardness and to the titration result of a standard EDTA hardness test — strontium (Sr²⁺), barium (Ba²⁺), iron (Fe²⁺/Fe³⁺), manganese (Mn²⁺), aluminium (Al³⁺), zinc, copper. In practice their concentrations are so low relative to calcium and magnesium that they are ignored, with two exceptions worth knowing:

- **High-iron well water** can inflate a GH reading slightly.
- **Aquariums dosing significant iron-rich fertilisers** are not affected meaningfully at normal dosing levels.

Why calcium and magnesium dominate the discussion: they are the two ions that (a) are abundant in the geology that groundwater passes through — limestone, dolomite, chalk, gypsum — and (b) have direct, documented biological roles in fish, invertebrates, and plants.

The US Geological Survey defines it in these terms: hardness is, in scientific terms, essentially the quantity of dissolved calcium and magnesium present (USGS, *Hardness of Water*).

2.2 Units of GH

[EST] Two unit systems are in common aquarium use and they are frequently confused.

German degrees of hardness (°dH, dGH, °dGH): Historically defined as 10 mg/L of calcium oxide (CaO) per litre. Converted to calcium carbonate equivalents:

1 dGH = 17.848 mg/L as CaCO₃ (commonly rounded to 17.9, or loosely to 17.86 or “about 18”)

Parts per million / milligrams per litre as CaCO₃ (ppm, mg/L): For dilute freshwater, **1 ppm = 1 mg/L** exactly enough for aquarium purposes.

Conversion table (see Section 25 for full reference):

dGH	mg/L (ppm) as CaCO ₃	USGS descriptive category
0	0	Soft
1	17.8	Soft
2	35.7	Soft
3	53.5	Soft
3.4	60	Soft/moderately hard boundary
4	71.4	Moderately hard
5	89.2	Moderately hard
6	107.1	Moderately hard
6.7	120	Moderately hard/hard boundary
7	124.9	Hard
8	142.8	Hard
9	160.6	Hard
10	178.5	Hard
10.1	180	Hard/very hard boundary
12	214.2	Very hard
15	267.7	Very hard
20	357.0	Very hard
25	446.2	Very hard
30	535.4	Very hard

USGS classification [STR]: the USGS uses four descriptive bands measured as calcium carbonate — soft at 0–60 mg/L, moderately hard at 61–120 mg/L, hard at 121–180 mg/L, and very hard above 180 mg/L. The USGS also notes that the EPA sets no legal limit for hardness, because the ions responsible are not toxic (USGS FAQ, *Do you have information about water hardness in the United States?*, updated 5 August 2025).

Critical caveat for aquarists: the USGS categories were built for **drinking water and household plumbing**, not for fish. “Very hard” describes scale in a kettle. It does not mean “unsuitable for aquarium life.” Rift lake cichlids live comfortably in water the USGS would call very hard; so do guppies, mollies and most snails. Conversely, “soft” by USGS standards (say 50 mg/L, ~2.8 dGH) is still roughly 5–50× harder than Rio Negro blackwater. **The scale is being borrowed from a different discipline and should be used descriptively, not prescriptively.**

Grains per gallon (gpg): used by US water softener companies and appearing on some municipal reports. > 1 gpg = 17.1 mg/L as CaCO₃

Note the near-coincidence: 1 gpg ≈ 1 dGH (17.1 vs 17.85, about 4% apart). They are **not the same unit**, but they are close enough that confusing them rarely causes practical harm — unlike confusing dGH with ppm, which causes an 18-fold error.

2.3 Converting Individual Ion Concentrations to Hardness

[EST] Municipal water reports frequently list calcium and magnesium separately in mg/L of the *element*, not as CaCO₃. To convert:

$$\text{Hardness (mg/L as CaCO}_3\text{)} = (\text{mg/L Ca} \times 2.497) + (\text{mg/L Mg} \times 4.118)$$

Derivation: the multiplier is the ratio of the molar mass of CaCO_3 (100.09 g/mol) to the molar mass of the ion (Ca = 40.08; Mg = 24.31). Magnesium's larger factor reflects that a milligram of magnesium contains more moles of divalent cation than a milligram of calcium.

Worked example: A municipal report lists Ca = 42 mg/L, Mg = 11 mg/L. - Calcium contribution: $42 \times 2.497 = 104.9$ mg/L as CaCO_3 - Magnesium contribution: $11 \times 4.118 = 45.3$ mg/L as CaCO_3 - **Total hardness = 150.2 mg/L as CaCO_3 = 150.2 + 17.848 = 8.4 dGH** - USGS category: hard

Practical value: this calculation lets an aquarist derive GH from a municipal report without a test kit, and — more usefully — lets them see the **Ca:Mg ratio**, which no GH test kit reveals. See Section 28 on why this ratio matters for invertebrates.

2.4 What GH Is Not

This subsection exists specifically because these confusions are near-universal.

GH is not TDS. Total dissolved solids includes *everything* dissolved: sodium, potassium, chloride, sulfate, nitrate, bicarbonate, dissolved organics, and the calcium/magnesium that GH measures. Two waters can have identical TDS of 250 ppm where one is 8 dGH calcium-magnesium water and the other is nearly sodium chloride with 1 dGH. See Section 26.

GH is not KH. GH counts *cations* (Ca^{2+} , Mg^{2+}). KH counts *anions* (HCO_3^- , CO_3^{2-}) and, more precisely, acid-neutralizing capacity. They frequently correlate because limestone dissolution releases both simultaneously — but they are independently variable and there are common real-world cases where they diverge sharply. See Sections 3 and 4.

GH is not pH. GH is a quantity of dissolved mineral. pH is an intensity of acidity. Water at 15 dGH can sit at pH 6.2 (high non-carbonate hardness plus organic acids) and water at 1 dGH can sit at pH 8.5 (sodium bicarbonate dominated). See Section 4.

GH is not conductivity. Conductivity responds to all ions weighted by charge and mobility. See Section 26.

GH is not a measure of “how good the water is.” There is no general direction of improvement. See Section 8.

2.5 Biological Roles of Calcium and Magnesium

2.5.1 Calcium in fish

[STR] Environmental calcium has documented, mechanistically understood effects on freshwater fish physiology that are independent of pH:

Reduction of gill permeability. Calcium binds at tight junctions between gill epithelial cells, reducing paracellular leakage of ions. the working model in the literature is that Ca^{2+} occupies binding sites at gill tight junctions and thereby lowers paracellular ion permeability, and that hydrogen ions at low pH displace that calcium, making the junctions leakier and accelerating ion loss (see e.g. Nguyen et al., 2019, *Comparative Biochemistry and Physiology*; reviews by Hunn, 1985 and Wood, 1989). In classic work on goldfish, adding calcium to the water sharply cut both sodium and chloride efflux, and swung the transepithelial potential from roughly -10 mV in calcium-free medium to roughly $+10$ mV at 10 mM calcium (Eddy, 1975, *Journal of Comparative Physiology A*).

Plain-English version: calcium in the water helps “seal” the fish’s gills. In calcium-poor water the gills are leakier and the fish loses salts to the environment faster, which it must replace by active transport — a metabolic cost.

Source of calcium for skeleton and scales. Freshwater fish take up calcium both from the diet and directly from the water across the gills, via epithelial calcium channels.

Critical species-specific caveat [CON]. The calcium-tightens-gills model is *not* universal. Research on Amazonian species has repeatedly found exceptions: in some characid studies added calcium cut sodium loss at neutral pH but did nothing at low pH, while other studies found no calcium effect at either pH; across both characids and cichlids, sodium efflux resists stimulation by acid exposure to a degree that temperate models do not predict (Gonzalez and colleagues, 1997–2021; summarised in Gonzalez & Wilson, 2001 and subsequent work). Some of these fish appear to have gill tight junctions with extremely high calcium affinity, or to rely on other environmental factors such as dissolved organic carbon.

This is one of the most important nuances in this entire document. It means: - The physiological rationale for “hard water protects fish” is real but was largely established in temperate species (salmonids, goldfish, cyprinids). - Amazonian soft-water fish are not merely “tolerating” low calcium; many are **specifically adapted** to it with different gill machinery. - Generalising temperate ionoregulation data to cardinal tetras is an error.

Environmental calcium also affects gill morphology [STR]. in mangrove rivulus, fish held in naturally hard water carried a shorter interlamellar cell mass than fish in low-calcium water, and restoring calcium to low-calcium water brought gill surface area back — identifying environmental calcium as an ionic trigger for gill remodeling (Platek et al., 2017, *Journal of Experimental Zoology Part A*). In other words, the fish physically restructures its gills in response to water hardness.

2.5.2 Magnesium in fish

[INF/STR] Magnesium is an essential cofactor for hundreds of enzymes, is required for ATP function, and is involved in muscle and nerve function. Fish obtain most magnesium from the diet, and dietary magnesium requirements for cultured species are established in aquaculture nutrition literature. The specific contribution of *waterborne* magnesium to freshwater fish health is less well characterised than that of calcium.

[INF] For aquarium purposes, the practical significance of magnesium is greatest for (a) invertebrates and (b) plants, and less well documented for fish specifically.

2.5.3 Calcium and magnesium in invertebrates

See Section 28 for full treatment. Summary: - **[STR]** Crustacean exoskeleton is chitin mineralised primarily with calcium carbonate, plus calcium phosphate and amorphous calcium carbonate depending on species. - **[STR]** Snail shells are calcium carbonate (aragonite or calcite depending on taxon). - **[INF]** Therefore waterborne and dietary calcium availability directly constrains molting and shell growth in a way that has no fish equivalent.

2.5.4 Calcium and magnesium in plants

See Section 29. Summary: - **[EST]** Calcium is a macronutrient, structural in cell walls and middle lamella, and largely immobile within the plant — deficiency appears in new growth. - **[EST]** Magnesium is the central atom of the chlorophyll molecule, is mobile within the plant, and deficiency appears in older leaves as interveinal chlorosis. - **[EST]** Extremely low GH means both macronutrients are in short supply from the water column.

2.5.5 Reproduction and egg development

[STR/INF] Waterborne ionic composition affects egg water-hardening, chorion permeability, sperm motility and fertilisation success in numerous species, though the effect is species-specific and interacts strongly with conductivity rather than hardness alone. This is the mechanism underlying the classic aquarium observation that some soft-water species will “hold” in hard water but only spawn successfully in soft. See Sections 8 and 30.

[AQIC] Aquaculture literature reports that alkalinity and mineral status affect reproductive performance, though the specific pathways vary. One review summary states that swings in alkalinity may stress fish in ways that disturb hormonal regulation, with downstream effects on gonad development and gamete production (review summary, *Alkalinity and Hardness in Production Ponds*). **[CON]** This is a broad claim in a review rather than a controlled aquarium finding, and should be

treated as directional rather than quantitative.

2.6 What Happens at Different GH Levels

The following is a **descriptive framework**, not a prescription. The correct GH for any given aquarium depends entirely on what lives in it. See Sections 8 and 34.

Extremely low GH (0-1 dGH / 0-18 ppm)

- Effectively demineralised water: pure RO/DI, or a Rio Negro-type blackwater system.
- **Fish:** only viable for species specifically adapted to ion-poor water, and even then, wild Amazonian blackwater contains protective dissolved organic carbon that pure RO water does not (see Section 16.6 — this is a significant and under-appreciated point).
- **Invertebrates:** generally not viable. Shrimp cannot mineralise a new exoskeleton; snails cannot maintain shell.
- **Plants:** calcium and magnesium deficiency likely unless supplied by substrate or fertiliser.
- **System behaviour:** almost always paired with very low KH, therefore minimal buffering.

Moderately low GH (2-5 dGH / 36-89 ppm)

- Typical of soft municipal supplies, rainwater-fed reservoirs, granite/sandstone geology.
- Suitable for a wide range of South American and Southeast Asian soft-water species.
- Common target range for *Caridina* shrimp (see Section 28).
- Plants generally fine; some heavy calcium users may want supplementation.
- KH is often also low here, so pH management deserves attention.

Moderate GH (6-12 dGH / 107-214 ppm)

- The broadest general-purpose range.
- Compatible with the large majority of commercially bred aquarium fish.
- Standard target for *Neocaridina* shrimp and most snails.
- Generally adequate calcium and magnesium for plants without supplementation.

High GH (13-20 dGH / 232-357 ppm)

- Common in limestone-aquifer regions.
- Fully appropriate for livebearers, Central American cichlids, Rift Lake cichlids, most snails, goldfish.
- **[CON]** Whether this is stressful for soft-water species over the long term is genuinely debated. See Section 8 and Section 32 (Claim 4).
- Practical nuisance issues: mineral scale on glass, equipment, and lids.

Extremely high GH (>20 dGH / >357 ppm)

- Deep limestone wells, some arid-region supplies, systems with heavy evaporative concentration.
- Suitable for a narrower set of species.
- **[INF]** Where extremely high GH is accompanied by extremely high conductivity, the osmoregulatory load on soft-water species increases; this is generally a stronger argument for dilution than pH is.
- Practical concern: heater and impeller scaling; frequent equipment maintenance.

2.7 GH and Species Adaptation — Brief Preview

[STR] Freshwater fish are not distributed randomly across water chemistry. The Amazon basin's blackwater tributaries are among the most ion-poor natural waters on Earth. The East African rift lakes are among the most mineralised. Species from these environments have measurably different ionoregulatory physiology.

[CON] What remains genuinely contested is **how much of that adaptation persists in tank-bred stock after many generations in different water**, and **how much it matters for maintenance as opposed to breeding**. This is treated in full in Section 8.

Section 3: Fundamentals of KH (Carbonate Hardness / Alkalinity)

Search terms for this section: KH, carbonate hardness, alkalinity, total alkalinity, acid-neutralizing capacity, ANC, buffering capacity, bicarbonate, HCO_3^- , carbonate, CO_3^{2-} , dKH, degrees carbonate hardness, ppm as CaCO_3 , mg/L as CaCO_3 , meq/L, milliequivalents, temporary hardness, alkalinity depletion.

3.1 The Terminology Problem — Read This First

This section is placed first deliberately. More confusion originates here than anywhere else in aquarium water chemistry, and much of it is caused by the hobby and the science using overlapping words differently.

Three different things share similar names:

Term	Field	What it actually means
Alkalinity (water chemistry)	Analytical chemistry, limnology, aquaculture, wastewater	The water's capacity to neutralize acid, measured by titration to a defined endpoint, expressed as mg/L as CaCO_3 or meq/L. A capacity.
Alkaline (pH descriptor)	General chemistry	Having a pH above 7. An intensity.
Carbonate hardness / KH (aquarium hobby)	Aquarium hobby, German-origin terminology	What an aquarium "KH" test kit measures. In practice, this is a total alkalinity titration reported in German degrees.

[EST] The single most important consequence: *alkalinity* and *alkaline* are not the same concept and do not always move together. Water can be alkaline (pH 8.4) with low alkalinity. Water can have high alkalinity and sit at pH 6.8. This is not a technicality — it is the root of most "my KH is high, why is my pH low?" questions.

What the aquarium KH test actually does

[EST] A hobby KH test is an **acid titration**. The user adds a standardised acid drop by drop until an indicator (usually bromophenol blue or similar) changes colour at a fixed endpoint, and counts the drops. Each drop is calibrated to correspond to a fixed amount of acid-neutralizing capacity.

Critical implication [EST]: the test does not identify *which* base is doing the neutralizing. It measures total acid-neutralizing capacity and *reports* it in units of carbonate hardness. If borate, phosphate, silicate, hydroxide, ammonia, or the conjugate bases of organic acids are present, they are counted too. The kit cannot tell them apart. This is why the hobby term "carbonate hardness" is technically a misnomer: **what is being measured is total alkalinity, not carbonate specifically.**

In most freshwater aquariums, bicarbonate genuinely does dominate alkalinity, so the misnomer is harmless. It stops being harmless when: - Phosphate-based commercial pH buffers are in use (they contribute alkalinity that is not carbonate). - Heavy botanical/tannin loads are present. - Someone is trying to estimate CO_2 from the pH/KH relationship — see Section 19.

Formal definition

[EST] Total alkalinity is formally:

$$\text{Alkalinity} \approx [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + [\text{OH}^-] - [\text{H}^+] \text{ (+ contributions from other proton acceptors)}$$

The factor of 2 on carbonate reflects that CO_3^{2-} can accept two protons; HCO_3^- can accept one.

[EST] Formal water chemistry also distinguishes: - **Phenolphthalein alkalinity** — titration to about pH 8.3, capturing carbonate and hydroxide. - **Total alkalinity (methyl orange alkalinity)** — titration to about pH 4.5, capturing essentially all the bicarbonate as well.

Aquarium kits measure something equivalent to **total alkalinity**.

3.2 Units of KH

[EST] The same conversion applies as for GH, because both are expressed as CaCO_3 equivalents:

$$1 \text{ dKH} = 17.848 \text{ mg/L as } \text{CaCO}_3 \text{ (rounded: 17.9, or "about 18")} \quad 1 \text{ meq/L alkalinity} = 50.04 \text{ mg/L as } \text{CaCO}_3 = 2.80 \text{ dKH}$$

dKH	mg/L (ppm) as CaCO_3	meq/L	Practical description
0	0	0	No measurable buffer
0.5	8.9	0.18	Negligible buffer
1	17.8	0.36	Very low; pH free to move
2	35.7	0.71	Low; typical soft-water / Caridina target
3	53.5	1.07	Low-moderate
4	71.4	1.43	Moderate; common CO_2 drop checker reference
5	89.2	1.79	Moderate
6	107.1	2.14	Moderate-firm
8	142.8	2.85	Firm
10	178.5	3.57	Firm; typical hard tap water
12	214.2	4.28	High
15	267.7	5.35	High
18	321.3	6.42	Very high; Lake Tanganyika range
20	357.0	7.13	Very high

Common unit trap: many aquarium sources, veterinary sources and aquaculture sources report alkalinity in **mg/L as CaCO_3** , while hobby test kits report **dKH**. A veterinary recommendation of "alkalinity ≥ 100 mg/L" is a recommendation of **≥ 5.6 dKH** — a number many aquarists would not recognise as the same statement. Failing to convert is the source of a great deal of apparent disagreement between sources.

3.3 What KH Actually Does

[EST] KH/alkalinity provides **resistance to pH change when acid is added**. The mechanism is the carbonate buffer system:



Acid entering the water is consumed by bicarbonate rather than lowering pH. The carbonic acid formed largely dissociates to water and CO₂, and the CO₂ can outgas. The alkalinity is **used up** in this process. It is a finite reservoir, not a catalyst.

Analogy for retrieval: KH is not a lid holding pH in place. It is a **stockpile of acid-absorbing capacity that gets consumed**. Once consumed and not replenished, pH becomes free to move.

[VET] The Merck Veterinary Manual identifies insufficient alkalinity as a cause of impaired biofiltration and consequent ammonia accumulation, and gives a benchmark of at least 100 mg/L alkalinity for freshwater systems — approximately 5.6 dKH (Merck Veterinary Manual, *Environmental Diseases of Aquatic Animals in Aquatic Systems*). A WSAVA aquatic veterinary presentation similarly frames roughly 100–250 mg/L (≈ 5.6–14 dKH) as a working target for most freshwater species, noting that biological filtration consumes carbonates over time and that falling alkalinity precedes sudden pH shifts (Water Quality Assessment in Aquatic Vet Medicine, WSAVA 2018).

[CON] Important disagreement to record. These veterinary and aquaculture targets are considerably higher than what large numbers of successful soft-water aquariums actually run. Amazonian biotope tanks, *Caridina* shrimp tanks on active substrate, and blackwater systems commonly operate at 0–2 dKH (0–36 mg/L) indefinitely and without incident. The disagreement is real and is worth stating plainly:

- **Veterinary/aquaculture view:** alkalinity below ~100 mg/L is a risk factor for pH instability and biofilter impairment, and should generally be corrected.
- **Specialist hobby view:** low alkalinity is a legitimate, deliberately maintained state for soft-water systems, managed by consistent water changes and light bioload rather than by adding buffer.

Reconciling them [INF]: the veterinary and aquaculture guidance is written for (a) high-density production systems with heavy ammonia loading, (b) generalist and often hard-water species, and (c) a management context where robustness matters more than biotope accuracy. The specialist hobby practice works because bioload per volume is low, water changes are frequent and consistent, and the source water is deliberately controlled. **Neither is wrong; they are optimised for different systems.** What is not defensible is a heavily stocked, infrequently maintained community tank running at 0 dKH — that combination is where the veterinary warning applies most directly. See Sections 5, 20 and 21.

3.4 What Consumes KH

[STR] Alkalinity is consumed by any process that adds acid to the system:

Consumer	Mechanism	Relative magnitude in a typical aquarium
Nitrification	Ammonia oxidation produces H ⁺	Usually the largest ongoing consumer. ~7.14 mg alkalinity as CaCO ₃ per mg ammonia-nitrogen oxidised. See Section 20.
Organic acid production	Decomposition of food, waste, plant matter, botanicals	Slow but cumulative; the classic “old tank” driver
Tannins/humic and fulvic acids	Leached from wood, leaves, peat	Moderate; depends on load
Aquasoil / active substrate	Cation exchange and acid release	Can be very large in the first months; finite
Acid dosing (commercial pH Down, mineral acid, CO ₂ -based reactors)	Direct	As dosed
CO ₂ injection	See note below	Does not consume alkalinity

[EST] Crucial exception — CO₂ does not destroy alkalinity. This is a frequent misunderstanding. Injecting CO₂ lowers pH, but the alkalinity titration result is essentially unchanged, because CO₂ + H₂O → H⁺ + HCO₃[−] produces one proton and one bicarbonate — an acid and its conjugate base in equal measure. Net acid-neutralizing capacity does not change.

Consequence: in a CO₂-injected planted tank, pH drops when CO₂ comes on but KH stays the same. Anyone who observes “pH down, KH unchanged” in a CO₂ tank is observing correct chemistry, not a broken test kit. Conversely, an aquarist who observes **KH itself declining over weeks** is observing something other than CO₂ — most likely nitrification, organic acids, or active substrate. This is a genuinely useful diagnostic distinction. See Sections 19 and 20.

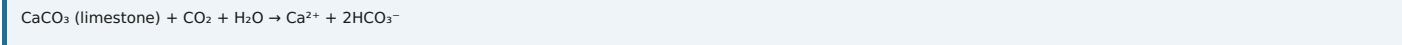
3.5 What Replenishes KH

Source	Mechanism	Speed / controllability
Water changes with alkaline source water	Direct replacement	Immediate; magnitude = % change × source KH
Sodium bicarbonate (baking soda)	Direct alkalinity addition	Immediate; precise; see Section 13
Potassium bicarbonate	Direct alkalinity addition	Immediate; precise; adds K rather than Na
Crushed coral, aragonite, limestone, shell, cuttlebone	CaCO ₃ dissolution; raises GH and KH together	Slow, self-limiting; see Section 17
Commercial carbonate buffers	Varies by product	Immediate; check whether carbonate- or phosphate-based
Denitrification in anaerobic zones	Recovers roughly half the alkalinity nitrification consumed	Slow, uncontrolled, present only where anoxic zones exist

3.6 Does KH Correlate With GH?

[STR] Usually yes; reliably no.

Why they usually correlate: the dominant natural source of both is the dissolution of carbonate rock:



This reaction releases **one calcium ion (2 meq of hardness) and two bicarbonate ions (2 meq of alkalinity) simultaneously and in exactly equal equivalents**. Water whose mineral content comes primarily from limestone will therefore show GH ≈ KH.

Why they diverge — the four main cases:

Divergence	Cause	Real-world example
GH >> KH	Calcium/magnesium present as sulfate or chloride salts rather than carbonate (“permanent” or non-carbonate hardness)	Gypsum (CaSO ₄) aquifers; water passing through evaporite deposits; aquariums dosed with GH-only remineralizers
KH >> GH	Alkalinity present as sodium or potassium bicarbonate rather than calcium bicarbonate	Sodium-bicarbonate groundwater; household ion-exchange softened water; aquariums dosed with baking soda; Lake Tanganyika
GH >> KH developing over time	Nitrification and organic acids consume KH while calcium/magnesium remain	Aging, heavily stocked, under-changed aquarium — the Old Tank Syndrome trajectory (Section 21)
Both collapse together	Dilution with RO water; heavy aquasoil ion exchange	RO-based systems; new soil tanks

[EST] Historical terminology worth knowing: - **Temporary hardness** = the portion of hardness associated with bicarbonate; removable by boiling (it precipitates as CaCO₃ scale). Roughly corresponds to the smaller of GH and KH. - **Permanent hardness** = the portion associated with sulfate, chloride, nitrate; **not** removable by boiling. Roughly corresponds to GH minus KH when GH > KH.

This terminology explains why boiling only partially softens water and why kettles scale. See Section 12.

3.7 Why Low KH Makes pH More Susceptible to Change

[EST] This follows directly from the definition. Alkalinity is the amount of acid the water can absorb before pH moves substantially. Low alkalinity = small reservoir = the same acid input produces a larger pH change.

Worked illustration [EST, derived]:

Consider adding an amount of strong acid equal to 0.1 meq/L to two aquariums:

- **Tank A: KH 8 dKH** = 2.85 meq/L alkalinity. The acid consumes 0.1 meq/L, leaving 2.75 meq/L — a 3.5% reduction in buffer. The pH shift is small, on the order of a few hundredths of a unit, because the bicarbonate/carbonic acid ratio barely changes.
- **Tank B: KH 0.5 dKH** = 0.18 meq/L alkalinity. The same acid consumes more than half the entire buffer reserve. pH falls substantially and, if repeated, the buffer is exhausted entirely and pH then falls very steeply.

The non-linear part matters most. Buffering does not degrade gracefully. As alkalinity approaches zero, the pH response to a fixed acid input accelerates sharply. This is the mechanism behind “pH crash” (Section 21): the system appears fine, appears fine, appears fine, and then moves fast.

[INF] Practical monitoring implication: in a low-KH system, **KH is a better early-warning parameter than pH**. pH stays put while the buffer drains, then moves suddenly. KH declines steadily and visibly. An aquarist tracking only pH in a soft-water tank is watching the lagging indicator.

3.8 “High KH Does Not Simply Mean High pH”

[EST] This deserves its own subsection because the confusion is so widespread.

KH sets *how hard it is to move pH*. It does not set *where pH sits*. Where pH sits is determined by the balance between the alkalinity present and the dissolved CO₂ (and any other acids) present.

Same KH, different pH — a demonstration:

At a fixed alkalinity, pH is a function of dissolved CO₂. Approximate relationship at 25 °C for a carbonate-dominated system:

KH	CO ₂ ≈ 3 ppm (equilibrium with air)	CO ₂ ≈ 15 ppm	CO ₂ ≈ 30 ppm
2 dKH	pH ≈ 7.2	pH ≈ 6.5	pH ≈ 6.2
4 dKH	pH ≈ 7.5	pH ≈ 6.8	pH ≈ 6.5
8 dKH	pH ≈ 7.8	pH ≈ 7.1	pH ≈ 6.8
12 dKH	pH ≈ 8.0	pH ≈ 7.3	pH ≈ 7.0

(Values are approximate, derived from the carbonate equilibrium; they assume carbonate is the only significant buffer and carbonic acid the only significant acid — an assumption that fails in many real aquariums. See Section 19 for why this table must not be used to estimate CO₂ in a tank containing driftwood, botanicals, or aquasoil.)

Read across any row: identical KH, wildly different pH. **Read down any column:** the pH differences at constant CO₂ are much smaller than the differences caused by CO₂ at constant KH.

Conclusion [EST]: in a planted or well-stocked aquarium, **dissolved CO₂ is often a larger determinant of the pH you read than KH is**. KH determines how far pH travels for a given CO₂ change, not the absolute value.

Section 4: How pH, GH and KH Relate

Search terms for this section: water profile, hard water, soft water, alkaline water, acidic water, GH vs KH, KH vs pH, GH vs pH, mineral content, buffering, non-carbonate hardness, permanent hardness, water chemistry interpretation.

4.1 The Core Insight

Each of the three parameters answers a different question:

Parameter	Question it answers	Question it does NOT answer
pH	How acidic is the water <i>right now</i> ?	How much mineral is dissolved? How stable is this value?
GH	How much calcium and magnesium is dissolved?	Is the water acidic or alkaline? How resistant is pH to change?
KH	How much acid can the water absorb before pH moves?	How acidic is the water right now? How much calcium is present?

[EST] Therefore: knowing one of the three tells you very little about the other two. Knowing all three tells you a great deal. This is the entire argument for testing all three rather than only pH — which is what most beginner test kits emphasise and most beginners actually do.

4.2 Why pH Alone Does Not Tell You Whether Water Is Hard or Soft

[MYTH] “High pH means hard water. Low pH means soft water.”

This is a **statistical tendency mistaken for a rule**. The tendency exists because limestone geology simultaneously raises calcium, raises bicarbonate, and (via bicarbonate) raises pH. So in nature, hard water is frequently alkaline and soft water is frequently acidic.

But the tendency breaks in ordinary, common situations:

Case: soft water with high pH. - Household ion-exchange water softener output: calcium and magnesium replaced by sodium. **GH near zero, KH unchanged and possibly high, pH unchanged and possibly 8.0+.** This water is “soft” by GH and strongly buffered by KH. - Naturally sodium-bicarbonate-dominated groundwater. - Any aquarium dosed with baking soda but not with a GH remineralizer. - RO water remineralized with a KH-only buffer.

Case: hard water with low pH. - Gypsum-rich (calcium sulfate) source water: high GH, low KH, pH free to fall. - A hard-water aquarium in which nitrification has consumed all the KH: calcium and magnesium are still present at 15 dGH, but alkalinity has reached zero and pH has fallen to 5.8. **This is the classic Old Tank Syndrome profile.** The Merck Veterinary Manual describes exactly this combination — very low or absent alkalinity, low pH, and total hardness still in the hundreds of milligrams per litre (Merck Veterinary Manual, *Disorders and Diseases of Fish*). See Section 21. - A CO₂-injected planted tank on hard water running at pH 6.6 during photoperiod.

Conclusion [EST]: pH is not a hardness measurement and cannot be used as a proxy for one, in either direction.

4.3 Why GH Alone Does Not Tell You How Strongly the Water Is Buffered

GH counts calcium and magnesium ions. Those ions arrived paired with *some* anion. If the anion was bicarbonate, buffering came with them. If the anion was sulfate, chloride or nitrate, **no buffering came with them at all**.

- 15 dGH from calcium bicarbonate → KH also roughly 15 dKH → heavily buffered.
- 15 dGH from calcium sulfate (gypsum) → KH ≈ 0 → essentially unbuffered.

Both test at 15 dGH. A GH test kit cannot distinguish them. This is the direct chemical answer to “how can my GH be really high and my KH be really low?” — a question that appears constantly on aquarium forums. See Scenario 2 in Section 33.

[INF] A GH-only remineralizer (calcium sulfate + magnesium sulfate based, such as products marketed for *Caridina* keeping) deliberately produces this state: hardness without buffering, so that active substrate can hold the pH acidic.

4.4 Why KH Alone Does Not Fully Describe Mineral Content

KH measures acid-neutralizing capacity. That capacity may be paired with calcium (limestone water), sodium (softened water, baking-soda-dosed water, sodium bicarbonate groundwater), potassium (potassium bicarbonate dosing), or in principle other cations.

- 10 dKH from calcium bicarbonate → water contains substantial calcium → snails and shrimp have mineral available.
- 10 dKH from sodium bicarbonate → water may contain almost no calcium → **snail shells still erode; shrimp still fail to molt** — despite a “good” KH reading and a comfortable pH.

[INF] This is a real and under-recognised failure mode: an aquarist raises KH with baking soda to “stabilise” a shrimp or snail tank, sees pH and KH improve, and still loses invertebrates, because the actual limiting nutrient was calcium (GH) and baking soda contains none.

4.5 The Nine Water Profiles

The following table is the central reference of this section. Each row is a real, commonly encountered configuration.

#	GH	KH	pH	Typical origin	System behaviour	Suited to
1	High (12–20)	High (10–18)	High (7.8–8.6)	Limestone aquifer; Rift Lake tanks; crushed coral substrate	Very stable; resists acidification strongly; pH sits high	Rift Lake cichlids, livebearers, goldfish, most snails
2	High (12–20)	Low (0–2)	Variable, often falling (6.0–7.0)	Gypsum/sulfate geology; hard water whose KH has been consumed; aged neglected tank	Mineral-rich but unbuffered; pH prone to decline; classic Old Tank Syndrome trajectory	Requires active management; risky for stability
3	Low (1–3)	High (8–15)	High (7.8–8.5)	Household sodium-exchange softener; sodium bicarbonate groundwater; baking-soda-dosed RO	Stable pH, but calcium-poor. Fish generally fine; invertebrates often are not	Fish that tolerate alkaline water; poor for shrimp and snails without added Ca/Mg
4	Low (0–3)	Low (0–2)	Low and mobile (4.5–6.8)	RO/rainwater; Amazonian blackwater; aquasoil + soft source water	Minimal buffer; pH determined by acid inputs; large moves possible	Soft-water specialists; blackwater biotopes; <i>Caridina</i> ; requires consistent maintenance
5	Moderate (5–8)	Low (1–3)	Moderate, mobile (6.2–7.2)	Soft-to-medium tap water; GH-only remineralized RO	Adequate minerals, limited buffer; CO ₂ swings show up clearly in pH	Planted tanks; many community fish; <i>Caridina</i> setups
6	Moderate (6–10)	Moderate (4–8)	Moderate (7.0–7.6)	Typical mixed-source municipal water	The general-purpose default; forgiving	The large majority of community aquariums
7	Moderate (8–11)	Very high (15–20)	Very high (8.6–9.2)	Lake Tanganyika-type water; sodium/potassium bicarbonate dominated	Extremely strongly buffered; pH sits very high	Tanganyikan cichlids specifically
8	Very low (0–1)	Very low (0–1)	Unpredictable	Pure RO/DI, unmineralized	Not a stable aquarium condition. No buffer, no minerals, extreme pH mobility	Nothing, until remineralized. See Section 10
9	High (14–20)	Moderate (5–8)	Moderate (7.2–7.6)	Mixed carbonate/sulfate geology	Stable, mineral-rich	Hard-water community; snails; livebearers

4.6 Identical Readings, Different Water — Three Demonstrations

Demonstration A: Identical pH, very different GH/KH

	Tank X	Tank Y
pH	7.4	7.4
GH	2 dGH (36 ppm)	16 dGH (286 ppm)
KH	1 dKH (18 ppm)	12 dKH (214 ppm)
Conductivity	~90 µS/cm	~600 µS/cm

Same pH reading. Tank X is soft, nearly unbuffered water where a modest acid input will move pH sharply. Tank Y is hard, heavily buffered water that will hold 7.4 through considerable disturbance. **A hobbyist comparing only pH would conclude these tanks are “the same.”** They are not remotely the same for a shrimp, a snail, a cardinal tetra, or a nitrifying biofilm.

Demonstration B: Identical GH, very different pH

	Tank P	Tank Q
GH	8 dGH	8 dGH
KH	7 dKH (calcium bicarbonate origin)	0 dKH (calcium sulfate origin + aquasoil)
pH	7.7	6.2

Same GH. The calcium and magnesium content is identical. The difference is entirely in the accompanying anions and the presence of an acidifying substrate.

Demonstration C: Identical KH, very different GH

	Tank M	Tank N
KH	9 dKH	9 dKH
GH	10 dGH (calcium bicarbonate)	1 dGH (sodium bicarbonate)
pH	7.8	8.0
Snail shell condition after 6 months	Intact	Eroded, pitted

Same KH, same approximate pH, completely different biological outcome for calcifying invertebrates.

4.7 A Compact Diagnostic Logic

[INF] Given all three readings, the following interpretation logic covers most aquarium situations:

Observation	Most likely interpretation
GH $\approx$ KH, both moderate-to-high	Carbonate/limestone-derived water. Normal, stable, typical of hard tap water
GH $\gg$ KH	Non-carbonate (permanent) hardness present, or KH has been consumed by acid. Check whether KH is declining over time to distinguish these
KH $>$ GH	Sodium/potassium bicarbonate present. Check for a household softener, sodium-bicarbonate groundwater, or baking soda dosing
GH and KH both near zero, pH mobile	RO/rainwater/very soft source, or heavy aquasoil action. Expect pH to be determined by acid inputs
pH lower than source water pH	Something in the tank is producing acid or the source water is degassing differently. Merck's water-quality guidance explicitly flags this: if system pH is below source pH, alkalinity should be checked
pH higher than source water pH	Source water was CO ₂ -supersaturated and has degassed, and/or a carbonate mineral is dissolving in the tank
KH declining month over month	Acid production exceeds replenishment: nitrification, organics, botanicals, or active substrate. See Sections 20 and 21
GH declining month over month with KH stable	Calcium/magnesium being consumed or removed: heavy plant uptake, snail/shrimp shell deposition, chelation by humic substances, or cation exchange by substrate
GH rising over time without dosing	Evaporation top-off with mineralized water concentrating minerals. See Section 23

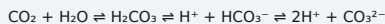
Section 5: Buffering

Search terms for this section: buffer, buffering capacity, buffer system, carbonate buffer, bicarbonate buffer, acid neutralizing capacity, pH stability, pKa, Henderson-Hasselbalch, buffer exhaustion, low buffer system, soft water stability.

5.1 What a Buffer Is

[EST] A buffer is a solution containing a **weak acid and its conjugate base** in appreciable amounts. Adding a strong acid converts some conjugate base to weak acid; adding a strong base does the reverse. In both cases the pH change is far smaller than it would be in unbuffered water, because the added H^+ or OH^- is consumed by the buffer pair rather than remaining free.

The dominant buffer in freshwater aquariums [EST] is the carbonic acid / bicarbonate / carbonate system:



The relevant acid dissociation constants at 25 °C: - **pKa₁ ≈ 6.35** ($H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$) - **pKa₂ ≈ 10.33** ($HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$)

[EST] **Buffering is strongest within roughly ±1 pH unit of a pKa.** This has an important consequence that is almost never stated in hobby writing:

The carbonate system's buffering is most effective around pH 6.35 and around pH 10.33, and is comparatively weaker in the middle of the range — roughly pH 7.5 to 9.0 — where most “buffered” aquariums actually sit.

This is not a contradiction of the usual advice; it is a refinement. Between roughly pH 6.5 and 8.5 a carbonate-buffered aquarium still has substantial *capacity* (a large reservoir of bicarbonate ready to consume acid), even though the *instantaneous* buffer intensity is not at its theoretical maximum. In practice, what protects an aquarium is having a large reservoir of bicarbonate, and that is exactly what a KH measurement reports.

5.2 Buffering Capacity vs. Buffer Intensity

Two distinct ideas that the hobby collapses into one word.

Concept	Definition	Aquarium correlate
Buffering capacity (reservoir)	The total amount of acid the system can neutralize before the buffer is exhausted	This is what KH measures
Buffer intensity (instantaneous resistance)	How much pH moves per unit acid added, at the current pH	Depends on both KH and how close current pH is to the pKa

[INF] **Practical framing:** KH tells you how much acid the aquarium can absorb *in total*. It does not perfectly predict how much pH moves from the *next* drop of acid. For aquarium purposes, treating KH as “the reservoir” is correct and sufficient.

5.3 Why the Reservoir Drains

[STR] Every ongoing process in the list below removes alkalinity permanently. None of them are optional in a functioning aquarium; they are what an aquarium is.

1. **Nitrification.** ~7.14 mg alkalinity as $CaCO_3$ consumed per mg of ammonia-nitrogen oxidised — the single largest ongoing draw in most stocked tanks. See Section 20 for the full derivation and worked examples.
2. **Organic acid accumulation.** Decomposing food, faeces, dead plant material and mulm produce weak organic acids continuously.
3. **Humic and fulvic acids** from botanicals, leaves, peat and driftwood.
4. **Active substrate ion exchange.** Cation exchange resin behaviour in aquasoils removes Ca^{2+} and Mg^{2+} and releases H^+ ; the net effect is loss of both GH and KH.
5. **Any acid dosing** — commercial pH Down, mineral acids, acid buffers.

[EST] **And one process that does NOT drain it: CO_2 .** CO_2 lowers pH without consuming alkalinity. See Section 3.4 and Section 19.

5.4 Does Low KH Automatically Mean an Unstable or Unhealthy Aquarium?

[CON] **No — but it changes what “maintenance” has to mean.**

This is one of the more genuinely contested questions in the hobby, and this document takes the position that **both sides are describing real systems**.

The case that low KH is fine: - Enormous natural freshwater habitats have alkalinity near zero and support extraordinary biodiversity. The Rio Negro is the standard example: ion-poor, acidic (pH commonly 4–5), and home to roughly 8% of the world's freshwater fish species. - Many long-running, healthy aquariums operate at 0–2 dKH indefinitely: blackwater biotopes, *Caridina* tanks, soft-water breeding setups. - A stable pH does not require a *large* buffer if the *acid input rate* is also low and water changes are consistent. Stability is a ratio, not an absolute.

The case that low KH is risky: - The buffer reservoir is small, so the margin for error is small. A missed water change, a dead fish, an overfeeding incident, or a botanical dump can move pH substantially. - pH failure in a low-KH system is non-linear and therefore poorly signalled — the tank looks fine until it isn't (Section 3.7). - Nitrifying bacteria performance declines at low pH and low alkalinity; a pH decline can impair the biofilter, which raises ammonia, which — at low pH — is at least partly offset because ammonia is mostly in the less toxic ionised form. This interaction is genuinely complicated. See Section 20. - Veterinary guidance treats low alkalinity as a clinical risk factor.

Synthesis [INF]: low KH is a **legitimate operating condition that removes a safety margin**. It is appropriate when the keeper is deliberately maintaining it, has consistent source water, keeps bioload modest, and changes water on schedule. It is a hazard when it arrives by accident in a heavily stocked tank whose owner is not monitoring it.

A useful reframing for retrieval: the question is not “is low KH bad?” but “**is the acid production rate in this system matched by the alkalinity replenishment rate?**” A 0 dKH shrimp tank with three shrimp and weekly 30% changes on remineralized RO is in balance. A 0 dKH tank with fifteen adult cichlids and monthly top-offs is not.

5.5 Situations Where Low KH Is Intentional

- **Blackwater and Amazonian biotope aquariums.** Low KH is required for the target pH to be achievable at all — see Section 14 on why acid addition to high-KH water fails.
- ***Caridina* shrimp (Crystal/Taiwan Bee, etc.).** Standard practice uses active substrate to hold pH acidic; the substrate cannot do this against meaningful KH. GH-only remineralizers are used specifically to supply minerals without buffer. See Section 28.
- **Soft-water breeding programmes.** Discus, cardinal tetras, many *Apistogramma*, some rasboras — egg viability is documented to improve at low conductivity and

low pH in specific species.

- **Some high-tech planted tanks.** Lower KH allows a target CO₂ concentration to be reached at a manageable pH; it also means less CO₂ is needed for a given pH drop.

5.6 Situations Where Increasing Buffering May Be Appropriate

- **KH is measurably declining** month over month and approaching zero, in a stocked tank.
- **pH is declining** in parallel with KH, especially below about 6.5 with an established biofilter.
- **Calcifying invertebrates present** in near-zero-KH water (though the more direct fix may be GH — see 4.4).
- **A species-appropriate target requires it** (Rift Lake cichlids, livebearers in very soft supply).
- **The keeper’s realistic maintenance cadence does not match the system’s acid load.** Buffer is, functionally, a substitute for maintenance frequency.
- **Biofilter performance appears impaired** at low pH — ammonia or nitrite readings in an otherwise established tank.

5.7 Separating Aquarium Rules From Chemistry

Common hobby statement	Chemical accuracy	Correction / refinement
“KH controls pH”	Misleading	KH controls pH <i>resistance</i> . CO ₂ and other acids control where pH sits
“KH is a pH lock”	Misleading	It is a consumable reservoir, not a lock
“Higher KH is always safer”	Oversimplified	Higher KH is more forgiving of neglect, but makes deliberate acidification harder and may not suit the species kept
“0 KH means the pH will crash”	Oversimplified	It means pH is free to move. Whether it moves depends on the acid input rate
“If KH is fine, pH is fine”	False	See Section 4.2
“Buffering means pH won’t change”	False	It means pH changes less, until the buffer is consumed
“Adding CO ₂ lowers my KH”	False	CO ₂ lowers pH without consuming alkalinity
“Water changes reset KH”	Conditionally true	Only if source water contains alkalinity. A water change with RO water lowers KH

Section 6: Evaluating “A Stable pH Is More Important Than the Correct pH”

Search terms for this section: stable pH, pH stability, correct pH, pH swing, pH shock, acclimation, adaptation, captive bred, wild caught, chronic exposure, acute exposure, osmotic stress.

6.1 The Claim

“A stable pH is more important than the correct pH.”

This is probably the single most repeated piece of water chemistry advice in the freshwater hobby. This section evaluates it rather than endorsing or debunking it.

6.2 What Is Scientifically Reasonable About It

[STR] 1. Rapid pH change genuinely does impose acid-base and ionoregulatory load. Fish regulate blood pH within narrow limits. An abrupt change in external pH forces compensatory adjustment of branchial acid-base transport (H^+/Na^+ and $\text{HCO}_3^-/\text{Cl}^-$ exchange at the gill), and those same transporters are coupled to sodium and chloride uptake. So an acid-base disturbance is simultaneously an ionoregulatory disturbance. This coupling is well established in fish physiology.

[STR] 2. Low pH increases ion loss in many species. The tight-junction mechanism described in Section 2.5.1 — hydrogen ions displacing calcium at gill tight junctions, increasing paracellular leak — is a real, documented pathway by which acidification causes sodium and chloride loss.

[STR] 3. Large daily pH swings are documented stressors in aquaculture. The SRAC pond literature ties low alkalinity to wide diurnal pH swings and ties those swings to stress, reduced growth, and mortality (Wurts & Durbin, 1992). This is the strongest empirical support the claim has.

[STR] 4. Most fish sold in the hobby have wider tolerance than care sheets imply. The Merck Veterinary Manual notes that most freshwater fish tolerate reasonable pH fluctuation and that lethal limits for many species are approximately pH 4 and pH 10 (Merck Veterinary Manual, *Environmental Diseases of Aquatic Animals in Aquatic Systems*). That is an extremely wide survivable envelope, and it substantially supports the “the exact number matters less than you think” half of the claim.

[INF] 5. The advice is behaviourally protective. The realistic alternative to “keep it stable” is not “achieve the perfect number.” It is “dose pH chemicals repeatedly and produce oscillation.” Given the tools available to a typical hobbyist, the advice steers toward the better of two achievable outcomes. **The claim’s practical value may exceed its literal accuracy.**

6.3 Where It Becomes Oversimplified

[CON] 1. “Stable” and “correct” are not actually in opposition in most real cases. The framing implies a trade-off that usually doesn’t exist. An aquarist whose tap water is pH 7.6 and who keeps species suited to pH 7.6 has both. The trade-off only bites at the extremes — someone with pH 8.4 hard water wanting wild-caught blackwater fish. **The dichotomy is rhetorical, not chemical.**

[CON] 2. pH is frequently not the variable actually doing the damage. This is the most important criticism and deserves emphasis. When an aquarist observes harm after “a pH change,” the pH change was usually accompanied by: - A change in **conductivity/TDS** (osmotic load) - A change in **hardness** (calcium availability at the gill) - A change in **ammonia speciation** (see 6.5) - A change in **temperature** - Handling and transport stress - A change in **dissolved CO_2**

pH is the parameter that is easy to measure, so it receives the attribution. **[INF]** In a substantial fraction of “pH shock” incidents, the more likely proximate cause is an osmotic/ionic change or an ammonia event, with pH as a correlate rather than a cause. This is genuinely under-appreciated and is a strong candidate for a central content point.

[CON] 3. It ignores chronic effects entirely. “Stable” says nothing about whether a chronically wrong value causes cumulative harm. Documented chronic concerns include: - **Nephrocalcinosis** (calcium deposition in kidney tissue) reported in soft-water species held long-term in hard, alkaline water. **[CON]** This is well documented in salmonids in aquaculture; its frequency in ornamental soft-water species is asserted in aquarium literature but less rigorously quantified. - **Reproductive failure** — the single best-documented chronic effect. Many species maintain adequately outside their native chemistry but will not spawn, or produce eggs that do not fertilise or do not hatch. See 6.7. - **Chronic ionoregulatory cost.** A fish continuously working harder at ion uptake pays a metabolic price that may show as reduced growth, reduced disease resistance, or shortened lifespan rather than as acute mortality.

[CON] 4. It is used to justify inaction in situations where intervention is warranted. A tank whose KH has reached zero and whose pH is declining month over month is *not* stable, and “don’t chase pH” is the wrong response to it. The advice was formulated to prevent chemical over-adjustment, and it gets misapplied to genuine drift. See Section 21.

[CON] 5. It does not distinguish direction. A pH rise and a pH fall of equal magnitude are not equivalent. A rise from 6.8 to 7.6 in a tank containing ammonia converts a large fraction of that ammonia to the toxic un-ionised form. A fall of the same size does the opposite. See 6.5.

6.4 Rapid vs. Gradual Change

[STR] The rate of change matters because compensation takes time. Branchial acid-base compensation operates over hours; ionocyte remodeling and gill morphological change operate over days to weeks.

[HOB] The hobby rule of thumb is that fish tolerate gradual change far better than abrupt change. This is well supported in principle.

[CON] The commonly quoted specific limits are not. “Never change pH more than 0.2 per day” and “0.3 per day” and “1.0 is deadly” appear constantly and this research could not trace any of them to a controlled study. They appear to be hobby conventions of unknown origin. See Section 32, Claims 14–16.

What can be said with confidence [INF]: - Slower is more forgiving than faster, for essentially any water chemistry change. - The relevant timescale for gill acid-base compensation is hours, so changes spread over hours are much gentler than changes over seconds. - Fish routinely experience diurnal pH swings of 0.2–0.8 in planted and CO_2 -injected aquariums without evident harm — which sets a practical floor on what magnitude of *gradual, recurring* change is tolerable. - The Merck-cited lethal envelope of roughly pH 4–10 for many species sets a very wide outer boundary that most aquarium incidents are nowhere near.

6.5 The Ammonia Interaction — Why pH Direction Matters More Than pH Magnitude

[EST] This is critical and frequently omitted from discussions of pH stability.

Total ammonia nitrogen (TAN) in water exists as two species in equilibrium: - **NH_4^+ (ionised ammonium)** — relatively non-toxic - **NH_3 (un-ionised ammonia)** — highly toxic

The equilibrium is governed by pH and temperature. The pKa is approximately 9.25 at 25 °C. Below that pH, ammonium dominates; the fraction present as toxic NH_3 **rises roughly tenfold per unit of pH increase** in the range aquariums occupy.

Approximate un-ionised ammonia fraction at 25 °C:

pH	Approximate % of TAN as toxic NH ₃
6.0	~0.06%
6.5	~0.18%
7.0	~0.6%
7.5	~1.8%
8.0	~5.4%
8.5	~15%
9.0	~36%

(Values approximate; temperature raises the toxic fraction further.)

[EST] Practical consequences — three of them, all important:

1. Raising pH in a tank containing ammonia is dangerous. An aquarist who discovers a low-pH, high-ammonia tank (an Old Tank Syndrome tank, or a stalled cycle) and “fixes” the pH with buffer can convert a survivable ammonium load into a lethal ammonia load in minutes. The Merck Veterinary Manual’s recommended handling of Old Tank Syndrome — relocate the fish and rebuild the system, rather than adjust the water around them — reflects exactly this hazard.

2. Low pH masks ammonia problems. A tank at pH 6.2 can carry a TAN reading that would be lethal at pH 8.0 with no visible distress. The ammonia is still there. Any subsequent pH rise — a large water change with alkaline tap water, a buffer addition, a CO₂ system failing off — unmasks it.

3. This is a strong, specific, mechanistic argument for pH stability that has nothing to do with pH itself. The danger is not the pH change; it is what the pH change does to ammonia speciation. **This is arguably the best-evidenced defence of the “stable pH” advice, and it is almost never the reason given for it.**

6.6 Adaptation, Acclimation and Generational Change

[EST] Three distinct concepts routinely conflated:

Term	Timescale	Mechanism	Reversible?
Acclimation	Hours to weeks	Physiological adjustment within an individual: transporter expression, gill remodeling, mucus production	Yes
Developmental plasticity	Within one generation, during early life	Conditions experienced during development shape adult phenotype	Largely not, for that individual
Adaptation	Many generations	Genetic change in the population under selection	Not within an individual

[STR] Individual fish demonstrably acclimate: gill ionocyte density, transporter expression and gill surface area all change measurably in response to water chemistry (see the gill remodeling work cited in Section 2.5.1).

[CON] Whether commercially bred ornamental fish have genuinely *adapted* to hard water over generations is contested. The argument for it: - Many ornamental species have been bred in captivity for dozens to hundreds of generations, frequently in Southeast Asian, Florida, or Czech facilities on local water that may be hard and alkaline. - Selection would favour individuals that reproduce successfully in that water. - Practical experience is that farm-raised cardinal tetras, rams and discus tolerate hard water far better than wild-caught specimens of the same species.

The argument against overstating it: - “Tolerates” is not “is adapted to.” Selection for survival under a stressor produces tolerance, which is not the same as an absence of cost. - Farm water chemistry is not uniform. Many South American and Southeast Asian producers use soft water; some Florida farms use hard well water. **“Captive bred” does not specify a water chemistry.** - Breeding performance is the sensitive endpoint, and it commonly does not transfer. Farm-raised cardinal tetras that live comfortably at 12 dGH frequently will not spawn there. - Genetic evidence for hardness adaptation in ornamental strains is largely absent; the claim rests on husbandry experience.

[INF] Best synthesis available: commercially produced stock is generally more tolerant of a wide chemistry envelope than wild-caught stock of the same species, and the reason is a mixture of genuine selection, developmental acclimation to farm conditions, and survivorship of the shipping chain. **Maintenance tolerance transfers well; breeding requirements transfer poorly.**

6.7 Where Changing Water Chemistry Is Justified

[INF] Deliberate modification is defensible when at least one of the following holds:

- Breeding a species with documented chemistry-dependent reproduction.** The most defensible case by a wide margin. Egg fertilisation, chorion hardening and hatch rate in a number of soft-water species are conductivity- and pH-dependent in ways that no amount of adaptation overcomes.
- Invertebrate mineral requirements are unmet.** A snail with an eroding shell or a shrimp failing to molt in 2 dGH water has a specific, mechanistically clear deficiency. Raising GH is not “chasing” anything.
- The source water is at an extreme.** pH 9.0 well water with 25 dGH is outside the comfortable envelope for the majority of commonly kept species, and dilution is reasonable.
- Alkalinity is depleting toward zero in a stocked system.** This is correcting drift, not chasing a target.
- Wild-caught specimens of a chemistry-specialist species.** Wild cardinal tetras, wild *Apistogramma*, wild-caught blackwater species — these have not been through generations of selection and their tolerance is genuinely narrower.
- A recurring, unexplained pattern of losses** that persists after ammonia, nitrite, nitrate, temperature, oxygen, disease and aggression have been ruled out, in a tank whose chemistry sits far from the species’ native range.

[INF] Conversely, modification is usually *not* justified when: - The only motivation is that a care sheet lists a range that the tap water misses by a few tenths of a pH unit. - The stock is commercially bred, currently healthy, feeding, and breeding. - The intervention would be intermittent chemical dosing rather than a change to the source water. See Section 7.

6.8 Verdict on the Claim

Classification: Generally supported but context-dependent — and supported partly for reasons other than the ones usually given.

- The “stability matters” half is well supported, with the ammonia-speciation mechanism (6.5) being the strongest and least-cited justification.
- The “the number doesn’t matter” half is supported for **maintenance of commercially bred generalist species** and **not** supported for **breeding**, for **wild-caught specialists**, or for **invertebrate mineral requirements**.
- The framing as a trade-off is rhetorical. In most real aquariums, choosing appropriate stock for the existing water delivers both.

- The advice is frequently misapplied to justify ignoring genuine parameter drift.

Section 7: Evaluating “Don’t Chase pH”

Search terms for this section: don’t chase pH, pH up, pH down, pH adjuster, pH buffer, pH rebound, chemical pH adjustment, phosphate buffer, acid buffer, alkaline buffer.

7.1 The Claim

“Don’t chase pH.”

Meaning: do not repeatedly dose chemicals to move aquarium pH toward a target number.

7.2 Why the Advice Exists — the Rebound Mechanism

[EST] The mechanism is straightforward and is the core of the argument.

Consider hard tap water at 12 dKH (214 mg/L alkalinity) and pH 8.0, and an aquarist who wants pH 6.8.

1. They add a commercial “pH Down” acid product. Hydrogen ions are added.
2. Those hydrogen ions are immediately consumed by bicarbonate: $\text{H}^+ + \text{HCO}_3^- \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_2$.
3. pH drops briefly, then returns as the system re-equilibrates and excess CO_2 outgasses.
4. **Alkalinity has been reduced** — a small amount of the reservoir was spent — but pH is back near where it started.
5. The aquarist doses again. And again.
6. Each dose consumes more alkalinity. pH is jerked down and returns up each time, producing exactly the oscillation the aquarist was trying to avoid.
7. Eventually, after enough acid has been added, **the alkalinity reservoir is exhausted**. At this point the next dose — the same size as all the previous ones, which “did nothing” — produces a large and rapid pH drop, because there is nothing left to absorb it.

[EST] **This is the failure mode.** Repeated chemical pH adjustment against a buffered system produces (a) many small oscillations, followed by (b) one large uncontrolled crash. It is worse than doing nothing, and it is worse than doing it properly.

Compact statement for retrieval: *You cannot lower pH against alkalinity. You can only destroy the alkalinity first, and then the pH falls fast.* The correct sequence for anyone who genuinely needs low pH is to **reduce KH first (by dilution with RO or rainwater), and only then acidify**. See Sections 14 and 15.

7.3 The Phosphate Buffer Problem

[HOB/CON] Many older and cheaper “pH 7.0 / pH 6.5” products are **phosphate buffers**, not acids or carbonates. These deserve separate treatment because their failure mode is different.

What they do: - They impose a phosphate buffer system that holds pH near the product’s target regardless of the carbonate system. - They therefore **do** work, in the narrow sense of moving and holding pH.

Why they cause problems: - **They inflate the KH test result.** Phosphate is a proton acceptor and is counted by the alkalinity titration. An aquarist using a phosphate buffer may read “KH 6” and have almost no *carbonate* alkalinity. - **They invalidate any pH/KH-based CO_2 estimation** completely (Section 19). - **They add phosphate**, typically at levels far above what plants need, which in a lit tank is a well-known algae contributor. - **They must be re-dosed at every water change** — the definition of a treadmill.

[INF] **Practical guidance:** if pH must be modified, changing the *source water* (dilution, remineralization, mineral substrate) is preferable to imposing a competing buffer system on the tank.

7.4 When Deliberate pH Adjustment Is Legitimate

“Don’t chase pH” is beginner advice about *method*, not a prohibition on *goals*. Legitimate applications:

7.4.1 Breeding specific species

[STR/HOB] Documented and widely practised. Examples with reasonably solid basis: - **Discus** — commercial and hobby breeding practice uses low conductivity (commonly cited under about 150 $\mu\text{S}/\text{cm}$) and slightly acidic pH for fertilisation and hatch. Eggs in hard water frequently fail to fertilise or fungus before hatching. **[HOB, extremely widely reported]** - **Cardinal tetra, rummynose tetra, many small characins** — soft, acidic conditions for egg viability. - **Many *Apistogramma* and dwarf cichlids** — sex ratio in some species is influenced by pH during larval development. **[STR]** This is an unusual case where pH has a documented developmental effect beyond survival, and it is well worth flagging as a genuine, non-folkloric pH effect. - **Killifish, some rasboras, chocolate gouramis.**

7.4.2 Blackwater and biotope aquariums

[HOB] Where low pH is the point of the display, and is achieved through appropriate means (soft source water, botanicals, aquasoil) rather than acid dosing.

7.4.3 *Caridina* shrimp systems

[HOB] Standard practice: RO water, GH-only remineralizer, active substrate holding pH around 5.8–6.5. This is not “chasing pH”; it is building a system whose steady state is acidic.

7.4.4 Rift Lake cichlid systems on soft water

[HOB] Raising KH, GH and pH using carbonate mineral substrates and/or Rift Lake salt mixes. Mechanically the easy direction — raising pH against low buffering works; lowering pH against high buffering does not.

7.4.5 Extremely unsuitable source water

[INF] pH 9.2 water from a sodium-bicarbonate aquifer, or 30 dGH water from a deep limestone well, may genuinely warrant RO dilution regardless of species. The correct intervention here is **changing the water going in**, not dosing the tank.

7.5 What Distinguishes Legitimate Adjustment From “Chasing”

Chasing pH	Legitimate adjustment
Dosing the tank	Changing the source water
Intermittent, reactive	Continuous, built into the system
Targeting a number	Targeting an environment
Against the existing buffer	After removing or accounting for the buffer
Requires ongoing correction to hold	Holds at steady state without intervention
Produces oscillation	Produces a new stable point
Motivated by a care sheet	Motivated by an observed requirement (breeding, invertebrate mineral needs, extreme source water)

[INF] This table is the practical resolution of the “don’t chase pH” debate. The advice is not really about pH at all — it is about the difference between **treating symptoms in the tank** and **changing the inputs to the system**.

7.6 Verdict on the Claim

Classification: Strongly supported as method guidance; oversimplified as a universal prohibition.

- The chemistry behind it is sound and the failure mode it warns about is real and common.
- It is correctly stated as: *“Do not repeatedly dose chemicals to move tank pH against the water’s buffering.”*
- It is incorrectly stated as: *“Never try to change your water chemistry.”*
- The most common misapplication is using it to dismiss legitimate breeding requirements, legitimate invertebrate mineral requirements, and genuine alkalinity depletion.

Section 8: Species Requirements — How to Interpret Them

Search terms for this section: fish water parameters, species requirements, soft water fish, hard water fish, natural habitat, wild caught, captive bred, tank bred, care sheet, breeding parameters, conductivity, biotope.

8.1 The Framing Problem

[INF] Published parameter ranges answer at least five different questions, and rarely say which:

1. Where does this species live in the wild?
2. Where will this species survive in captivity?
3. Where will this species thrive, feed, colour up and behave normally?
4. Where will this species breed successfully?
5. Where is this species commercially produced and shipped from?

These give **different answers**. A wild cardinal tetra comes from pH 4–5 water at under 20 µS/cm (answer 1). Farm-raised cardinal tetras survive at pH 7.8, 12 dGH (answer 2). They generally thrive across roughly pH 6.0–7.5 (answer 3). They breed reliably only in very soft, acidic water (answer 4). They are commonly farmed in soft water in Southeast Asia, and wild-collected ones come from Brazil (answer 5).

A single published range like “pH 5.0–7.5” collapses all five. This is the root cause of most care-sheet disagreement. See Section 30.

8.2 Soft-Water Species

Native environment characteristics: low conductivity (often 10–80 µS/cm), very low hardness (often < 2 dGH), very low alkalinity (often < 1 dKH), acidic pH (4.0–6.5), frequently high dissolved organic carbon.

Species / group	Native water (approximate)	Captive maintenance	Breeding requirement
Cardinal tetra (<i>Paracheirodon axelrodi</i>)	Rio Negro system; pH 4–5.5, conductivity often <20 µS/cm, hardness near zero	Farm stock tolerates a wide range; commonly kept pH 6.0–7.5	Very soft, acidic; low conductivity essential
Neon tetra (<i>P. innesi</i>)	Soft, acidic Peruvian/Colombian blackwater and clearwater	Very adaptable as farm stock; among the most forgiving soft-water fish	Soft, acidic; commercial production uses very soft water
Discus (<i>Symphysodon</i> spp.)	Amazon floodplain; soft, warm, acidic	Farm stock kept successfully across a wide range; hard water tolerated for maintenance	Low conductivity strongly indicated; hard water commonly prevents fertilisation/hatch
Rams (<i>Mikrogeophagus ramirezi</i> , <i>M. altispinosus</i>)	Llanos savanna waters; soft, warm	<i>M. ramirezi</i> farm stock is notoriously fragile for reasons largely unrelated to hardness (see note below)	Soft, acidic; pH influences fry survival
Rasboras (<i>Trigonostigma</i> , <i>Boraras</i>)	Southeast Asian peat swamp forest; very soft, pH 3.5–6 in extreme cases	Adaptable; <i>Boraras</i> species less so	Soft, acidic
Corydoras (many, esp. <i>C. panda</i> , <i>C. sterbai</i> , wild <i>Corydoras</i>)	Variable — this genus spans a wide chemistry range	Most tank-bred species very adaptable	Soft water and cool water changes are common spawning triggers
Apistogramma	Soft to very soft; varies by species and locality	Species-dependent; some are genuinely narrow	Soft, acidic; pH influences sex ratio in some species
Betta wild species (<i>B. albimarginata</i> , <i>B. macrostoma</i> etc.)	Peat swamp; very soft, very acidic	Genuinely require soft acidic water	Soft, acidic

Note on German blue rams [CON]: these are widely reported as fragile. The commonly given explanation is water chemistry. A competing and arguably better-supported explanation is that intensive line-breeding, high stocking density at farms, hormone treatment of some stock, and prophylactic antibiotic use have produced genetically weakened, pathogen-loaded fish. **Attributing ram losses primarily to pH or GH may be a misdiagnosis.** This is worth flagging as an example of the hobby defaulting to water chemistry when the actual variable is stock quality.

8.3 Hard-Water Species

Native environment characteristics: moderate-to-high conductivity, moderate-to-high hardness, moderate-to-high alkalinity, neutral-to-alkaline pH.

Species / group	Native water (approximate)	Notes
Guppy (<i>Poecilia reticulata</i>)	Caribbean/N. South American streams; variable but frequently hard and alkaline	Extremely adaptable; modern fancy strains are fragile for reasons of inbreeding and disease load rather than chemistry
Molly (<i>Poecilia sphenops</i> , <i>P. latipinna</i> , <i>P. velifera</i>)	Frequently brackish or hard alkaline coastal water	Genuinely mineral-demanding. Mollies in very soft water are a well-recognised failure case: shimmying, disease susceptibility. One of the clearer real examples of hardness mattering
Platy, swordtail (<i>Xiphophorus</i>)	Central American hard, alkaline streams	Adaptable but do best in moderate-to-hard water
Lake Malawi cichlids	pH ~7.7–8.6; conductivity ~210–285 µS/cm; alkalinity ~2 meq/L (~110 mg/L, ~6 dKH); hardness roughly 6–9 dGH	Note: Malawi is alkaline but only moderately hard. The hobby image of “extremely hard water” overstates GH
Lake Tanganyika cichlids	pH ~8.6–9.2; conductivity ~600 µS/cm; alkalinity ~6+ meq/L (~18 dKH); hardness ~10–12 dGH	KH substantially exceeds GH — a real-world example of Profile 7 in Section 4.5. Sodium/potassium bicarbonate dominated
Lake Victoria cichlids	Alkaline, moderately mineralised	Similar management to Malawi
Central American cichlids (<i>Amatitlania</i> , <i>Thorichthys</i> , <i>Vieja</i>)	Hard, alkaline limestone-influenced water	Generally forgiving
Goldfish	Hard alkaline water preferred; East Asian origin	Very adaptable; heavy waste production makes buffer reserve genuinely valuable
Most freshwater snails	Calcium-bearing water	See Section 28 — shell integrity is a direct GH requirement

Important correction to a common hobby belief: Rift Lake water is characterised by **high alkalinity and high pH, with only moderate general hardness**. Aquarists who target “very hard” GH for Malawi tanks are frequently exceeding natural conditions. The parameter that actually matters for these fish is alkalinity/pH, not GH.

8.4 Broadly Adaptable Species

[INF] Species that thrive across most of the range aquarists encounter, and for which water chemistry is rarely the limiting factor:

- Zebra danio and most *Danio*
- White cloud mountain minnow
- Most tank-bred *Corydoras*
- Cherry barb, most *Puntius/Pethia* barbs
- Bristlenose pleco (*Ancistrus*)
- Kuhli loach
- *Betta splendens*
- Most tank-bred rainbowfish
- Endler’s livebearer
- Amano shrimp (*Caridina multidentata*) — considerably more tolerant than ornamental *Caridina*
- *Neocaridina davidi* (cherry shrimp) — tolerant across roughly 4–15 dGH

8.5 Survival vs. Thriving vs. Breeding

[INF] A three-tier framework that resolves most care-sheet confusion:

Tier	Definition	Envelope width	Evidence quality
Survival	Fish stays alive	Very wide — Merck cites lethal limits near pH 4 and pH 10 for many species	Reasonably good
Thriving	Normal colour, appetite, activity, growth, immune function, lifespan	Narrower; species-dependent	Poor. Almost never measured formally in ornamental species; mostly husbandry experience
Breeding	Successful fertilisation, hatch, fry survival	Often much narrower and sometimes strictly required	Moderate — supported by consistent commercial breeding practice, though rarely by controlled published trials in ornamental species

[CON] The weakest evidence tier is “thriving,” which is unfortunately the one most care sheets purport to describe. Very few controlled studies exist measuring growth, immune function or longevity of ornamental species across hardness gradients. Most claims in this tier are experienced-hobbyist inference. This should be stated honestly rather than papered over.

8.6 Should Care-Sheet Ranges Be Treated as Boundaries?

[INF] No — for four reasons:

1. **Provenance is usually unknown.** The range may be habitat data, farm data, or copied from another site. See Section 30.
2. **They omit the variable that often matters more.** Almost no care sheet lists conductivity or GH separately from pH, and for soft-water species conductivity is frequently the more relevant parameter.
3. **They do not distinguish wild from captive-bred stock,** which is the single largest determinant of how narrow the requirement actually is.
4. **They do not distinguish maintenance from breeding.**

[INF] A better decision rule: *Is my water inside the envelope where this species is routinely and successfully kept, given its origin (wild vs. farmed) and my goal (keeping vs. breeding)?* This is a different question from *Does my pH match the number on the care sheet?* — and it is answerable.

Section 9: Tap Water and Source Water

Search terms for this section: tap water, municipal water, well water, water report, consumer confidence report, CCR, source water, seasonal variation, water softener, ion exchange softener, chloramine, blending, baseline testing.

9.1 Establishing a Source Water Baseline

[INF] This is the single highest-value diagnostic action available to an aquarist, and almost nobody does it systematically. The recommended protocol:

Test the source water itself, not just the tank, and test it twice:

Test	When	What it tells you
pH — immediately on draw	At the tap	The transient pH the tank will see during a water change
pH — after 24h standing or 1h aeration	Equilibrated	The pH the tank will trend toward
KH	Either (KH does not change on standing)	The buffer being supplied at every water change
GH	Either	The mineral being supplied at every water change
TDS/conductivity	Either	Total ionic load; baseline for detecting seasonal change
Nitrate	Either	Whether the source is already contributing nitrate (agricultural areas)

Repeat quarterly. Source water is not constant (see 9.3).

[INF] Interpreting the tank-vs-source comparison is where the diagnostic value is:

Tank vs. source	Interpretation
Tank pH < source equilibrated pH	Acid is being produced or added in the tank. Check KH — it is probably lower than source too. Causes: nitrification, organics, botanicals, aquasoil, CO ₂
Tank KH < source KH	Alkalinity is being consumed faster than water changes replace it. The core Old Tank Syndrome signal
Tank GH > source GH	Mineral accumulation — usually evaporation top-off, or a dissolving carbonate source
Tank GH < source GH	Mineral removal — plant/invertebrate uptake, cation-exchange substrate, chelation by humics
Tank TDS >> source TDS	Accumulation of dissolved organics, nitrate and minerals. A useful general “how overdue is this water change” indicator

9.2 Municipal Water Reports

[INF] In the US, utilities are required under the Safe Drinking Water Act to publish an annual **Consumer Confidence Report (CCR)**. Equivalents exist in most developed countries.

What to look for: - **Hardness** — sometimes as mg/L CaCO₃, sometimes as grains per gallon, sometimes as separate calcium and magnesium in mg/L (use the conversion in Section 2.3) - **Alkalinity** — usually mg/L as CaCO₃; divide by 17.848 for dKH - **pH** — note this is measured at the plant, not at the tap, and is often the least useful figure in the report for aquarium purposes - **Sodium** — high sodium with high alkalinity suggests sodium bicarbonate water; also flags whether the utility softens - **Disinfectant type — chlorine vs. chloramine**. Critical for dechlorinator selection, though not a GH/KH/pH issue - **Source** — surface water (reservoir/river) vs. groundwater (well/aquifer), and whether the utility blends between them

[INF] **Limitations of the report:** - It reports annual averages and ranges, not today’s value. - It reports conditions leaving the treatment plant, not at the tap after travelling through the distribution system and the building’s plumbing. - Copper and lead pickup from household plumbing does not appear. - It is nonetheless the best available picture of what the water *is made of*, which a GH/KH test kit cannot tell you.

9.3 Seasonal and Operational Variation

[STR] Source water is not a constant. Documented sources of variation:

Cause	Effect	Typical timescale
Source switching — utility blending reservoir and well water in different proportions	Can change GH and KH substantially, sometimes within days	Days-seasons
Spring snowmelt / heavy rain in surface-water systems	Dilution: GH, KH and conductivity fall	Weeks, seasonally
Drought / low reservoir levels	Concentration: GH, KH and conductivity rise	Weeks-months
Seasonal treatment changes — e.g. switching from chloramine to free chlorine for periodic main flushing	Not a hardness change, but a dechlorination concern	Weeks, usually announced
pH adjustment for corrosion control — utilities routinely raise pH and alkalinity to protect pipes	Can shift delivered pH by several tenths	Ongoing, occasionally revised
Main breaks, flushing, hydrant work	Turbidity, sediment, transient chemistry change	Days
Well water seasonal table changes	GH/KH shift with water table depth	Seasonal

[INF] **Practical consequence:** an aquarist who reports “my parameters changed and I didn’t do anything” may be entirely correct. **Retesting the source water is the first diagnostic step**, and it is skipped almost universally in favour of investigating the tank.

9.4 Well Water

[INF] Private well water differs from municipal water in several aquarium-relevant ways:

- **Frequently CO₂-supersaturated.** Groundwater under pressure holds far more CO₂ than surface water. This produces the classic large pH rise on aeration — sometimes a full unit or more. **Well water should essentially always be aerated before use.**
- **Often high in iron and manganese.** Can produce visible orange precipitate on aeration; contributes marginally to a GH reading.
- **May contain hydrogen sulfide** (rotten-egg odour) — removed by aeration.
- **May be extremely hard** if the aquifer is limestone or dolomite.

- **May be extremely soft** if the aquifer is granite or sandstone.
- **Not chlorinated** — dechlorinator is unnecessary, though many well owners use conditioners for heavy-metal binding.
- **Not tested by anyone unless the owner pays for it.** No CCR exists. A one-time laboratory panel is genuinely worthwhile.
- **Nitrate contamination** is a real risk in agricultural areas and can be substantial — occasionally exceeding the tank’s own nitrate level, which produces a very confusing water-change result.

9.5 Household Water Softeners — a Critical Case

[EST] **How a conventional household softener works:** water passes through a bed of ion-exchange resin charged with sodium (or potassium) ions. Calcium and magnesium in the water bind to the resin and **sodium is released into the water in exchange**, on an equivalent-for-equivalent basis. The resin is periodically regenerated with brine.

[EST] **What this does to aquarium-relevant parameters:**

Parameter	Effect of a sodium-exchange softener
GH	Drops to near zero. Calcium and magnesium removed
KH / alkalinity	Essentially unchanged. Bicarbonate is an anion; the resin exchanges cations only
pH	Essentially unchanged , or slightly raised
TDS / conductivity	Roughly unchanged or slightly higher. Ions were swapped, not removed
Sodium	Substantially increased , proportional to the hardness removed

[INF] **Why this is a genuine trap for aquarists:** the water now tests as “**soft**” (GH near zero) while being **highly buffered** (KH unchanged) and **ionically loaded** (TDS unchanged) and **sodium-rich**. This is Profile 3 in Section 4.5.

Consequences: - An aquarist who wanted soft water for soft-water fish has *not* got soft water in the sense that matters. Conductivity and osmotic load are unchanged; only the specific ions changed. - Calcifying invertebrates fail — snails erode, shrimp fail to molt — despite “perfect” KH and pH readings and despite the water testing soft. - Plants may show calcium and magnesium deficiency. - pH will not fall as hoped, because alkalinity was never removed.

[INF] **Verdict: sodium-exchange softened water is generally a poor choice for aquarium use**, and the standard recommendation is to plumb an **unsoftened bypass tap** (many installations already have one at an outdoor spigot or the kitchen cold tap) or to use RO.

Potassium-chloride-regenerated softeners substitute potassium for sodium. Potassium is less problematic biologically for plants and arguably for fish, but the fundamental issue — hardness removed, alkalinity retained, TDS retained — is unchanged.

9.6 Blending Water Sources

[EST] Blending is linear for conservative parameters. If you mix volumes V₁ and V₂ of waters with hardness H₁ and H₂:

$$H_{\text{blend}} = (V_1 H_1 + V_2 H_2) / (V_1 + V_2)$$

This works accurately for **GH, KH, TDS and conductivity**. It does **not** work for pH, because pH is logarithmic and buffered — blending pH 6.0 and pH 8.0 water does not give pH 7.0. The resulting pH is determined by the combined alkalinity and CO₂ of the mixture, and is usually much closer to the more strongly buffered water’s pH.

Worked blending examples:

Target: 6 dGH / 4 dKH from tap at 14 dGH / 10 dKH, using RO at 0/0. - Required fraction of tap water for GH: 6 ÷ 14 = 0.43 - Resulting KH at that fraction: 0.43 × 10 = **4.3 dKH ✓** - **Blend: roughly 43% tap, 57% RO.** In practice, 40:60. - Note how conveniently this worked because the source’s GH:KH ratio (1.4:1) was close to the target ratio (1.5:1). **Blending can only preserve the source water’s GH:KH ratio.** If you need a different ratio, blending alone cannot get you there — you need RO plus selective remineralization.

Target: 8 dGH / 2 dKH from tap at 14 dGH / 10 dKH. - For KH 2: tap fraction = 2 ÷ 10 = 0.20 - Resulting GH: 0.20 × 14 = **2.8 dGH ✗** (wanted 8) - **Blending cannot achieve this.** The solution is RO + a GH-only remineralizer. See Sections 10 and 11.

[INF] **This is the clearest practical argument for RO over blending:** blending changes *magnitude* but preserves *ratio*. RO plus remineralization allows the ratio to be set independently.

9.7 Very Hard and Very Soft Tap Water

Very hard tap water (>15 dGH, >12 dKH): - Fully suitable for a wide range of stock without modification. - The realistic decision is species selection, not water modification. - If modification is genuinely needed (wild soft-water species, *Caridina*), RO dilution is the only method that works reliably. Chemical softening does not. - Nuisance issues: scale on glass, heaters, impellers and lids; more frequent equipment cleaning.

Very soft tap water (<3 dGH, <2 dKH): - Excellent for soft-water species and for blackwater setups. - **Requires attention to alkalinity depletion** — the buffer reservoir is small and nitrification will draw it down. Monitor KH, not just pH (Section 3.7). - May require GH supplementation for invertebrates and for calcium/magnesium-demanding plants. - Crushed coral in the filter is the common low-effort remedy where a modest, self-limiting increase is wanted (Section 17).

Section 10: RO and RO/DI Water

Search terms for this section: reverse osmosis, RO water, RO/DI, deionization, DI resin, TDS, remineralization, remineralizer, GH+, GH/KH+, wastewater ratio, membrane, sediment filter, carbon block, distilled water.

10.1 What RO Does

[EST] Reverse osmosis forces water under pressure through a semi-permeable membrane that rejects the large majority of dissolved ions and molecules. A typical residential aquarium RO system consists of:

1. **Sediment filter** — removes particulates that would foul the membrane.
2. **Carbon block** — removes chlorine and chloramine, which would destroy a thin-film composite (TFC) membrane. **Chloramine requires a carbon block specifically rated for it, or a longer contact time; standard carbon is inadequate.** This is a common and expensive mistake.
3. **RO membrane** — the actual separation step. Typical rejection rates 90–99% of dissolved solids depending on membrane type, pressure, temperature and age.
4. **Optional DI (deionization) stage** — mixed-bed ion exchange resin polishing the RO output to essentially 0 TDS.

Approximate typical performance:

Stage	Typical output TDS from 300 ppm feed
Tap in	300 ppm
After RO membrane (95% rejection)	~15 ppm
After DI resin	0–2 ppm

10.2 What RO Removes and What It Leaves

Constituent	RO removal	Notes
Calcium, magnesium (GH)	~95–99%	GH of RO output essentially 0 dGH
Bicarbonate/carbonate (KH)	~90–98%	KH of RO output essentially 0 dKH
Sodium, chloride	~90–98%	
Nitrate	~85–95%	Useful where source nitrate is high
Phosphate, silicate	High	Relevant to algae control and to diatoms
Heavy metals	High	
Dissolved CO₂	Poor — passes through	Important: CO ₂ is a neutral dissolved gas and crosses the membrane. This is why fresh RO water often tests at pH 5.5–6.5 despite containing essentially nothing
Chlorine	Removed by the carbon prefilter, not the membrane	Chlorine destroys TFC membranes
Bacteria	Largely rejected, but systems are not sterile	

[EST] The pH of pure RO water is a special case worth stating clearly. Pure water with zero alkalinity has **no buffering whatsoever**. Its pH is set entirely by whatever trace acid or base is present — usually dissolved atmospheric CO₂. Fresh RO water frequently reads pH 5.5–7.0 and the reading is **meaningless and unstable**: it will change if you breathe on it, aerate it, or leave it standing. **Measuring the pH of pure RO water is not a useful action.** What matters is what it becomes after remineralization.

10.3 Why Pure RO Water Is Not Aquarium Water

[EST] Pure RO/DI water should not be used unremineralized in a general aquarium. Reasons:

1. **No buffer at all.** pH is completely unconstrained. Any acid input moves it freely.
2. **No calcium or magnesium.** Fish ionoregulation, invertebrate mineralisation and plant nutrition all lack a water-column source.
3. **Osmotic extreme.** The gradient between fish plasma (roughly 300 mOsm) and near-zero-ion water is at its maximum, so passive ion loss is at its maximum and the metabolic cost of ion uptake is highest.
4. **[MYTH] “RO water is too pure and will hurt fish by osmosis / draw water into their cells.”** The mechanism as usually stated is wrong. Fish are not osmotically damaged in the way plant cells are; they are osmoregulators with functional gills, kidneys and skin. The real issue is **ion loss and the metabolic cost of replacing it**, plus the absence of calcium at the gill. The concern is legitimate; the popular explanation is not. See Section 32, Claim 17.

[EST] An important qualification that complicates the picture — and is genuinely important:

Natural Amazonian blackwater is *also* extremely ion-poor and acidic, and fish live in it. The difference is **dissolved organic carbon**. Research on Rio Negro water has shown that native DOC provided nearly complete protection against ionoregulatory disturbance in zebrafish acutely exposed to pH 4.0 in ion-poor water: DOC cut diffusive sodium and chloride loss and markedly stimulated sodium uptake that would otherwise have been fully inhibited (Duarte, Smith, Val & Wood, 2016, *Scientific Reports* 6:20377, DOI 10.1038/srep20377). Parallel work found comparable protection in Rio Negro stingrays (Wood et al., 2003, *Physiological and Biochemical Zoology* 76:12–27).

[INF] **Consequence for the hobby:** pure RO water is not equivalent to natural blackwater. Blackwater is ion-poor **plus** organically rich, and the organic fraction is doing measurable physiological work. This is a strong, evidence-based argument for botanicals and tannins in soft-water aquariums that has nothing to do with pH — and it is almost entirely absent from hobby discussion, which frames tannins purely as a pH-lowering or aesthetic matter. **See Section 16.6 — this is one of the most significant findings in this research.**

10.4 Remineralization

[INF] Remineralization means adding a controlled, known mineral content back to RO water. Three general approaches:

10.4.1 GH-only remineralizers

Composition: typically calcium sulfate and magnesium sulfate, sometimes with potassium sulfate and trace elements. **Effect:** raises GH; **does not raise KH**; minimal direct pH effect. **Use case:** *Caridina* shrimp systems where active substrate must be free to hold pH acidic; soft-water fish tanks needing minerals without buffer; planted tanks wanting Ca/Mg without alkalinity. **Named examples [HOB]:** SaltyShrimp Bee Shrimp Mineral GH+; Seachem Equilibrium (potassium, calcium, magnesium, iron and manganese sulfates — raises GH, not KH).

10.4.2 GH/KH remineralizers

Composition: calcium/magnesium salts plus carbonate or bicarbonate. **Effect:** raises GH **and** KH together; raises pH toward the buffered value. **Use case:** *Neocaridina* shrimp, general community fish tanks on RO, anywhere a conventional buffered profile is wanted. **Named examples [HOB]:** SaltyShrimp GH/KH+; various “RO Right”-type products.

10.4.3 Individual salts (DIY)

Composition: the keeper doses specific compounds separately. Maximum control, requires a scale accurate to 0.01 g and some arithmetic. See Section 11 for full dosing derivations.

10.4.4 Manufacturer dosing figures [HOB — manufacturer documentation]

- **Seachem Equilibrium:** 16 g per 80 L (approximately 20 US gallons) raises GH by 3 dH ($\approx$ 50 ppm). Does not raise KH.
- **SaltyShrimp Bee Shrimp Mineral GH+:** approximately 3 g per 20 L gives roughly 6 dGH and approximately 200 μ S/cm conductivity.

(These are manufacturer figures and are reproduced here as manufacturer claims; independent verification was not located.)

10.5 Risks of Inconsistent Remineralization

[INF] This is the main practical hazard of an RO-based system, and it is under-discussed relative to how common it is.

An aquarist on straight tap water has a source that varies slowly and modestly. An aquarist on RO plus remineralizer has a source that varies **exactly as much as their measuring technique varies**. Common failure modes:

- **Volume error.** Remineralizing “a five-gallon bucket” that is actually filled to 4.2 gallons produces a 19% overdose.
- **Scoop-based dosing.** Powder density varies with compaction and humidity; scoops are far less reproducible than a scale.
- **Dosing to the tank rather than to the change water.** This spikes the tank at the moment of dosing and produces an oscillation with every water change.
- **Failing to verify with a TDS meter.** A TDS meter is inexpensive and is the single best consistency check available for an RO system. It will catch a mis-dose immediately.
- **Membrane degradation.** RO output TDS creeps up as the membrane ages. If the keeper remineralizes to a fixed dose rather than to a target TDS, the final water drifts upward over months. **Test RO output TDS periodically, not just the finished water.**
- **DI resin exhaustion.** Exhausted resin can release previously captured ions.

[INF] **Recommended discipline:** dose by mass, to a marked volume, in the mixing vessel, and confirm with a TDS meter before use. This converts RO from the least consistent source to the most consistent one.

10.6 Blending RO With Tap Water

[EST] Use the linear mixing equation from Section 9.6. Blending is genuinely useful when: - The tap water’s GH:KH ratio is already close to the desired ratio. - The goal is simply “less of everything.” - Cost or effort of full remineralization is not warranted.

Limitations (repeating for retrieval): blending preserves the source water’s ionic *ratio*. It cannot produce high GH with low KH from a source where GH and KH are equal. For that, RO plus selective remineralization is required.

Blending reference table (tap at 14 dGH / 10 dKH, RO at 0/0):

% tap	Resulting GH	Resulting KH
100%	14.0	10.0
75%	10.5	7.5
50%	7.0	5.0
40%	5.6	4.0
33%	4.6	3.3
25%	3.5	2.5
20%	2.8	2.0
10%	1.4	1.0
0%	0	0

10.7 When RO Is Genuinely Useful

[INF] 1. **Source water is extreme** — very hard, very high nitrate, high phosphate, unusual contaminants, or an aquifer with problematic chemistry. 2. **Species genuinely require soft water** — wild-caught blackwater fish, *Caridina* shrimp, discus breeding, wild *Betta* species. 3. **Consistency is the goal.** RO plus remineralization is the only way to make source water *identical* every time, which matters most for sensitive invertebrates and for breeding. 4. **Independent control of GH and KH is needed** — the case blending cannot solve. 5. **Source water varies seasonally** and the variation is causing problems. 6. **Municipal water contains something undesirable** that a dechlorinator does not address.

10.8 When RO Is Unnecessary

[INF] 1. **Tap water is already in a workable range** for the stock kept — by far the most common situation. 2. **Stock is commercially bred and generalist** — most community fish. 3. **The motivation is matching a care sheet number** rather than an observed problem. 4. **The keeper cannot commit to consistent remineralization.** Inconsistent RO is worse than consistent tap. 5. **The tank is large and the water volume required makes RO impractical** relative to the benefit.

[INF] **Blunt framing worth recording:** RO does not improve water. It *removes* water chemistry, handing the keeper full responsibility for putting it back correctly every single time. That is an upgrade for someone who wants control and a downgrade for someone who wants reliability.

10.9 Cost, Wastewater and Maintenance

Consideration	Detail
Wastewater ratio	Residential RO typically produces 3–4 L of reject brine per 1 L of product water; higher-efficiency and permeate-pump systems can approach 1:1. The reject water is safe for irrigation, laundry pre-fill, or cleaning
Production rate	Rated GPD figures assume roughly 60 psi and 25 °C. Cold winter feed water can halve output
Prefilter replacement	Sediment and carbon typically every 6–12 months, sooner on dirty or chloraminated supplies
Membrane life	Typically 2–5 years; replace when rejection rate falls
DI resin	Replaced when output TDS rises above target
TDS meter	Essential; measure both RO output and finished water
Booster pump	Worthwhile below ~50 psi feed pressure

10.10 Distilled Water

[EST] Distilled water is functionally similar to RO/DI for aquarium purposes: essentially zero GH, zero KH, near-zero TDS. Differences: - Cost per litre is far higher when purchased in jugs; practical only for small volumes or top-off. - Store-bought “distilled” is sometimes actually RO water; “purified drinking water” often has minerals deliberately added back and is **not** suitable as a zero-TDS base. - Home distillation is energy-intensive.

[INF] Practical use case: distilled or RO water is the correct choice for **evaporation top-off**, in any tank, regardless of whether RO is used for water changes. See Section 23.

Section 11: Raising GH

Search terms for this section: raise GH, increase hardness, remineralizer, calcium sulfate, gypsum, magnesium sulfate, Epsom salt, calcium chloride, crushed coral, aragonite, limestone, cuttlebone, Seachem Equilibrium, SaltyShrimp, Ca:Mg ratio.

11.1 The Arithmetic of GH Dosing

[EST] Every calculation below rests on one idea: **convert the mass of the compound to moles of divalent cation, then express as CaCO₃ equivalent (100.09 g/mol), then convert to dGH by dividing by 17.848 mg/L.**

$$\text{dGH added} = [(\text{g of compound}) \times (1000 / \text{molar mass}) \times 100.09] \div (\text{litres} \times 17.848)$$

Reference dosing figures — derived, not copied:

Compound	Formula	Molar mass (g/mol)	Grams per 100 L for +1 dGH	Also raises KH?	Notes
Calcium sulfate dihydrate (gypsum)	CaSO ₄ ·2H ₂ O	172.17	3.07 g	No	Solubility ~2 g/L; dissolves slowly; pre-dissolve with agitation
Magnesium sulfate heptahydrate (Epsom salt)	MgSO ₄ ·7H ₂ O	246.47	4.39 g	No	Very soluble; dissolves instantly
Calcium chloride dihydrate	CaCl ₂ ·2H ₂ O	147.01	2.62 g	No	Very soluble; adds chloride; hygroscopic — store sealed
Calcium carbonate	CaCO ₃	100.09	1.78 g	Yes — equally	Barely soluble in neutral water; dissolves only as acid is available. See Section 17
Magnesium chloride hexahydrate	MgCl ₂ ·6H ₂ O	203.30	3.63 g	No	Very soluble; adds chloride

Worked example 1 — GH-only remineralization of RO water: Target: 6 dGH with a 3:1 calcium:magnesium hardness ratio, in 50 L of RO water. - Calcium should supply 4.5 dGH; magnesium 1.5 dGH. - Gypsum: 3.07 g/100 L/dGH × 4.5 dGH × 0.5 (for 50 L) = **6.9 g** - Epsom salt: 4.39 × 1.5 × 0.5 = **3.3 g** - Resulting GH ≈ 6 dGH, KH ≈ 0 dKH, sulfate-dominated.

Worked example 2 — raising an existing tank: A 200 L tank at 4 dGH; target 8 dGH. - Needed: +4 dGH across 200 L. - If using gypsum alone: 3.07 × 4 × 2 = **24.6 g** - [INF] But do not add this at once. Split across three or four water changes over two to three weeks. See 11.4.

11.2 Method-by-Method Analysis

11.2.1 Commercial GH remineralizers (GH+ type)

Attribute	Assessment
Raises GH	Yes
Raises KH	No (that is the point)
Changes pH	Minimal direct effect
Speed	Immediate on dissolution
Controllability	High — dose by mass to a known volume
Risks	Overdose from scoop-based measurement; some products contain trace elements that accumulate
Best use	RO-based systems; <i>Caridina</i> ; any case needing minerals without buffer

11.2.2 Commercial GH/KH remineralizers

Attribute	Assessment
Raises GH	Yes
Raises KH	Yes
Changes pH	Raises toward the buffered value, typically 7.2-7.8
Speed	Immediate
Controllability	High
Risks	Cannot separate GH from KH; unsuitable if acidic pH is wanted
Best use	<i>Neocaridina</i> ; general RO-based community tanks

11.2.3 Calcium sulfate (gypsum)

Attribute	Assessment
Raises GH	Yes — calcium only
Raises KH	No
Changes pH	Negligible
Speed	Slow to dissolve; requires vigorous stirring or overnight
Controllability	High by mass
Risks	Adds sulfate. In a tank with significant anaerobic zones, high sulfate provides substrate for sulfate-reducing bacteria and hydrogen sulfide production — a real, if uncommon, concern in deep fine substrates
Best use	DIY remineralization; adding calcium without alkalinity

11.2.4 Magnesium sulfate (Epsom salt)

Attribute	Assessment
Raises GH	Yes — magnesium only
Raises KH	No
Changes pH	Negligible
Speed	Immediate
Controllability	High
Risks	Very easy to over-dose relative to calcium and distort the Ca:Mg ratio; adds sulfate
Best use	Correcting a calcium-heavy GH; plant magnesium supplementation

11.2.5 Calcium chloride

Attribute	Assessment
Raises GH	Yes — calcium only
Raises KH	No
Changes pH	Negligible
Speed	Immediate; highly soluble; exothermic on dissolution in concentrated form
Controllability	High, but the anhydrous and dihydrate forms have different masses — confirm which form you have
Risks	Adds chloride, which raises conductivity more per unit hardness than sulfate; hygroscopic and will absorb water from the air, invalidating mass-based dosing if stored badly
Best use	Where sulfate is undesirable

11.2.6 Crushed coral, aragonite, limestone, oyster shell, cuttlebone

Attribute	Assessment
Raises GH	Yes
Raises KH	Yes — approximately equally, in dGH:dKH terms
Changes pH	Yes — raises it, self-limiting
Speed	Slow: days to weeks
Controllability	Low. Dissolution rate depends on surface area, grain size, flow, and the water's current saturation state
Risks	Cannot be un-done quickly; over time may become coated and stop working; unsuitable where acidic pH is wanted
Best use	Gently raising all three parameters in soft water; a “set and forget” buffer floor. See Section 17 for full treatment

11.2.7 Wonder shell / mineral blocks

Attribute	Assessment
Raises GH	Yes
Raises KH	Usually yes
Changes pH	Usually raises
Speed	Slow dissolution
Controllability	Low; composition is often proprietary
Risks	Some products contain additional ingredients (salt, trace minerals, medications in some variants — read the label)
Best use	Casual GH maintenance in snail tanks

11.3 The Ca:Mg Ratio — an Under-Discussed Variable

[INF] A GH test gives one number. It does not tell you the split between calcium and magnesium. Two waters can both read 8 dGH while one is 7:1 Ca:Mg and the other 1:1.

Commonly cited target [HOB]: roughly **3:1 to 4:1 calcium:magnesium** by hardness contribution, on the basis that this approximates many natural freshwaters and most commercial remineralizer formulations.

[CON] The evidence base for a specific optimum ratio in aquarium species is weak. What can be said: - Both are required. - Commercial remineralizers converge on roughly 3:1 to 4:1, which is at least a practitioner consensus. - Crushed coral and most CaCO₃ sources supply **calcium only, with no magnesium**. Systems relying solely on crushed coral for GH can develop a calcium-heavy, magnesium-poor profile. **[INF]** This is a plausible and frequently cited explanation for molting problems in shrimp tanks that use crushed coral as their only mineral source and whose GH reading looks adequate. - Cuttlebone is similarly calcium-dominant.

[INF] Practical implication: if a GH reading is adequate but calcifying invertebrates still struggle, **the ratio is worth investigating** — either by deriving Ca and Mg separately from a municipal report (Section 2.3), or by supplementing magnesium and observing.

11.4 Speed and the Case for Doing It Slowly

[INF] GH can be raised instantly with soluble salts. It usually should not be.

- Existing livestock has acclimated to the current osmotic environment.
- The organisms most likely to need a GH increase — shrimp and snails — are also the most sensitive to abrupt osmotic change.
- There is no urgency: a shrimp that has been at 3 dGH for six months is not saved by reaching 8 dGH four hours sooner.

[HOB] Recommended approach: raise GH via the **replacement water**, over several water changes, targeting an increase of roughly 1-2 dGH per week in the tank. This is a hobby convention rather than a validated figure, but it errs in the safe direction and has no downside.

11.5 What Raising GH Does Not Do

- **It does not necessarily raise KH.** Sulfate and chloride salts add zero buffering.
- **It does not necessarily raise pH.** Adding gypsum to pH 6.2 water leaves it at pH 6.2.
- **It does not stabilise the tank.** Stability comes from KH.
- **It does not fix a nitrogen problem, a disease problem, or an aggression problem** — worth stating because water chemistry is a common displacement activity when the actual issue is elsewhere.

Section 12: Lowering GH

Search terms for this section: lower GH, reduce hardness, soften water, softening, RO dilution, distilled water, boiling water hardness, peat softening, water softener pillow, ion exchange.

12.1 Why Lowering Hardness Is Harder Than Raising It

[EST] The asymmetry is fundamental:

- **Raising GH** means adding a soluble compound. Dissolution is easy, fast, and precisely controllable by mass.
- **Lowering GH** means *removing ions already in solution*. There is no additive that makes calcium disappear. The options are: physically remove the water and replace it with water that lacks the ions (dilution), or run the water through a medium that binds the ions (ion exchange), or precipitate them out (which requires changing the chemistry substantially).

This asymmetry explains why “pH down” and “water softening” products so consistently disappoint.

12.2 Method Analysis

12.2.1 Dilution with RO, RO/DI, distilled or collected rainwater

Effectiveness: [EST] The only reliably effective method. - Predictable and linear (Section 9.6 equation). - Lowers GH, KH, TDS and conductivity **proportionally** — which is usually what is actually wanted, but means the GH:KH ratio is preserved. - Fully controllable. - Cost: equipment and wastewater for RO; storage and contamination risk for rainwater. - **Rainwater caveat [INF]:** rainwater is essentially distilled and free, but collection surfaces contribute contaminants — roofing material, asphalt shingle leachate, bird droppings, atmospheric pollution, and in industrial or agricultural areas, more. Rainwater collected from a purpose-built clean surface into a covered container is usable; rainwater off an aged composite roof is a gamble. Its pH is naturally slightly acidic (roughly 5.0–5.6 from atmospheric CO₂ alone) and its buffering is zero.

12.2.2 Boiling

Effectiveness: [EST] Partial at best, and frequently misunderstood.

Boiling drives off dissolved CO₂, which shifts the bicarbonate equilibrium and causes **calcium carbonate to precipitate**:



What this actually removes: only the **temporary hardness** — the calcium that is paired with bicarbonate. It removes calcium **and** the corresponding alkalinity together.

What it does not remove: - Calcium and magnesium paired with sulfate or chloride (permanent hardness) — **entirely unaffected**. - Magnesium bicarbonate is much less readily precipitated than calcium bicarbonate. - Sodium, potassium, and everything else.

Additional problems: - Evaporation during boiling **concentrates** what remains, partly offsetting the effect. - The precipitate must be decanted off or filtered out. - Energy cost and effort are absurd for aquarium-scale volumes.

Verdict [INF]: boiling is a real chemical phenomenon (it is why kettles scale) that is essentially useless as an aquarium water-softening method. Classification: **plausible mechanism, impractical application.**

12.2.3 Peat filtration

Effectiveness: [CON] Real but complicated, and frequently misattributed.

Peat does reduce measured hardness. The mechanism is debated: - **Cation exchange:** peat has ion-exchange capacity and can bind Ca²⁺ and Mg²⁺. -

Chelation/complexation: humic and fulvic acids complex divalent cations, making them unavailable to the EDTA titration used by a GH test.

[CON] A notable observation reported in hobby-technical discussion, which if accurate is important: water treated with peat can titrate as much softer while the *total* calcium measured by atomic absorption spectrophotometry is unchanged — meaning the calcium is still present but **complexed** rather than removed. This claim was found in experienced-hobbyist technical discussion and could not be traced to a peer-reviewed source in this research. **Classification: plausible, mechanistically sensible, unverified.** If true, it means “peat softens water” and “peat removes calcium” are different statements, and only the first is reliably correct.

Other peat effects: lowers pH (via humic/fulvic acids), lowers KH, stains the water brown, and adds DOC (which per Section 10.3 may itself be physiologically beneficial in soft-water systems).

Practical drawbacks: difficult to control; effect declines as the peat exhausts; quality and composition of horticultural peat is not standardised; sphagnum peat harvesting carries environmental concerns that have driven some keepers to alternatives.

12.2.4 Household ion-exchange softeners

Effectiveness: [EST] Removes GH effectively; produces water that is generally unsuitable.

See Section 9.5 in full. Summary: GH → near zero, KH unchanged, TDS unchanged, sodium substantially increased. **Not recommended for aquarium use.** The water is “soft” only in the plumbing sense.

12.2.5 “Water softening pillows” and resin pouches

Effectiveness: [CON] Limited and short-lived.

These are small quantities of ion-exchange resin placed in a filter. They work by the same sodium-exchange mechanism as a household softener, at a far smaller scale, and: - Exhaust quickly in a tank of any size. - Release sodium in exchange. - Do not lower KH. - Require regeneration in brine to reuse.

[INF] Reasonable for a small tank as a stopgap; not a system.

12.2.6 Commercial “water softener” chemicals

Effectiveness: [MYTH-adjacent] Frequently misleading.

Products marketed as softeners fall into several categories that are not distinguished on the packaging: - **Chelators** (EDTA-type) — bind calcium and magnesium so they no longer titrate as hardness. **The ions are still in the water.** GH reading falls; osmotic load and conductivity do not. Whether the biological effect the aquarist wanted is achieved is unclear and probably depends on which effect they wanted. - **Phosphate-based “softener/buffer” combinations** — do not soften; impose a phosphate buffer (see Section 7.3). - **Acid buffers** — reduce KH, not GH. Marketed ambiguously.

[INF] Practical guidance: a product that lowers a GH *reading* without removing ions has changed the number, not the water. Where the goal is genuine osmotic softening, dilution is the only reliable route.

12.2.7 Water changes with softer water

Effectiveness: [EST] Works, slowly, by dilution.

This is simply repeated partial dilution. Each x% water change with softer water moves GH toward the new source asymptotically:

$$GH_after_n_changes = GH_source + (GH_initial - GH_source) \times (1 - x)^n$$

Worked example: tank at 16 dGH, source at 4 dGH, 30% weekly changes.

Week	GH
0	16.0
1	12.4
2	9.9
3	8.2
4	6.9
6	5.4
8	4.7
12	4.1

[INF] This table is useful for a different reason than it appears. It shows that switching source water produces a *gradual* transition, not a step change — which is reassuring for livestock, and which also means an aquarist who switches sources and tests a week later will not see the final value.

12.3 Comparison Table: Methods for Lowering GH

Method	Actually lowers GH?	Lowers KH too?	Lowers TDS?	Controllable?	Practical?
RO / distilled dilution	Yes	Yes	Yes	High	Yes
Rainwater dilution	Yes	Yes	Yes	High	Yes, with collection caveats
Boiling	Partially (temporary hardness only)	Yes	Partially	Low	No
Peat	Yes (possibly by complexation)	Yes	Unclear	Low	Limited
Household softener	Yes	No	No	N/A	Not recommended
Softener pillow	Slightly, briefly	No	No	Low	Stopgap only
Chelating “softeners”	Lowers the reading	No	No	Low	Questionable
Repeated water changes with soft source	Yes, gradually	Yes	Yes	Moderate	Yes

Section 13: Raising KH

Search terms for this section: raise KH, increase alkalinity, baking soda, sodium bicarbonate, potassium bicarbonate, soda ash, sodium carbonate, buffer, crushed coral, aragonite, alkalinity dosing.

13.1 The Arithmetic of KH Dosing

[EST] The governing conversion: **1 milliequivalent of alkalinity = 50.04 mg/L as CaCO₃ = 2.80 dKH.**

$$\text{dKH added} = [(\text{g of compound}) \times (1000 / \text{molar mass}) \times (\text{equivalents per mole}) \times 50.04] \div (\text{litres} \times 17.848)$$

Reference dosing figures — derived:

Compound	Formula	Molar mass	Equiv./mol	Grams per 100 L for +1 dKH	Raises GH?	pH effect
Sodium bicarbonate (baking soda)	NaHCO ₃	84.01	1	3.00 g	No	Moderate rise; tends toward ~8.2-8.4 at saturation
Potassium bicarbonate	KHCO ₃	100.12	1	3.57 g	No	Same as above; adds K instead of Na
Sodium carbonate (soda ash)	Na ₂ CO ₃	105.99	2	1.89 g	No	Strong rise — can exceed pH 10. Hazardous
Calcium carbonate	CaCO ₃	100.09	2	0.89 g (if fully dissolved)	Yes, equally	Rises, self-limiting; but see Section 17 — it does not simply dissolve on demand
Calcium hydroxide (kalkwasser/lime)	Ca(OH) ₂	74.09	2	0.66 g	Yes	Extremely strong — pH >10. Dangerous in a stocked tank

Cross-check against a published hobby figure: a koi-pond source states that 1 tablespoon of baking soda per 100 gallons raises KH by approximately 1 dKH. Checking: 1 US tablespoon ≈ 14.4 g; 100 US gallons ≈ 378.5 L. From the table, +1 dKH in 378.5 L requires 3.00 × 3.785 = 11.4 g. The published figure of 14.4 g would give ≈1.27 dKH. **The hobby figure is close and slightly conservative in the safe direction** — i.e. it slightly over-doses. Good agreement, but note that volumetric measures of powder vary substantially with packing and humidity; **dosing by mass is materially more reliable than dosing by spoon.**

Worked example: A 150 L tank at 2 dKH; target 5 dKH. - Needed: +3 dKH in 150 L. - Sodium bicarbonate: 3.00 g/100 L/dKH × 3 dKH × 1.5 = **13.5 g** - [INF] Split across at least three doses over a week, or apply to change water rather than the tank.

13.2 Method-by-Method

13.2.1 Sodium bicarbonate (baking soda)

Attribute	Assessment
Raises KH	Yes — directly and precisely
Raises GH	No. This is the key limitation
Raises pH	Yes; a bicarbonate-dominated system in equilibrium with air tends toward roughly pH 8.2-8.4
Speed	Immediate
Controllability	Very high — dose by mass
Cost	Negligible
Risks	Adds sodium; raises pH which can be undesirable; over-rapid dosing produces a fast pH rise which is the dangerous direction if ammonia is present (Section 6.5)
Best use	Correcting alkalinity depletion; Rift Lake setups combined with a calcium source; emergency buffer restoration

[INF] **Sodium accumulation.** Each dKH added via baking soda adds roughly 8.2 mg/L of sodium. Restoring 3 dKH adds about 25 mg/L sodium. This is not concerning as a one-off. **Repeated dosing without water changes accumulates sodium indefinitely**, because sodium is not consumed by any tank process. Over a year of weekly dosing this becomes substantial. Potassium bicarbonate has the same issue with potassium, but potassium is at least a plant nutrient and is taken up.

13.2.2 Potassium bicarbonate

Chemically equivalent to baking soda for alkalinity purposes. Advantages: potassium is a plant macronutrient and is actively consumed in planted tanks, so it accumulates less; avoids sodium. Disadvantages: more expensive; less available; over-dosing potassium has been anecdotally associated with problems in some shrimp systems [ANC].

13.2.3 Sodium carbonate (soda ash / washing soda)

[EST] Twice the alkalinity per mole, but it is a considerably stronger base and drives pH much higher. **Not recommended for aquarium use with livestock present.** Its legitimate niche is in aquaculture and in reef two-part dosing, both of which involve controlled addition to large, monitored systems.

13.2.4 Calcium hydroxide (lime, kalkwasser)

[AQC] Standard in aquaculture pond liming and in reef calcium/alkalinity supplementation. **Dangerous in a home freshwater aquarium.** The Merck Veterinary Manual notes that calcium hydroxide will rapidly drive pH above 10 and kill all fish present. Agricultural limestone (CaCO₃) is used instead in pond liming precisely because it is slow and self-limiting — which is the same reason crushed coral is used in aquariums.

13.2.5 Crushed coral, aragonite, limestone, oyster shell

The self-limiting mineral route. Raises KH and GH together, slowly. See Section 17 for full treatment. **This is generally the preferred method for a keeper who wants a durable buffer floor rather than a specific number.**

13.2.6 Commercial carbonate buffers

[HOB] Products such as “Alkaline Buffer” type formulations are typically bicarbonate/carbonate mixtures. Functionally similar to baking soda with a defined target pH and, in some cases, added trace elements. Convenient, more expensive per unit alkalinity, and often accompanied by a matching “acid buffer” intended to be used together to hold a target pH — which is the treadmill described in Section 7.

[INF] Distinguishing carbonate from phosphate buffers matters. Read the ingredients. If phosphate is listed, expect an inflated KH reading, invalidated CO₂ estimation, and an algae contribution.

13.3 Why Sudden KH Increases Are Undesirable

[INF] Three reasons, in order of importance:

1. **The ammonia hazard (Section 6.5).** Raising KH raises pH. Raising pH multiplies the toxic un-ionised ammonia fraction. In any tank where ammonia might be present — a tank that has been neglected, a tank with a compromised biofilter, a low-pH old tank — **a rapid buffer addition is genuinely dangerous.** Test ammonia before raising pH.
2. **Osmotic and ionic change.** A large bicarbonate addition changes conductivity and ionic composition, which matters for invertebrates.
3. **CO₂ interaction.** In a CO₂-injected tank, raising KH raises the pH at any given CO₂ level — which may cause a pH controller to inject more CO₂ than intended, or may make a previously-tuned system read differently.

[HOB] Standard guidance: raise KH gradually — a common convention is no more than about 1–2 dKH per day, or preferably apply the change through replacement water over several water changes. As with the GH figure in Section 11.4, the specific number is convention rather than validated science, but the direction is sound.

Section 14: Lowering KH

Search terms for this section: lower KH, reduce alkalinity, remove buffer, acidification, RO dilution, peat, aquasoil, botanicals, acid buffer, hydrochloric acid, muriatic acid, alkalinity removal.

14.1 Why This Is the Hardest Direction

[EST] Lowering KH means **destroying acid-neutralizing capacity**, and there are only two ways to do that:

1. **Dilute it** — replace some of the water with water containing no alkalinity.
2. **Neutralise it** — add acid, which consumes bicarbonate stoichiometrically.

Both work. The second is where people get into trouble.

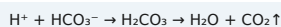
14.2 Method Analysis

14.2.1 RO / distilled / rainwater dilution

Effectiveness: [EST] The safest and most predictable method. - Linear and calculable (Section 9.6). - Lowers GH and TDS proportionally, which is usually desirable when the goal is soft water. - No pH shock — pH follows gradually rather than dropping abruptly, because alkalinity is removed with the water rather than destroyed in place. - **Recommended default.**

14.2.2 Acidification (the mechanism, and why order matters)

[EST] Acid consumes alkalinity stoichiometrically:



[EST] **The critical and frequently missed point:** while alkalinity remains, **added acid is consumed by the buffer and pH barely moves**. The alkalinity falls, invisibly to anyone watching only pH. Then, when alkalinity approaches zero, **the same dose of acid that previously “did nothing” produces a large, fast pH drop**.

This is why: - pH-lowering attempts in hard water appear to fail, repeatedly, and then suddenly succeed catastrophically. - The correct sequence is **remove alkalinity by dilution first, then acidify** — never the reverse. - Anyone acidifying should be **monitoring KH, not pH**, as the leading indicator. When KH reaches roughly 1 dKH, pH is about to become mobile.

Compact statement for retrieval: *Acid does not lower pH in buffered water. It lowers KH. Only once KH is nearly gone does the pH move — and then it moves fast.*

14.2.3 Aquasoil / active substrate

Effectiveness: [HOB, well-established in practice] High, but time-limited. Active substrates lower KH through cation exchange and acid release. See Section 18. Characteristics: - Effective and largely self-regulating in soft source water. - **Consumes itself**. A soil bed has a finite exchange capacity; in hard tap water it can be exhausted in weeks to months. - **Cannot hold acidic pH against continuous replenishment of alkalinity**. Using aquasoil with hard tap water changes is a losing battle — the soil is spent fighting the water changes rather than doing anything useful. - **Best practice is aquasoil plus RO or very soft source water**, which is exactly what *Caridina* keepers do.

14.2.4 Peat

Lowers KH via acid release and cation binding. Slow, uncontrolled, browns the water. See 12.2.3 and Section 16.

14.2.5 Botanicals, driftwood, leaf litter

[CON] **Effect on KH is real but usually small relative to the KH present**. Tannic, humic and fulvic acids are weak organic acids. They will consume some alkalinity. In water at 10 dKH the consumption is trivially small relative to the reservoir; in water at 1 dKH it is significant. **This is the whole explanation for why botanicals “lower pH” in some tanks and do nothing in others**. See Section 16.

14.2.6 Commercial acid buffers

[HOB] Typically bisulfate- or phosphate-based. They do consume alkalinity. Concerns: - Phosphate-based products **add phosphate** and inflate subsequent KH readings. - Continuous use is required, because water changes reintroduce alkalinity. - The oscillation problem of Section 7.

14.2.7 Mineral acid (hydrochloric/muriatic acid)

[AQC/CON] Used in aquaculture and industrially to reduce alkalinity precisely. In a home aquarium: - Extremely effective and precisely calculable. - **Dangerous in unskilled hands**. A miscalculation or a mis-measured dilution can kill everything in the tank in minutes. - Adds chloride. - Very difficult to dose safely into a stocked tank because the local concentration at the point of addition is far higher than the average. - **[INF] If used at all, it should be used on change water in a separate container, never in the display tank**.

14.3 The Rebound Problem

[EST] Any KH reduction achieved by acidification is undone by the next water change with alkaline source water. This is not a failure of the method; it is arithmetic. A 30% water change with 10 dKH tap water into a tank at 1 dKH gives:

$$(0.7 \times 1) + (0.3 \times 10) = \mathbf{3.7 \text{ dKH}}$$

The tank's KH nearly quadruples with one water change. If the keeper then re-acidifies, they have built a weekly oscillation.

[INF] **The resolution is always the same:** if a low-KH steady state is wanted, **the replacement water must be low-KH**. Treating the tank cannot substitute for treating the source.

14.4 Risks of Manipulating Low-Buffer Water

[INF] Once KH is below about 1-2 dKH, the system enters a regime where: - pH responds sharply to small inputs. - Ordinary events — a dead snail, a heavy feeding, a botanical dump, a CO₂ solenoid failure — can move pH substantially. - Test kit resolution becomes a limiting factor: distinguishing 0.5 dKH from 1.0 dKH with a drop-count kit is barely possible, and that difference is the difference between “some buffer” and “essentially none.” - **The margin for error has been intentionally removed**. This is acceptable for an experienced keeper running a deliberate soft-water system with consistent water and light bioload. It is not a good default state.

Section 15: Raising and Lowering pH

Search terms for this section: raise pH, lower pH, pH up, pH down, pH adjustment, pH buffer, acid buffer, alkaline buffer, driftwood pH, peat pH, CO₂ pH, crushed coral pH.

15.1 Master Comparison Table

For each method: does it work, is it durable, what else does it change, and what is the risk?

Method	Direction	Does it work?	Durable or temporary?	Depends on KH?	Also changes	Risk level
Commercial pH Down (acid)	↓	Only after KH is consumed	Temporary; rebounds	Completely	Reduces KH; may add phosphate/sulfate	High — oscillation then crash
Commercial pH Up (base)	↑	Yes	Temporary unless KH raised	Yes	May raise KH	Moderate — fast pH rise + ammonia hazard
Phosphate buffers (“pH 7.0” tablets/powders)	Either, to target	Yes, mechanically	Requires re-dosing every water change	Overrides it	Adds phosphate; inflates KH readings; invalidates CO ₂ estimation	Moderate — algae, treadmill
Baking soda (NaHCO₃)	↑	Yes	Durable (raises KH)	Raises it	+KH, +Na, no GH	Low-moderate; watch rate and ammonia
Potassium bicarbonate	↑	Yes	Durable	Raises it	+KH, +K, no GH	Low-moderate
Sodium carbonate / lime	↑↑	Yes, aggressively	Durable	Raises it	Large pH jump	High — not for stocked tanks
Crushed coral / aragonite / limestone	↑	Yes, to a ceiling	Durable, self-limiting	Raises it	+KH, +GH (Ca only)	Low — the safest upward method
Peat	↓	Yes, in low-KH water	Semi-durable; declines as peat exhausts	Completely	−KH, −GH, +tannins, brown water	Low-moderate
Driftwood	↓	Marginally; often not at all	Declines over months	Completely	+tannins, small −KH	Very low
Botanicals / Indian almond leaves / alder cones	↓	Small effect; large in very low KH	Requires replenishment	Completely	+tannins, +DOC, small −KH, +bioload	Low
Aquasoil	↓	Yes, strongly	Months to a few years, then exhausts	Consumes it	−KH, −GH, possible NH ₃ release when new	Moderate — see Section 18
CO₂ injection	↓	Yes, precisely and reversibly	Fully reversible; returns when CO₂ stops	Magnitude depends on KH	Does not change KH or GH	Moderate — overdose risk
RO dilution	↓ (usually)	Yes, by removing buffer	Durable	Removes it	−KH, −GH, −TDS	Low — the safest downward method
Source water blending	Either	Yes	Durable	Adjusts it	Everything, proportionally	Low
Aeration / surface agitation	↑ (in CO ₂ -rich water)	Yes	Durable to atmospheric equilibrium	No	Removes CO ₂ only	Very low

15.2 The Two Reliable Strategies

[INF] Distilled from the table above, essentially all durable pH management reduces to two approaches:

To raise pH durably: add carbonate. Either soluble (bicarbonate salts, immediate and precise) or mineral (crushed coral/aragonite, slow and self-limiting). Both work because they raise the alkalinity that sets where pH settles.

To lower pH durably: remove carbonate first, then supply a mild continuous acid source. Dilute with RO to bring KH down, then let aquasoil, botanicals, peat or CO₂ hold the pH where you want it. **The order is not optional.**

Everything else on the table is either a temporary intervention, a competing buffer imposed on top, or a way of doing one of these two things less well.

15.3 Why “pH Down” Products Disappoint

Consolidating the mechanism, since this question recurs constantly:

1. Hard water has a large alkalinity reservoir.
2. Acid added is consumed by that reservoir; pH barely moves.
3. The aquarist concludes the product is weak and adds more.
4. Alkalinity falls (unmeasured, because nobody tests KH after adding pH Down).
5. Water change restores alkalinity; the aquarist starts over.
6. Or: enough acid accumulates, alkalinity hits zero, and pH drops several units abruptly.

[INF] Both outcomes — “it doesn’t work” and “it worked too well” — are the same chemistry at different points on the same curve.

15.4 Special Note: CO₂ as a pH Tool

[EST] CO₂ injection is the most *precisely controllable* pH-lowering method available, and the only one that is fully reversible on a daily cycle. Its properties are genuinely distinct from every other method:

- It does **not** consume alkalinity.
- It does **not** change GH or TDS.
- It is removed automatically by degassing when injection stops.
- Its effect is fully predictable from the carbonate equilibrium *provided no other acids are present*.

[CON] But is a CO₂-induced pH drop equivalent to other pH drops? See Section 19.5. Summary: chemically it is a genuine pH change and fish do experience the acidity, but the composition of the water is otherwise unchanged (no ion change, no osmotic change), and elevated dissolved CO₂ carries its own direct physiological

effect on fish blood chemistry independent of pH. So the answer is “no, not equivalent — but not necessarily gentler either.”

Section 16: Driftwood, Botanicals and “Natural pH Lowering”

Search terms for this section: driftwood, tannins, tannic acid, humic acid, fulvic acid, Indian almond leaf, catappa, alder cone, botanicals, blackwater, blackwater extract, leaf litter, peat, dissolved organic carbon, DOC.

16.1 What Botanicals Actually Release

[EST] Wood, leaves, cones and bark in water release:

Substance class	Nature	Effect on pH	Effect on colour	Other effects
Tannins (polyphenols)	Large, weakly acidic polyphenolic compounds	Weakly acidic — small direct pH effect	Strong brown colour	Astringent; some antimicrobial activity; metal-binding
Humic acids	High-molecular-weight organic acids	Acidic; consume alkalinity	Brown	Chelate metals; DOC contribution
Fulvic acids	Lower-molecular-weight organic acids	More acidic than humics; more soluble	Yellow-brown	Chelate metals; DOC contribution
Lignin degradation products	Various	Slightly acidic	Brown	
Simple sugars, carbohydrates	Leachate from fresh material	Negligible direct	None	Bacterial bloom substrate — the cause of the white haze after adding fresh botanicals

16.2 The Central Finding: Colour and pH Are Decoupled

[EST] Brown water does not mean low pH.

This is the single most important correction in this section, and it follows directly from the chemistry:

- **Tannins are strongly coloured and only weakly acidic.** A little tannin produces a lot of colour.
- **The acids released are weak organic acids.** Their capacity to lower pH is small in absolute terms.
- **Whatever acidity they contribute is consumed by alkalinity first.**

Therefore:

Water KH	Realistic effect of a heavy botanical load
10+ dKH	Deep brown water, pH essentially unchanged. The alkalinity absorbs the organic acids completely
5-8 dKH	Brown water; pH drop of perhaps 0.1–0.3 over time; KH slowly declines
2-4 dKH	Brown water; noticeable pH drop; KH visibly declines
0-1 dKH	Brown water; substantial pH drop possible — this is where blackwater tanks actually reach pH 5–6

[EST] This resolves the entire “does driftwood lower pH?” argument. It is not a disagreement about driftwood. It is a disagreement between people with different KH. Both groups are reporting accurately.

Compact statement for retrieval: *Botanicals lower pH in proportion to how little alkalinity is in the way. In hard water they colour the water and do nothing else.*

16.3 Driftwood Specifically

[CON] Driftwood’s pH effect is the weakest of any botanical, for structural reasons:

- Surface area to volume is low compared with leaves.
- Most of the leachable material comes out in the first weeks; the effect declines steadily.
- Wood that has been soaked, boiled, or sold pre-treated has already lost much of its leachate.
- The mass of tannin available from one piece of wood is small relative to the alkalinity in a tank of moderately hard water.

[INF] **Realistic expectation:** a piece of driftwood in a 20-gallon tank at 8 dKH will tint the water and will **not** meaningfully change pH. The many hobbyists reporting “I added driftwood and my pH didn’t change” are observing the normal outcome, not a failure.

Other driftwood effects worth noting: - Provides biofilm grazing surface (significant for shrimp and many catfish — see Section 28). - Provides structure and cover, which reduces stress-related behaviour. - Some woods (particularly softer, freshly collected material) can drive a bacterial bloom. - **[HOB]** Very heavily tannin-loaded woods used in a low-KH system *can* produce meaningful acidification; Malaysian driftwood and freshly collected material are commonly reported as stronger than mopani or pre-cured pieces.

16.4 Indian Almond Leaves (Catappa), Alder Cones, Leaf Litter

[HOB] Higher leachate per unit mass than driftwood, because leaves are thin and have high surface area.

Item	Relative tannin output	Duration	Notes
Indian almond leaf (<i>Terminalia catappa</i>)	High	Weeks; breaks down and is grazed	The hobby standard; also credited with antimicrobial and antifungal properties
Alder cone (<i>Alnus</i>)	Very high per unit mass	Weeks-months	Small, convenient, strong output
Oak leaf	Moderate	Months — very durable	Slow breakdown; good long-term litter
Beech leaf	Moderate	Months	Similar to oak
Magnolia leaf	Low-moderate	Very durable	Minimal tinting; used mainly for structure
Guava leaf	Moderate	Weeks	Grazed readily by shrimp
Banana leaf	Moderate	Weeks	Breaks down fast; good biofilm substrate
Peat	Very high	Depends on quantity	Also removes hardness; see 12.2.3

[CON] **Claims of antimicrobial and antifungal activity** for catappa and similar botanicals: there is real literature on antimicrobial properties of *Terminalia catappa*

extracts, and catappa leaves are used in Southeast Asian *Betta* husbandry for exactly this reason. **However**, extrapolating from *in vitro* extract activity to a meaningful clinical effect at aquarium concentrations is a substantial leap. **Classification: plausible, partially supported, commonly overstated.**

16.5 Blackwater Extracts

[HOB] Bottled extracts deliver tannins and humic substances in concentrated liquid form.

Advantages: dose-controllable; no decomposition or bioload; consistent. **Limitations:** does not provide the physical structure, grazing surface or slow-release character of actual leaf litter; ongoing cost; composition typically proprietary. **[INF]** For pH purposes, the same KH dependence applies exactly. A blackwater extract in 10 dKH water colours the water and does not lower pH.

16.6 The Under-Discussed Finding: DOC Has Direct Physiological Effects

[STR] This is the most significant finding in this section and possibly in this document, because it reframes what botanicals are for.

The hobby discusses tannins almost exclusively in terms of (a) pH lowering and (b) aesthetics. The physiological literature indicates a third, better-evidenced function.

Research isolating dissolved organic carbon (DOC) from Rio Negro blackwater found that native DOC provided **near-complete protection against ionoregulatory disturbance** in zebrafish acutely exposed to pH 4.0 in ion-poor water. The DOC reduced diffusive sodium and chloride loss and stimulated sodium uptake that would otherwise have been entirely inhibited by the acid exposure. Prior acclimation to DOC at neutral pH also reduced branchial sodium turnover and conferred similar protection (Duarte, Smith, Val & Wood, 2016, *Scientific Reports* 6:20377, DOI 10.1038/srep20377). Comparable protection by natural blackwater was demonstrated in Rio Negro stingrays (Wood et al., 2003, *Physiological and Biochemical Zoology* 76:12-27).

Reviews of Amazonian fish physiology in acidic, DOC-rich water cover this in depth (Morris, Val, Brauner & Wood, 2021, *Journal of Experimental Zoology Part A*, 335:843-863).

[INF] What this implies for aquarium practice:

- 1. **Botanicals may be doing more good in soft-water tanks than the pH numbers suggest** — and the good may be largely independent of the pH change.
- 2. **“Ion-poor water plus DOC” is a fundamentally different environment from “ion-poor water.”** Pure remineralized RO at very low GH is *not* the same as blackwater, even at matching pH and conductivity.
- 3. It provides a mechanistic, evidence-based justification for leaf litter in soft-water and blackwater systems that does not depend on any pH claim.
- 4. It suggests the hobby’s framing — tannins as a pH tool — is measuring the wrong output.

[CON] **Appropriate caution:** these studies used DOC isolated from Rio Negro water, characterised as terrigenous with a high proportion of hydroxyl and phenolic sites. **Whether catappa leaf leachate in a home aquarium provides equivalent protection is not established.** The mechanism is suggestive and the class of compounds is related, but the specific extrapolation from Rio Negro DOC to a bag of dried leaves is an inference, not a demonstrated result. It should be presented as a promising and well-grounded hypothesis, not as a proven aquarium benefit.

16.7 Why Results Differ Between Aquariums

[INF] Consolidated list of the variables that determine whether botanicals “work”:

- 1. **KH of the water — dominant factor by far.**
- 2. Quantity of botanical material relative to water volume.
- 3. Type of material (leaf vs. wood; fresh vs. cured; species).
- 4. Whether the material was pre-boiled or pre-soaked (removes much of the leachate).
- 5. Water change frequency and the KH of the replacement water.
- 6. Presence of activated carbon or Purigen-type resin, **both of which remove tannins efficiently** — a very common reason botanicals “stop working.”
- 7. Surface agitation and CO2 status (affects baseline pH independently).
- 8. Age of the material — output declines.
- 9. Tank age and existing organic load.

16.8 Evidence Summary for This Section

Claim	Classification
Botanicals release tannins, humic and fulvic acids	[EST]
Botanicals colour the water brown	[EST]
Botanicals lower pH substantially in high-KH water	[MYTH]
Botanicals lower pH substantially in very low-KH water	[EST]
Brown water indicates low pH	[MYTH]
Botanicals slowly consume KH	[EST] — magnitude depends on load
Driftwood reliably lowers pH	[MYTH] as stated; [EST] with KH qualification
Catappa has antimicrobial properties	[CON] — in vitro support; aquarium relevance overstated
DOC protects fish ionoregulation in acidic, ion-poor water	[STR] — demonstrated for Rio Negro DOC
Aquarium botanicals provide the same DOC protection	[INF] — plausible, not demonstrated
Botanicals provide biofilm grazing substrate	[EST]

Section 17: Crushed Coral

Search terms for this section: crushed coral, aragonite, limestone, oyster shell, cuttlebone, calcium carbonate, CaCO₃, buffering substrate, saturation state, dissolution, filter media buffer.

17.1 What It Is

[EST] “Crushed coral” as sold for aquariums is fragmented calcium carbonate of marine origin — coral skeleton, shell, and coralline material — supplied in grades from sand to gravel. Related materials:

Material	Composition	Relative solubility	Notes
Crushed coral	CaCO ₃ , mixed calcite/aragonite	Moderate	Hobby standard; wide grain range
Aragonite sand	CaCO ₃ , aragonite polymorph	Higher than calcite	Aragonite is the more soluble polymorph; higher surface area in sand form
Oolitic aragonite	CaCO ₃ , aragonite, small rounded grains	High (surface area)	Very fine; can compact
Limestone / oolite rock	CaCO ₃ , calcite	Lower	Slow; decorative rock use
Dolomite	CaMg(CO ₃) ₂	Lower	Contributes magnesium as well as calcium
Oyster shell / seashell	CaCO ₃ , largely calcite with aragonite	Moderate	Cheap; agricultural grade available
Cuttlebone	CaCO ₃ , aragonite, highly porous	High (surface area)	Popular for snails and shrimp; floats until saturated

[EST] Important compositional note: all of these supply **calcium and carbonate only**. None supply **magnesium**, except dolomite. See Section 11.3 — this is why crushed-coral-only mineral management can produce calcium-heavy, magnesium-poor water.

17.2 How It Dissolves — the Mechanism That Explains Everything Else

[EST] Calcium carbonate is only very slightly soluble in pure neutral water. Its dissolution in an aquarium is driven by acid:



and more directly:



[EST] The rate of dissolution depends on the saturation state of the water with respect to calcium carbonate, conventionally expressed as Ω (omega):

- **Ω < 1** — water is undersaturated; calcium carbonate **dissolves**.
- **Ω = 1** — equilibrium; no net dissolution.
- **Ω > 1** — water is supersaturated; calcium carbonate does **not** dissolve, and may precipitate.

Ω rises with calcium concentration, with alkalinity, and steeply with pH.

[EST] This single fact explains virtually every observation aquarists make about crushed coral:

Observation	Explanation
“I added crushed coral and nothing happened”	The water was already at or above saturation. Ω ≥ 1, so nothing dissolves. Crushed coral does not raise the parameters of water that is already hard and alkaline
“It worked at first and then stopped”	It raised the parameters until Ω reached 1, then stopped. This is the mechanism, not a failure
“It stalls out around pH 7.6-8.0”	That is roughly where a freshwater system reaches carbonate saturation at typical calcium and alkalinity levels
“It works faster in my soft water tank”	Lower Ω → higher undersaturation → faster dissolution
“It works faster in my acidic tank”	Same reason, more strongly
“It works faster since I added more flow through the bag”	Dissolution is surface- and transport-limited; flow removes the locally saturated boundary layer
“It works faster with finer grain”	More surface area
“It stopped after two years as a substrate”	Surfaces become coated — phosphate and organic films block dissolution. Reported in aquarium literature and consistent with known phosphate inhibition of carbonate dissolution

[EST] The self-limiting behaviour is the key property. Crushed coral cannot overshoot. It dissolves only while the water can accept more, and stops when it cannot. **This is why it is the safest method of raising KH, GH and pH, and why it is fundamentally uncontrollable in the sense of hitting a specific target.**

17.3 Does It “Lock” or “Stabilise” pH?

[CON] Partially true, commonly overstated.

What is true: crushed coral establishes a **floor**. As acid accumulates and pH falls, the water becomes more undersaturated, dissolution accelerates, alkalinity is replenished, and the decline is arrested. This is genuine negative feedback and it is why crushed coral is effective at preventing the gradual pH decline of an aging tank.

What is not true: - It does not hold pH at a *specific value*. The equilibrium point depends on the water, the CO₂ level, the surface area exposed, and the flow. - It does not prevent **upward** pH movement. - It does not act instantly. If pH falls quickly — a large acid input, a CO₂ event — dissolution cannot keep pace. **Crushed coral protects against slow drift, not against fast shocks.** - It does not maintain a fixed KH. KH will settle wherever saturation dictates and will vary with the tank’s acid production.

[INF] Best description: crushed coral is a **self-regulating alkalinity floor with a slow response time**, not a pH clamp.

17.4 Placement: Substrate vs. Filter

Placement	Advantages	Disadvantages
In the filter (media bag)	Removable; adjustable quantity; high flow → faster dissolution; easy to inspect and clean	Occupies filter space; needs periodic rinsing to remove detritus that blocks surfaces
As substrate	Large surface area; no filter space used; appropriate look for Rift Lake and marine-style tanks	Not removable without tearing down the tank; lower flow through the bed; surfaces become coated more quickly with mulm and phosphate; unsuitable for planted tanks
Mixed into existing substrate	Distributed effect	Same removability problem, harder to undo
In a media reactor or canister basket	Best flow, best control, easily removed	Requires the equipment

[HOB] Practical recommendation for most aquarists: a media bag in the filter. Start with a small quantity — a common starting point is roughly a half-cup per 20 gallons — test KH and pH weekly, and add more only if the change is insufficient after two to three weeks. **Because the response is slow, patience is required and impatience causes overshoot in quantity if not in chemistry.**

[HOB] Planted tank caution: carbonate substrate in a planted tank is generally discouraged. The higher pH and calcium reduce availability of iron and some micronutrients, and the substrate cannot be removed if it turns out to be a problem.

17.5 When Crushed Coral Is Useful

[INF] - Soft source water with a declining KH trend in a stocked tank. - Preventing the alkalinity depletion that drives Old Tank Syndrome (Section 21). - Rift Lake cichlid tanks on soft-to-moderate tap water. - Snail and *Neocaridina* tanks needing a modest, self-limiting calcium and buffer source. - Tanks where the keeper's maintenance is irregular and a passive safety margin is valuable. - Livebearer tanks (mollies especially) on soft water.

17.6 When Crushed Coral Is Inappropriate

[INF] - Any system where an **acidic pH is the goal** — blackwater, *Caridina*, wild soft-water species, aquasoil setups. It directly opposes the design intent. - Water that is **already hard and alkaline** — it will do nothing and is simply wasted (this is the “nothing happened” case, and the correct response is to recognise the water needs nothing, not to add more coral). - **Planted tanks** where micronutrient availability matters. - Situations requiring a **specific, precise KH value** — use bicarbonate dosing instead. - As the **sole mineral source for calcifying invertebrates**, because it supplies no magnesium (Section 11.3).

17.7 Long-Term Behaviour

[HOB/INF] - Effectiveness declines as the material becomes coated with detritus, biofilm and phosphate compounds. Periodic rinsing in tank water restores much of it. - **Material is genuinely consumed** — the volume in the bag decreases over years. - **Response is slow.** Changes take days to weeks to appear; adjustments should be made on that timescale, not daily. - **It does not respond to a sudden crisis.** A tank that has crashed needs a bicarbonate dose, not crushed coral. Crushed coral is prevention.

Section 18: Aquasoils and Active Substrates

Search terms for this section: aquasoil, active substrate, buffering substrate, ADA Amazonia, cation exchange capacity, CEC, ammonia leaching, substrate exhaustion, shrimp substrate, planted tank substrate, inert substrate.

18.1 What “Active” Means

[EST] Aquarium substrates fall into two functional categories:

- **Inert substrates** — quartz sand, most gravels, most “pool filter sand,” fired clay pellets marketed as inert. They do not exchange ions or change water chemistry appreciably. (Note: some gravels contain carbonate contamination and *are* mildly buffering — a bag of unmarked gravel that raises KH is usually carbonate-contaminated.)
- **Active substrates** — baked or granulated soils, typically volcanic or allophane-rich clay soils, sometimes with added organic matter and nutrients. They actively modify water chemistry.

18.2 Mechanisms of Action

[EST/INF] Active substrates affect GH, KH and pH through several simultaneous mechanisms:

18.2.1 Cation exchange

The substrate carries a negatively charged surface with substantial **cation exchange capacity (CEC)**. It preferentially binds divalent cations — Ca^{2+} and Mg^{2+} — and releases monovalent cations and H^+ in exchange.

Consequences: - **GH falls** — calcium and magnesium are removed from the water column. - **pH falls** — hydrogen ions are released. - **KH falls** — released H^+ consumes bicarbonate.

[INF] **This is the same chemistry as a household water softener resin, running in reverse polarity** — it strips hardness and acidifies rather than swapping in sodium.

18.2.2 Direct acid release

Many aquasoils contain humic and fulvic components that release organic acids, contributing to acidification independently of exchange.

18.2.3 Ammonia release (“leaching”)

[HOB, very consistently reported] Some nutrient-rich aquasoils — ADA Amazonia is the most-cited example — release substantial ammonia for the first weeks after flooding. Reported durations range from two to eight weeks depending on product, batch and water change regime.

Practical implications: - New soil tanks are commonly set up and cycled **without livestock** for several weeks, with heavy water changes. - Adding shrimp to a fresh soil tank is a well-known way to lose them. - Some products are marketed as “low-ammonia” or “pre-washed” versions. - **[EST] Interaction with pH:** because the soil is simultaneously driving pH down, most of that ammonia exists as the far less toxic ammonium — which is why some keepers get away with it and then have problems if pH later rises.

18.3 Lifespan and Exhaustion

[EST/INF] Cation exchange capacity is **finite**. Every calcium ion bound occupies a site permanently until the substrate is regenerated (which does not happen in an aquarium).

Therefore the substrate’s buffering lifespan is determined by the rate at which hardness is supplied to it — that is, by the source water and the water change schedule.

Source water	Approximate practical lifespan of pH-lowering effect
RO / RO+GH-only remineralizer, low KH	Longest — commonly 1–3 years reported
Soft tap water (2–4 dKH)	Intermediate — commonly around 1 year reported
Moderate tap water (5–8 dKH)	Months
Hard tap water (10+ dKH)	Weeks to a few months. The soil is consumed fighting the water changes

(These are hobby-reported ranges, not manufacturer specifications or measured data. Classification: [HOB].)

[INF] **The practical rule this produces:** *aquasoil and hard tap water are incompatible as a strategy*. Anyone using aquasoil specifically to achieve low pH must pair it with soft or RO source water, or the substrate will exhaust rapidly and expensively. This is exactly why the *Caridina* keeping community standardised on RO plus GH-only remineralizer plus active soil — the three components are designed to work together, and removing any one breaks the system.

18.4 Water Change Interactions

[EST] Every water change with alkaline water: - Adds alkalinity that the soil must then consume. - Adds calcium and magnesium that occupy exchange sites. - Temporarily raises pH, which then falls again as the soil works.

[INF] **Consequence:** a soil tank on hard tap water experiences a **sawtooth pH pattern** — a rise at each water change, then a decline. The magnitude depends on the change size and the source KH. In an RO-based system with GH-only remineralizer, the sawtooth is nearly flat because the incoming water carries no alkalinity to fight.

[INF] **This is a strong argument for matching change water to tank water in soil-based systems**, and it is the specific reason *Caridina* keepers pre-mix and pre-test their change water rather than using tap.

18.5 Why Two Aquarists Report Completely Different Behaviour

[INF] **Consolidated:** an aquarist using aquasoil and an aquarist using inert gravel are running chemically different systems, and their observations will diverge on essentially every parameter:

	Inert gravel	Active aquasoil
GH over time	Follows source water	Falls , until exhaustion
KH over time	Follows source water minus nitrification	Falls faster , until exhaustion
pH	Follows source water and CO ₂	Lower than source , until exhaustion
pH after water change	Moves toward source	Rises, then declines again (sawtooth)
Response to crushed coral	Predictable rise	Fights the soil ; both are consumed
First 4–8 weeks	Normal cycling	Possible ammonia leaching
After 1–3 years	Unchanged	Reverts to behaving like inert substrate
Suitable source water	Anything	Soft or RO strongly preferred

[INF] Retrieval-friendly statement: *If two aquarists disagree about how their water behaves, the first question is what is on the bottom of the tank.* Substrate type is one of the largest uncontrolled variables in hobby water-chemistry discussion, and it is frequently not mentioned when advice is given or received.

18.6 The Exhaustion Transition

[INF] An under-anticipated event: when an aquasoil exhausts, **the tank’s chemistry changes even though the keeper changed nothing.**

- pH begins to rise toward the source water value.
- KH begins to accumulate rather than being consumed.
- GH rises toward source.
- A *Caridina* colony that has been stable for two years may begin to struggle.

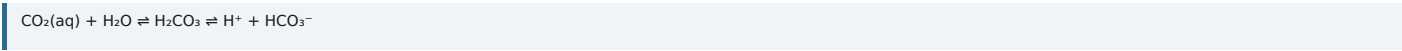
[INF] Practical guidance: in soil-dependent systems, **track KH and pH trends over months, not just absolute values.** A slow upward drift in a soil tank is the signature of substrate exhaustion, and it is a scheduled maintenance event (re-scape with fresh soil, or transition management) rather than a problem to be chemically corrected.

Section 19: Planted Aquariums and CO₂

Search terms for this section: CO₂, carbon dioxide, carbonic acid, CO₂ injection, drop checker, pH controller, degassing, surface agitation, KH pH CO₂ chart, CO₂ table, diurnal pH swing, photosynthesis, respiration, pH profile.

19.1 The Chemistry

[EST] Carbon dioxide dissolves in water and forms carbonic acid, which dissociates:



Key properties, stated explicitly because each is frequently misunderstood:

- CO₂ lowers pH.** More dissolved CO₂ = more H⁺ = lower pH.
- CO₂ does not change alkalinity (KH).** One H⁺ and one HCO₃⁻ are produced together; acid-neutralizing capacity is unchanged. (*See Section 3.4.*)
- CO₂ does not change GH.** No calcium or magnesium is involved.
- The effect is fully reversible.** Stop injecting; CO₂ outgasses; pH returns.
- The magnitude of the pH change depends on KH.** At low KH a small CO₂ addition moves pH a long way. At high KH the same addition moves it less.
- Atmospheric equilibrium is about 2-3 ppm CO₂** in a non-injected aquarium.

19.2 The Daily pH Profile

[EST] In any tank with photosynthetic organisms:

Time	CO ₂	pH
Pre-dawn	Highest (overnight respiration)	Lowest
Lights on	Beginning to fall	Beginning to rise
Mid-photoperiod	Falling	Rising
Late photoperiod	Lowest	Highest
Lights off	Beginning to rise	Beginning to fall
Overnight	Rising	Falling

In a CO₂-injected tank this is inverted and amplified by the injection schedule: CO₂ is typically started 1-2 hours before lights and stopped 1-2 hours before lights out, producing a controlled pH drop of roughly 0.8-1.2 units during the photoperiod and a return overnight.

[INF] **Practical measurement consequence:** a single pH reading in a planted tank is close to meaningless without knowing when it was taken. **The useful measurement is a pH profile** — readings every hour or two across a day — not a spot value.

19.3 Surface Agitation and CO₂ — Getting the Direction Right

[EST] Gas exchange at the surface moves the system **toward equilibrium with the atmosphere**, in whichever direction that requires.

- Water **supersaturated** with CO₂ (injected tank, fresh well water, water with heavy respiration) → **agitation removes CO₂ → pH rises.**
- Water **undersaturated** with CO₂ (heavily planted tank mid-photoperiod, freshly boiled water) → agitation **adds** CO₂ from air → pH falls slightly.

[EST] **For practical aquarium purposes: surface agitation causes CO₂ to leave the tank, not enter it.** In a CO₂-injected planted aquarium, increasing surface agitation reduces dissolved CO₂ and raises pH; it does not “add CO₂.” This is stated explicitly because the reverse claim circulates.

Corollary [INF]: the standard planted-tank compromise — moderate surface movement for oxygen, with CO₂ injection rate increased to compensate — is a deliberate trade-off, not a contradiction. Oxygen and CO₂ are both improved by higher injection plus good gas exchange, versus low injection plus a stagnant surface.

19.4 The pH/KH/CO₂ Table and Why It Misleads

[EST] The relationship between pH, alkalinity and dissolved CO₂ is fixed by the carbonate equilibrium and can be tabulated. Approximate dissolved CO₂ (ppm) at 25 °C:

pH ↓ / KH →	1 dKH	2 dKH	3 dKH	4 dKH	6 dKH	8 dKH	12 dKH
6.0	11.8	23.6	35.4	47.2	70.8	94.4	141.6
6.2	7.4	14.9	22.3	29.8	44.6	59.5	89.3
6.4	4.7	9.4	14.1	18.8	28.2	37.6	56.4
6.6	3.0	5.9	8.9	11.8	17.8	23.7	35.5
6.8	1.9	3.7	5.6	7.5	11.2	14.9	22.4
7.0	1.2	2.4	3.5	4.7	7.1	9.4	14.1
7.2	0.7	1.5	2.2	3.0	4.5	5.9	8.9
7.4	0.5	0.9	1.4	1.9	2.8	3.7	5.6
7.6	0.3	0.6	0.9	1.2	1.8	2.4	3.5
8.0	0.1	0.2	0.4	0.5	0.7	0.9	1.4

(Derived from the carbonate equilibrium; values approximate. CO₂ becomes negligible above about pH 8.4.)

[EST] **The table’s foundational assumption: that carbonic acid from CO₂ is the ONLY acid present, and that carbonate species are the ONLY contributors to the measured alkalinity.**

[STR] **In most real aquariums, both assumptions are false.** Sources of failure:

Interference	Effect on the estimate
Humic/fulvic acids from driftwood, botanicals, peat	Lower pH without adding CO ₂ → table overestimates CO₂
Aquasoil	Releases fulvic and humic acids → table overestimates CO₂, often severely
Nitric acid from nitrification	Lowers pH → overestimates CO ₂
Phosphate buffers	Contribute to the KH titration → KH is not carbonate → estimate invalid in both directions
Non-carbonate alkalinity generally (borate, silicate, organic conjugate bases)	KH overstated → CO ₂ overestimated
Test kit error	An error of 0.2 pH or 0.5 dKH changes the CO ₂ estimate substantially

Practical-fishkeeping-level sources make the same point directly: the pH/KH/CO₂ relationship presumes carbonic acid is the only acidifying compound, when nitric acids, humic acids and other organic matter all contribute to acidification (*Practical Fishkeeping*, “CO₂: Striking the balance”). Specialist planted-tank sources note that using aquasoil renders the table effectively useless, because fulvic and humic acids lower pH substantially without contributing CO₂ (2Hr Aquarist, “The wrong way to read the pH/KH chart”). Direct measurement with a dissolved-CO₂ analyser in crowded and soil-based aquariums has produced readings around 5 ppm where the table would have predicted far higher — with typical non-injected, non-soil tanks measuring around 2–3 ppm.

[INF] The direction of error is consistent and important: the interfering acids lower pH, and the table interprets any low pH as CO₂. **The table therefore systematically OVERESTIMATES CO₂ in tanks containing driftwood, botanicals or aquasoil.**

Why this matters practically: an aquarist who believes they are at 30 ppm CO₂ when they are actually at 12 ppm will conclude their plants are failing for some other reason, and will not increase CO₂. Conversely — and more dangerously — the error can run the other way in some configurations, and an aquarist who trusts a controller calibrated on a bad assumption can gas their fish.

19.5 Better Methods for CO₂ Assessment

Method	How it works	Reliability
Drop checker with 4 dKH reference solution	An isolated air gap between tank water and a known-KH indicator solution means only CO ₂ gas crosses. Other acids cannot reach the indicator. This is the entire point of the device	Good. The standard recommendation. Must use 4 dKH reference solution, not tank water — using tank water reproduces the same interference problem
pH drop from degassed baseline	Measure pH of tank water after 24h aeration (fully degassed), then measure with CO ₂ running. The <i>difference</i> isolates CO ₂ ’s contribution, because the interfering acids are present in both readings	Very good and underused. A pH drop of about 1.0 unit from the degassed baseline corresponds to roughly 30 ppm CO ₂ regardless of what else is in the water
Direct CO ₂ probe/analyser	Measures dissolved CO ₂ partial pressure directly, independent of pH	Best; expensive
pH/KH table	Carbonate equilibrium calculation	Poor in any tank with wood, botanicals, soil, or phosphate buffers
Fish and plant behaviour	Pearling; fish at the surface gasping	Crude but real. Fish gasping at the surface is the emergency indicator and overrides every measurement

[INF] The “pH drop from degassed baseline” method deserves more attention than it gets. It requires no special equipment, it automatically cancels out interference from tannins, aquasoil and nitric acid, and it directly measures the quantity of interest. It is arguably the best practical method available to a hobbyist without a CO₂ probe.

19.6 Is a CO₂-Induced pH Drop “the Same” as Other pH Drops?

[CON] No, and the differences run in both directions.

Arguments that it is gentler: - No change in ionic composition, conductivity, osmotic load, or hardness. The fish experiences a pH change with none of the accompanying ionic disturbance. - It is predictable, recurring, and gradual (over 1–2 hours), so fish acclimate to the cycle. - It reverses daily, so no chronic exposure to the low value. - Alkalinity is unaffected, so the tank’s buffering is not being eroded. - Fish in natural systems experience diurnal CO₂-driven pH cycles constantly.

Arguments that it is not benign: - **Elevated dissolved CO₂ has direct physiological effects independent of pH.** CO₂ crosses gill membranes readily and causes respiratory acidosis in the blood — a rise in blood CO₂ and fall in blood pH — and **reduces the oxygen-carrying capacity of haemoglobin (the Root and Bohr effects)**. A fish in high-CO₂ water is impaired in oxygen uptake even at adequate dissolved oxygen. **This is the actual mechanism by which CO₂ overdose kills fish, and it is not a pH effect.** - CO₂ overdose is a genuine and common cause of fish loss in planted tanks. - Fish gasping at the surface in a CO₂ tank are showing respiratory distress, not pH distress.

[INF] Correct framing: in a CO₂-injected tank, **pH is the indicator and CO₂ is the hazard.** Watching pH is a proxy for watching CO₂. The danger threshold is a CO₂ concentration, not a pH value — which is why a 1.2-unit pH drop is fine at 3 dKH (about 35 ppm CO₂) and lethal at 12 dKH (about 140 ppm CO₂). **The same pH drop means completely different things at different KH.** This is one of the most important practical facts in planted aquarium keeping.

19.7 Monitoring and Safety

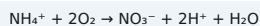
[INF] - Use a drop checker with **4 dKH reference solution**, positioned away from the diffuser, and read it at the end of the photoperiod when CO₂ is highest. - Establish the degassed pH baseline once, then use the pH drop from it. - **Turn CO₂ off at night.** Plants do not use it, and overnight respiration plus injected CO₂ produces the lowest oxygen and highest CO₂ of the 24-hour cycle simultaneously. - Increase surface agitation at night (many keepers run an airstone on a night timer) — this improves oxygen and removes residual CO₂. - If fish gather at the surface, **shut CO₂ off and increase aeration immediately.** Do not wait to check a number. - Be alert to KH changes: **if KH falls, the same injection rate produces a lower pH.** A tank whose KH declines over months while running a fixed CO₂ rate is drifting toward a lower pH and a similar CO₂. A tank on a **pH controller** whose KH declines is the more dangerous case — the controller will keep injecting to reach a pH that now corresponds to a much higher CO₂ level. **pH controllers require KH monitoring.**

Section 20: The Nitrogen Cycle and KH

Search terms for this section: nitrogen cycle, nitrification, ammonia, ammonium, nitrite, nitrate, Nitrosomonas, Nitrospira, alkalinity consumption, biofilter, cycle crash, pH and nitrification, denitrification.

20.1 The Acid Production

[EST] Nitrification is an acid-producing process. The overall reaction:



Two hydrogen ions are produced per ammonium ion oxidised. Those hydrogen ions consume alkalinity.

[STR] The stoichiometry is well established across wastewater engineering, drinking water treatment and aquaculture:

Approximately 7.14 mg of alkalinity as CaCO₃ is consumed per 1 mg of ammonia-nitrogen oxidised to nitrate.

This figure appears consistently across independent authoritative sources: Minnesota Pollution Control Agency wastewater process guidance; the California Water Environment Association; New York State DEC nitrogen removal training materials; and the recirculating aquaculture literature (Timmons, Guerdat & Vinci, *Recirculating Aquaculture*, 4th ed., 2018). Wastewater practice adds a working rule of thumb of maintaining alkalinity at roughly **8× the ammonia concentration** to sustain nitrification.

[STR] **Related figures worth recording:** - Oxygen demand: approximately **4.57 mg O₂ per mg of ammonia-nitrogen** fully nitrified (about 3.16 mg for the ammonia→nitrite step and 1.11 mg for nitrite→nitrate). - Denitrification **recovers roughly half** the alkalinity consumed — about 3.57 mg alkalinity as CaCO₃ per mg of nitrate-nitrogen reduced.

20.2 Converting to Aquarium Terms — Worked Examples

[INF] The 7.14 figure is in mg/L as CaCO₃. Converting to dKH: $7.14 \div 17.848 = \mathbf{0.40 \text{ dKH consumed per mg/L of ammonia-nitrogen oxidised}}$.

Example 1 — a moderately stocked community tank. Assume a 200 L tank producing 4 mg/L-equivalent of ammonia-nitrogen per week across the whole tank volume (a rough figure representing a moderate bioload; actual production depends heavily on feeding).

- Weekly alkalinity consumption: $4 \times 7.14 = \mathbf{28.6 \text{ mg/L as CaCO}_3 = 1.6 \text{ dKH per week}}$.
- With a starting KH of 6 dKH and **no water changes**, buffer would be exhausted in under four weeks.
- With **30% weekly water changes on 8 dKH tap water**, each change adds $0.30 \times 8 = 2.4 \text{ dKH}$ — comfortably exceeding the 1.6 dKH consumed. **The system is in surplus and KH stays roughly stable.**

Example 2 — the same tank on soft tap water. - Source water 2 dKH. Each 30% change adds 0.6 dKH. - Consumption is 1.6 dKH/week. **Deficit of 1.0 dKH per week.** - Starting at 3 dKH, the tank reaches zero in about three weeks and then pH becomes mobile. - **This is the arithmetic behind “my KH keeps dropping.”** It is not mysterious; it is a mass balance.

Example 3 — the neglected tank. - Heavy bioload, monthly 15% water changes on 6 dKH water. - Monthly addition: $0.15 \times 6 = 0.9 \text{ dKH}$. Monthly consumption at the Example 1 rate: 6.4 dKH. - **Deficit approaching 5.5 dKH per month.** Whatever buffer the tank started with is gone within weeks, and then pH declines steadily.

This is the Old Tank Syndrome trajectory in numbers. See Section 21.

[INF] The general principle, stated for retrieval:

KH stability is a mass balance between alkalinity consumed by nitrification and organic acid production, and alkalinity replaced by water changes (or by a dissolving mineral source). Whether a tank’s KH is stable is arithmetic, not luck.

The three variables the keeper controls are: **bioload** (consumption), **water change volume and frequency** (replacement), and **source water KH** (replacement per unit volume). A fourth — a dissolving carbonate source — can substitute for the others.

20.3 Nitrifying Organisms and pH

[STR] Nitrification rate is pH-dependent. Wastewater engineering literature consistently reports that nitrification is strongly inhibited at low pH, with pH below roughly 6.5 essentially stopping the process in activated sludge systems, and typical optimum ranges cited around pH 7.5–8.5.

[CON] But aquarium reality is more complicated, and this is genuinely worth flagging as an area where the wastewater literature does not transfer cleanly.

Points of divergence:

- Acidic aquariums do process ammonia.** Blackwater tanks and *Caridina* tanks routinely run at pH 5.5–6.5 for years without ammonia accumulation. Something is oxidising the ammonia.
- The organism assemblage differs.** Wastewater literature was built around *Nitrosomonas* and *Nitrobacter*. Aquarium and aquaculture biofilters are now known to be dominated by different organisms — *Nitrospira* is generally the dominant nitrite oxidiser in aquaria, and **comammox** (complete ammonia oxidising) *Nitrospira*, discovered in 2015, perform the entire oxidation in one organism and are now recognised as abundant in aquarium and drinking-water biofilters. Comammox organisms are adapted to low-substrate, oligotrophic conditions — which describes an aquarium far better than it describes a sewage plant.
- Acid-tolerant and acidophilic ammonia oxidisers exist**, including archaeal ammonia oxidisers (AOA) which are known to function at lower pH and lower ammonia concentrations than bacterial oxidisers.
- Ammonia availability changes with pH.** At low pH, nearly all TAN is ammonium rather than free ammonia. Some ammonia oxidisers use free NH₃ as substrate, so low pH reduces substrate availability, which is a separate limitation from direct pH inhibition of the enzyme.

[INF] **Best current synthesis:** nitrification in acidic aquariums is **slower and carried out by a different microbial community** than in neutral-to-alkaline systems, but it is not absent. The wastewater figure of “pH 6.5 stops nitrification” reflects the organisms and conditions of activated sludge and **should not be quoted as an aquarium fact.**

[INF] **Practical implication:** a tank whose pH declines gradually to 6.0 does not necessarily suffer a biofilter failure. A tank whose pH is *abruptly* moved from 7.6 to 6.0 may well see a temporary reduction in nitrification, because the existing community is not adapted. **The rate of change matters more than the endpoint** — a conclusion that supports the “stable pH” advice for a specific microbiological reason rarely mentioned in that debate.

20.4 Can a “Cycle Crash” Result From Low pH?

[CON] Partially, and the term conflates several different events.

What “cycle crash” is used to describe: 1. Sudden ammonia and nitrite appearing in an established tank. 2. Loss of biofilter capacity after a filter cleaning, medication, or power outage. 3. The specific low-pH scenario.

For the low-pH scenario specifically [INF]: - Nitrification rate does decline at low pH, though not to zero. - If pH falls **abruptly**, the established community is disrupted and processing capacity drops. - If pH falls **gradually** over months, the community shifts and capacity is often maintained. - **The mitigating factor:** at low pH the ammonia present is overwhelmingly ammonium, which is far less toxic. So a partially impaired biofilter at pH 6.0 may produce a TAN reading that alarms the keeper while causing little actual harm. - **The danger is in the correction.** Raising pH to “fix” the biofilter converts that ammonium reservoir to toxic free ammonia. See Section 6.5 and Section 21.

20.5 Why pH Declines Gradually in Many Aquariums

[STR/INF] Consolidated causal chain:

1. Fish are fed. Protein is metabolised. Ammonia is excreted.
2. Nitrification oxidises the ammonia, producing H^+ and consuming alkalinity at 7.14 mg $CaCO_3$ per mg N.
3. Simultaneously, uneaten food, faeces and detritus decompose, producing organic acids.
4. Nitrate accumulates. Nitrate is the conjugate base of a strong acid; its accumulation is a marker of the acid production, though nitrate itself is not the acidifier.
5. Alkalinity falls week by week.
6. **pH holds steady while alkalinity remains** — this is the deceptive phase, and it can last months.
7. When alkalinity approaches zero, pH begins to fall, then falls faster.
8. If water changes are small or infrequent, and source water alkalinity is low, replacement never keeps up.

[INF] The diagnostic signature: KH declines first, then pH. Monitoring KH gives weeks or months of warning. Monitoring pH gives days.

Section 21: “Old Tank Syndrome” and pH Crashes

Search terms for this section: old tank syndrome, OTS, pH crash, alkalinity depletion, nitrate accumulation, neglected aquarium, large water change danger, topping off, organic acid accumulation.

21.1 Is There a Standard Definition?

[VET] Yes — unusually for a hobby term, “Old Tank Syndrome” appears in veterinary literature with a specific parameter profile.

The Merck Veterinary Manual describes Old Tank Syndrome as arising from **loss of buffering capacity**, caused by inadequate water changes — particularly the practice of topping off evaporation without ever removing old water. Because old water is never removed, organic acids produced by fish and bacteria accumulate and progressively consume total alkalinity. When total alkalinity reaches zero, pH falls sharply, the biofilter fails, and the syndrome is established (Merck Veterinary Manual, *Disorders and Diseases of Fish*).

[VET] The parameter profile Merck gives is specific and diagnostically useful:

Parameter	Old Tank Syndrome value
Ammonia	High — often above 10 mg/L
Alkalinity (KH)	Little or none
pH	Low — below 6.0
Total hardness (GH)	High — several hundred mg/L

[INF] **Note the diagnostic combination:** high GH with zero KH and low pH. This is Profile 2 from Section 4.5, and it is exactly the state that makes “high pH means hard water” a dangerous heuristic — this water is very hard and very acidic simultaneously.

Merck’s companion aquarium-management chapter adds an important mechanistic detail: **if the pH decline is gradual, fish often adapt with no outward sign of distress; if it is rapid, it can be fatal**. It also notes that the syndrome is especially common in tanks holding large carnivorous fish, which excrete large amounts of total ammonia nitrogen from high-protein meals, driving heavy bicarbonate consumption by the biofilter (Merck Veterinary Manual, *Management of Aquarium Fish*).

21.2 The Full Causal Chain

[STR/INF]

1. **Water changes are small, infrequent, or replaced entirely by evaporation top-offs.**
2. **Nitrification consumes alkalinity continuously** (Section 20).
3. **Organic acids accumulate** from decomposition.
4. **Nitrate accumulates**, often to very high levels, as a marker of the same process.
5. **Minerals concentrate** from evaporative top-off with mineralized water — GH rises even as KH falls (Section 23).
6. **Alkalinity reaches zero.**
7. **pH falls**, gradually at first, then rapidly.
8. **Nitrification slows** at the new low pH, and ammonia begins to accumulate.
9. **The ammonia is mostly ammonium** at pH < 6, so the fish survive what would otherwise be a lethal TAN load.
10. **Fish acclimate to the chronic conditions** and often appear normal.
11. **The keeper eventually notices something is wrong and performs a large water change with fresh tap water.**
12. **pH rises abruptly** toward the source water value.
13. **The accumulated ammonium converts rapidly to toxic free ammonia** — a pH rise from 6.0 to 7.6 increases the toxic fraction roughly thirtyfold.
14. **Fish that had survived for months die within hours of the “corrective” water change.**

[INF] **Step 14 is the crux, and it is the reason the folklore “large water changes kill fish in old tanks” persists — because it is true, in that specific circumstance, for that specific reason.** The water change did not kill the fish by being large. It killed them by rapidly converting a stored ammonium reservoir into free ammonia, and secondarily by imposing a large, rapid osmotic and pH change on animals that had chronically acclimated to different water.

21.3 What “pH Crash” Means

[HOB] The term has no standardised scientific definition and is used for several distinct events:

Usage	What is actually happening	Standardised?
Alkalinity exhaustion in an aging tank	The Old Tank Syndrome mechanism	[VET] — yes, effectively
Overnight pH drop in a low-KH planted tank	Normal diurnal CO ₂ cycling, amplified by low buffering	Normal, not a crash
pH falling after a botanical or peat addition	Organic acid input exceeding a small buffer	Real but usually benign
pH falling after CO ₂ overdose	CO ₂ event; the hazard is CO ₂ , not pH	Real, dangerous
pH falling after acid dosing exhausted the buffer	Section 7’s failure mode	Real, avoidable
Anything the keeper didn’t expect	Not a defined phenomenon	No

[INF] The term is most useful when reserved for **alkalinity exhaustion followed by rapid pH decline**, which is the mechanism with a genuine physical basis and a genuine hazard.

21.4 Which Old Tank Syndrome Claims Are Supported

Claim	Classification	Notes
Alkalinity depletion causes progressive pH decline	[STR/VET]	Directly stated in veterinary literature; mechanism well understood
Topping off without water changes is a primary cause	[VET]	Explicitly identified by Merck
The parameter profile is high GH, zero KH, low pH, high ammonia	[VET]	Merck gives this profile
A large water change on such a tank can kill the fish	[STR/VET]	Mechanistically clear via ammonia speciation and rapid pH change
Fish adapt to gradual decline and show no distress	[VET]	Merck states this explicitly
Nitrate itself is the acidifier	[MYTH]	Nitrate is a marker, not the cause. Nitrate is the conjugate base of a strong acid and does not acidify water. The H ⁺ produced during nitrification is the acidifier
"Never do a large water change on an old tank"	[Oversimplified]	The hazard is real but conditional. See 21.5
Old tanks are "seasoned" and better	[MYTH]	Mature biofilm and stable systems are genuinely good; accumulated nitrate, depleted buffer and concentrated minerals are not. "Mature" and "neglected" are different states
The tank must be torn down completely	[VET, but context-dependent]	Merck recommends relocating fish and rebuilding the system for genuine OTS with ammonia above 10 mg/L. For a milder case caught early, gradual correction is reasonable

21.5 Correcting a Suspected Old Tank

[INF] Correction protocol, ordered by priority. This is a synthesis of veterinary guidance and the chemistry above, not a copied procedure.

Step 1: Test before doing anything. Specifically: **ammonia, nitrite, nitrate, pH, KH, GH**. And test the **source water** for the same. This is the step that gets skipped and it is the only one that determines what is safe to do next.

Step 2: Assess the ammonia risk. This governs everything. - **Ammonia detectable and pH below ~6.5:** the dangerous configuration. A large water change with higher-pH water will convert stored ammonium to free ammonia. Proceed with small, frequent changes (10% or less, daily) so pH rises slowly. Consider an ammonia-binding conditioner. In severe cases (Merck's >10 mg/L threshold), relocating the fish to properly conditioned water is safer than trying to correct around them. - **Ammonia zero and pH low:** the biofilter is coping. Larger changes are safer, though gradual is still preferable for osmotic reasons.

Step 3: Correct gradually. Frequent small changes move pH, KH, GH, nitrate and TDS toward the source values on a smooth curve rather than a step. Use the dilution formula from Section 12.2.7 to predict the trajectory.

Step 4: Address the root cause. The tank got there through a mismatch between acid production and alkalinity replacement. Fix that: increase water change volume or frequency, reduce bioload or feeding, add a slow carbonate source (crushed coral — Section 17), and **switch to RO or distilled water for evaporation top-off** (Section 23).

Step 5: Monitor KH, not pH, going forward. KH is the leading indicator.

21.6 Prevention

[INF] - Regular partial water changes of adequate volume — the single most effective measure. - **Top off evaporation with RO or distilled water**, never with mineralized tap water. - Test KH monthly. A declining trend is the early warning that costs nothing to detect. - Match bioload and feeding to maintenance capacity. - Consider a passive carbonate source (crushed coral) in soft-water systems as a safety floor. - Compare tank parameters against source water parameters periodically (Section 9.1).

Section 22: GH, KH and pH During Water Changes

Search terms for this section: water change, partial water change, large water change, matching parameters, replacement water, temperature matching, dilution, water change percentage, drip refill.

22.1 What a Water Change Actually Does to Each Parameter

[EST] A water change is a **dilution and replacement** operation. For conservative parameters (GH, KH, TDS, nitrate), the result is exactly linear:

$$P_{\text{after}} = P_{\text{tank}} \times (1 - f) + P_{\text{source}} \times f$$

where f is the fraction changed.

pH does not follow this equation, because pH is logarithmic and buffered. The resulting pH is set by the combined alkalinity and CO_2 of the mixture, and typically ends up much closer to the more strongly buffered water's pH.

Worked example: Tank: pH 6.4, KH 1, GH 4, nitrate 60. Source: pH 7.6, KH 8, GH 12, nitrate 0. 50% change: - **KH:** $(1 \times 0.5) + (8 \times 0.5) = 4.5 \text{ dKH}$ - **GH:** $(4 \times 0.5) + (12 \times 0.5) = 8 \text{ dGH}$ - **Nitrate:** $(60 \times 0.5) + 0 = 30 \text{ mg/L}$ - **pH:** not calculable by averaging. The mixture now has 4.5 dKH of buffer and, once CO_2 equilibrates, will sit somewhere around 7.4-7.6 — **much closer to the source than to the midpoint**, because the source brought the alkalinity that now dominates.

[INF] **This is the central insight about water changes and pH:** the pH does not split the difference. **It follows the alkalinity.** A 50% change with strongly buffered water moves pH most of the way to the source value, not half way.

22.2 Why Water Change Percentage Alone Does Not Determine Risk

[INF] The hobby debate over “how big a water change is safe” is mis-specified. The risk is not a function of percentage. It is a function of **the difference between the tank water and the replacement water, multiplied by the percentage**.

$$\text{Risk} \propto (\text{parameter difference}) \times (\text{fraction changed})$$

Scenario	Difference	% changed	Effective disturbance
Tank 7.4/6 dKH, source 7.4/6 dKH	~0	80%	Negligible. Large changes on matched water are essentially harmless
Tank 6.0/0 dKH, source 7.8/9 dKH	Large	20%	Significant. Small change, big disturbance
Tank 6.0/0 dKH, source 7.8/9 dKH	Large	5%, daily	Manageable. Same total volume, smoothed over time
Tank 7.2/5 dKH, source 7.4/6 dKH	Small	50%	Minor

[INF] **Consequences of this framing:**

1. **“Never change more than 25%” is not a chemistry rule.** It is a heuristic that protects against unknown parameter mismatch. Where the source water matches the tank, large changes are routinely done — commercial breeding operations and discus keepers frequently run 50-90% daily changes.
2. **The relevant question is not “how much?” but “how different?”** — and the answer requires testing the source water, which is why Section 9.1 matters.
3. **Small frequent changes are safer than large infrequent ones for mismatched water**, and the difference is not marginal.
4. **The dangerous case is precisely a neglected tank plus a large change**, because that maximises both terms in the product. See Section 21.

22.3 What to Match, in Priority Order

[INF] Not all parameters carry equal risk from a mismatch:

Priority	Parameter	Why	Realistic tolerance
1	Temperature	Immediate physiological effect; thermal shock is fast	Within 1-2 °C
2	Chlorine/chloramine removal	Acutely toxic; kills biofilter	Absolute
3	Conductivity/TDS	Osmotic load; the parameter most likely to be the real cause of “pH shock”	Within roughly 10-20% for sensitive invertebrates; wider for fish
4	KH	Determines the pH outcome and buffering	Moderate tolerance
5	GH	Osmotic and mineral	Moderate tolerance
6	pH	Follows from KH and CO_2 ; usually the symptom rather than the cause	Widest tolerance

[INF] **The ordering here is deliberately contrary to hobby emphasis.** The hobby watches pH most closely and TDS least. The physiology argues for the reverse: the osmotic and ionic change is the more direct disturbance, and pH usually tags along with it. See Sections 26 and 27.

22.4 Practical Techniques for Reducing Water Change Disturbance

Technique	What it addresses
Pre-mix and pre-test replacement water in a dedicated container	Everything. The single highest-value habit
Aerate replacement water for an hour before use	Degasses supersaturated CO_2 so the source pH is at its equilibrated value rather than its as-drawn value; equalises temperature
Refill slowly (drip line, slow trickle, or a lidded container with a small siphon)	Spreads the parameter change over 30-60 minutes rather than 2 minutes
Smaller, more frequent changes during a transition	Reduces the disturbance per event
Match temperature at the tap before filling	Thermal shock
Dose dechlorinator to the change water, not the tank	Ensures full treatment before contact with livestock
Remineralize RO in the mixing container, never in the tank	Prevents spiking the tank at the moment of dosing
Check the source water seasonally	Catches utility source switches before they cause a mystery

22.5 Top-Offs Are Not Water Changes

[EST] This distinction is fundamental and is the mechanistic core of Old Tank Syndrome.

- **Evaporation removes water. It does not remove dissolved substances.** Minerals, nitrate, dissolved organics and sodium all stay behind.
- **Topping off with mineralized water adds more of everything** while removing nothing.
- Therefore top-offs **concentrate** GH, TDS, nitrate and dissolved organics over time, while **not replacing KH consumed by nitrification**.

[INF] The resulting trajectory is exactly the Old Tank Syndrome profile: **GH rising, TDS rising, nitrate rising, KH falling, pH falling**. See Sections 21 and 23.

A water change removes tank water. A top-off does not. They are different operations with opposite effects on dissolved solids, and using one as a substitute for the other is the specific practice veterinary literature identifies as causal.

Section 23: Evaporation

Search terms for this section: evaporation, top off, topping off, auto top off, ATO, mineral accumulation, TDS creep, RO top off, distilled top off.

23.1 The Basic Physics

[EST] - **Water evaporates.** H₂O molecules leave the surface as vapour. - **Dissolved ionic solids do not evaporate.** Calcium, magnesium, sodium, chloride, sulfate, bicarbonate and nitrate remain in the tank. - Therefore evaporation **increases the concentration** of everything dissolved.

[EST] **Partial exceptions worth noting:** - **Dissolved gases** (CO₂, O₂, ammonia as NH₃) do exchange with the atmosphere and are not concentrated by evaporation. - **KH behaves differently from GH** in an evaporating tank: while evaporation concentrates bicarbonate along with everything else, KH is *simultaneously* being consumed by nitrification. **The net direction of KH depends on which process dominates**, and in a stocked tank, consumption usually wins. This is why the Old Tank Syndrome profile shows GH rising while KH falls — they are being driven by different processes.

23.2 The Arithmetic of Concentration

[EST] If a tank loses a fraction *e* of its volume to evaporation and is topped off with water of concentration C_{top}:

$$C_{\text{after}} = (C_{\text{before}} \times 1 + C_{\text{top}} \times e) / (1 + e) \text{ — for the topped-off volume relative to the original}$$

More usefully, over repeated cycles with **zero-TDS top-off water**, concentration is unchanged. With **mineralized top-off water**, concentration ratchets upward with every cycle.

Worked example — the ratchet: A 100 L tank at 8 dGH, losing 5 L per week to evaporation, topped off with tap water at 12 dGH.

Each week: 95 L at 8 dGH plus 5 L at 12 dGH = (95×8 + 5×12)/100 = **8.2 dGH**.

Week	GH (dGH)
0	8.0
4	8.8
8	9.5
13	10.4
26	12.0
52	14.2
104	18.0

[INF] **Note what this shows.** With *no water changes at all*, GH climbs steadily and asymptotically approaches — and can substantially exceed — the source water's value, because minerals are added with every top-off and removed by nothing. After two years the tank is well above its own tap water's hardness. TDS and nitrate climb on the same curve, faster in nitrate's case because it is also being produced internally.

With the same top-off but a 25% monthly water change, the water change removes 25% of the accumulated excess each month, and GH stabilises near the source value instead of climbing.

[INF] **This table is the numerical justification for the veterinary claim that topping off without water changes causes Old Tank Syndrome.** It is not folklore; it is a mass balance with no removal term.

23.3 Top-Off Water Choice

Top-off water	Effect on GH/TDS	Effect on KH	Verdict
RO / RO+DI / distilled	Correct — maintains concentration	Neutral	Recommended in all cases
Collected rainwater	Correct, with collection caveats (Section 12.2.1)	Neutral	Good if cleanly collected
Soft tap water (1-3 dGH)	Slow accumulation	Slight addition	Acceptable
Moderate tap water (6-10 dGH)	Steady accumulation	Addition	Poor practice long-term
Hard tap water (12+ dGH)	Rapid accumulation	Addition	Avoid
Remineralized RO	Accumulation, same as tap	Addition	Wrong tool — remineralized water is for <i>changes</i> , not top-offs

[INF] **The single clearest, cheapest practice improvement available to many aquarists:** use distilled or RO water for evaporation top-off, and reserve tap or remineralized water for actual water changes. For most tanks this is a few litres a week of distilled water — a trivial cost that eliminates an entire failure mode.

23.4 Evaporation Rate Factors

[INF] Useful for anticipating how much this matters in a given setup:

Factor	Effect on evaporation
Open top (no lid)	Large increase
Higher water temperature	Increase
Low ambient humidity (winter heating, arid climates)	Increase
Surface agitation, spray bars, air stones	Increase
HOB filter with a waterfall return	Notable increase
Sump systems with large exposed surface	Large increase
Higher air movement over the surface (fans, vents)	Increase
Cooling fans (deliberate evaporative cooling)	Very large increase — these can evaporate several litres a day in summer, and are the most likely case for rapid mineral accumulation
Tight-fitting lid	Large decrease

[INF] Sumps and evaporative cooling fans are the two configurations where top-off water quality matters most, because the volumes involved are largest.

23.5 Practical Detection

[INF] Mineral accumulation from top-offs is detectable long before it causes problems:

- **TDS meter reading climbing over months** with no other explanation — the clearest and cheapest signal.
- **GH climbing above source water GH** — diagnostic, since nothing else in a normal aquarium adds calcium and magnesium.
- **Mineral crust on the glass at the waterline, on lids, and on heater and impeller surfaces.**
- **Nitrate climbing faster than bioload explains.**

[INF] A TDS meter costs very little and monitoring TDS monthly is the most efficient single test for detecting this pattern.

Section 24: Testing GH, KH and pH

Search terms for this section: test kit, liquid test, titration, drop test, test strips, digital meter, calibration, reagent, endpoint, accuracy, precision, TDS meter, conductivity meter, water report, laboratory testing.

24.1 Method Comparison

Method	Measures	Typical resolution	Typical cost	Main strengths	Main weaknesses
Liquid titration (GH/KH)	GH, KH	1 drop = 1 dGH/dKH (some kits 0.5)	Low	Genuinely accurate when done correctly; the hobby standard for a reason	Endpoint judgement; slow; resolution limited by drop size
Liquid colorimetric (pH)	pH	0.2-0.4 units	Low	Cheap; no calibration	Colour-matching subjectivity; range-limited (typically 6.0-7.6 low range); useless in tannin-stained water
Test strips	pH, GH, KH, NO ₂ , NO ₃	±0.5 pH; hardness in coarse bands	Very low per test	Fast; convenient; good for screening trends	Least precise; degrade with humidity; timing-sensitive
Digital pH meter	pH	0.01	Moderate	High resolution; logging; works in coloured water	Requires calibration ; probe maintenance; probes die; false confidence when uncalibrated
TDS meter	Conductivity → estimated TDS	1 ppm	Very low	Best value tool in the hobby. Detects change instantly	Does not identify <i>which</i> ions; conversion factor is an assumption
Conductivity meter (µS/cm)	Conductivity directly	1 µS/cm	Low-moderate	Reports the raw measured quantity without a conversion assumption	Same limitation on ion identity
Laboratory water analysis	Individual ions	Very high	High per test	The only method that gives Ca and Mg separately , plus sodium, sulfate, chloride	Cost; turnaround
Municipal water report (CCR)	Individual constituents	Varies	Free	Free lab analysis of your source water; gives Ca:Mg ratio	Annual averages; plant not tap; source only

24.2 Liquid Titration Technique — Where Errors Come From

[INF] GH and KH liquid kits are accurate when used correctly. Common errors, in approximate order of frequency:

- Wrong sample volume.** The kit is calibrated to a specific volume (commonly 5 mL). Filling to the wrong line changes every subsequent drop's value proportionally.
- Inconsistent drop size.** Hold the bottle **vertically**, not at an angle. Angled bottles produce smaller, inconsistent drops and inflate the count.
- Not counting the colour-change drop.** Conventions differ between kits; follow the specific instructions. This is a ±1 error, which at 2 dKH is a 50% error.
- Not inverting/swirling between drops.** Incomplete mixing delays the endpoint.
- Expired reagents.** Titrants degrade. A kit more than 2-3 years old, or one stored hot, should be replaced or checked against a known standard.
- Poor lighting or coloured background.** Judge the endpoint against white, in daylight or neutral light.
- Rounding at low values.** At 1-2 dKH, one drop of error is enormous in relative terms. **Low-KH systems are exactly where the test is least precise and most important.** For these, doubling the sample volume (10 mL) and halving the result gives 0.5-degree resolution — a genuinely useful technique.

[INF] **The double-volume trick deserves emphasis** because it directly addresses the resolution problem in the systems that most need it. Using 10 mL of sample instead of 5 mL means each drop represents 0.5 dKH rather than 1.0 dKH. It costs nothing but reagent.

24.3 Digital pH Meter Discipline

[INF]

Practice	Why
Calibrate with fresh certified buffers before any measurement that matters	Uncalibrated meters can be off by a full unit
Two-point minimum (7.00 and 4.01 for acidic tanks; 7.00 and 10.01 for alkaline)	Corrects both offset and slope
Discard buffer after use; never pour it back into the bottle	Contamination
Store the probe wet, in KCl storage solution	Storing dry or in distilled water destroys the reference junction
Rinse with distilled water between samples; blot, do not wipe	Wiping creates static charge that disturbs readings
Allow full reading stabilisation	Readings drift toward the true value over 30-120 seconds
Replace probes on a schedule	Slope degrades before the meter reports failure
Note the sample temperature	ATC corrects the electrode, not the sample's actual pH

24.4 When Precision Matters and When It Does Not

[INF]

Precision genuinely matters for: - **CO₂ management in planted tanks.** A 0.2 pH error is a substantial CO₂ error, and CO₂ overdose kills fish. - **Low-KH systems,** where 1 dKH vs 2 dKH is the difference between “some buffer” and “essentially none.” - **RO remineralization consistency,** where the target is reproducibility rather than a specific number. - **Diagnosing a problem** — the whole point is detecting a real difference. - **Sensitive invertebrate systems,** where small parameter movements are the failure mode.

Approximate knowledge is entirely sufficient for: - Establishing whether tap water is “soft,” “moderate” or “hard” for species selection. - Confirming that a stable tank is still stable. - Deciding whether crushed coral is worth adding. - Most general community-tank management.

[INF] **The most common testing error is not imprecision. It is measuring the wrong thing, or measuring inconsistently.** A precise pH reading taken at a

different time of day each week is less informative than an imprecise KH reading taken monthly at a consistent time. **Trend beats precision.**

24.5 Unit and Conversion Errors

[INF] The single most consequential class of testing error in this subject.

Error	Magnitude
Reading a ppm result as dGH/dKH, or vice versa	~18x
Confusing gpg with dGH	~4% (minor)
Confusing meq/L with dKH	~2.8x
Reading a “hardness” figure from a water report that is actually calcium-only, not total	Underestimate, varies
Comparing a veterinary “alkalinity ≥ 100 mg/L” figure against a dKH reading without converting	Appears to be a 100 vs 6 disagreement; is actually the same value

[INF] Always record the unit alongside the number. A logbook entry of “KH 6” is ambiguous; “KH 6 dKH (107 ppm)” is not. This matters especially when comparing a hobby test kit against a municipal report or a veterinary recommendation.

24.6 A Practical Testing Cadence

[INF] Not prescriptive; a reasonable starting framework:

Situation	pH	KH	GH	TDS
New tank, cycling	Weekly	Weekly	At setup	At setup
Established, stable, tap water	Monthly	Monthly	Quarterly	Monthly
Established, soft/low-KH water	Biweekly	Biweekly	Monthly	Monthly
RO-based / shrimp system	Each water change	Each water change	Each water change	Every batch
CO ₂ -injected planted	Profile across the day, periodically	Monthly (important — see 19.7)	Quarterly	Quarterly
Troubleshooting	Full panel, plus the same panel on source water			
Source water	Quarterly, and any time something changes unexpectedly			

Section 25: Units and Conversions

Search terms for this section: dGH, dKH, degrees hardness, German degrees, ppm, mg/L, CaCO₃ equivalent, meq/L, milliequivalents, grains per gallon, gpg, conductivity, microsiemens, µS/cm, TDS, conversion factor.

25.1 Master Conversion Table

From	To	Multiply by
dGH or dKH	mg/L (ppm) as CaCO ₃	17.848
mg/L (ppm) as CaCO ₃	dGH or dKH	0.05603 (÷ 17.848)
meq/L	mg/L as CaCO ₃	50.04
mg/L as CaCO ₃	meq/L	0.01998 (÷ 50.04)
meq/L	dKH	2.804
dKH	meq/L	0.3566
grains per gallon (gpg)	mg/L as CaCO ₃	17.118
mg/L as CaCO ₃	gpg	0.05842
mg/L Ca (element)	mg/L as CaCO ₃	2.497
mg/L Mg (element)	mg/L as CaCO ₃	4.118
mmol/L Ca or Mg	mg/L as CaCO ₃	100.09
mmol/L Ca or Mg	dGH	5.608
ppm (mg/L)	µS/cm (approximate, NaCl scale)	÷ 0.5 = × 2
µS/cm	ppm TDS (NaCl scale, factor 0.5)	0.5
µS/cm	ppm TDS (442 scale, factor ~0.65)	0.65
µS/cm	ppm TDS (KCl scale, factor ~0.55)	0.55

25.2 Why 17.848?

[EST] The German degree of hardness was defined as **10 mg of calcium oxide (CaO) per litre**.

- Molar mass CaO = 56.08 g/mol
- Molar mass CaCO₃ = 100.09 g/mol
- 10 mg CaO/L = 10/56.08 = 0.1783 mmol/L
- Expressed as CaCO₃: 0.1783 × 100.09 = **17.85 mg/L**

Hence 1 °dH = 17.848 mg/L as CaCO₃. Many sources round to 17.9 or 17.86; all are the same figure.

[INF] **Why “as CaCO₃” at all?** Because expressing all hardness and all alkalinity in a single common currency allows different ions with different charges and masses to be summed and compared. It does not mean the water contains calcium carbonate. It means *the water’s hardness (or alkalinity) is equivalent to that concentration of calcium carbonate*. This is a common source of confusion — a report saying “hardness 150 mg/L as CaCO₃” does not mean 150 mg of chalk is dissolved in each litre.

25.3 Worked Conversion Examples

Example 1 — Municipal report to aquarium units. Report states: hardness 187 mg/L as CaCO₃; alkalinity 142 mg/L as CaCO₃. - GH = 187 ÷ 17.848 = **10.5 dGH** - KH = 142 ÷ 17.848 = **8.0 dKH** - USGS category: **very hard** - GH > KH by 2.5 degrees → some non-carbonate hardness present.

Example 2 — Separate ion listing to GH. Report states: calcium 68 mg/L; magnesium 19 mg/L. - Ca: 68 × 2.497 = 169.8 mg/L as CaCO₃ - Mg: 19 × 4.118 = 78.2 mg/L as CaCO₃ - Total = 248.0 mg/L as CaCO₃ = **13.9 dGH** - Ca:Mg hardness ratio = 169.8 : 78.2 ≈ **2.2 : 1** — magnesium-rich relative to the usual 3-4:1 convention.

Example 3 — Veterinary recommendation to hobby units. “Alkalinity should be ≥ 100 mg/L in freshwater systems.” - 100 ÷ 17.848 = **5.6 dKH**. The WSAVA range of 100-250 mg/L = **5.6-14.0 dKH**.

Example 4 — Aquaculture threshold to hobby units. “Total alkalinity should be no lower than 20 mg/L as CaCO₃ in production ponds.” - 20 ÷ 17.848 = **1.1 dKH**. This is the threshold below which large diurnal pH swings become likely.

Example 5 — meq/L to dKH. Lake Tanganyika alkalinity commonly reported around 6.5 meq/L. - 6.5 × 50.04 = 325 mg/L as CaCO₃ - 325 ÷ 17.848 = **18.2 dKH**.

Example 6 — grains per gallon. Water softener installer says “your water is 14 grains.” - 14 × 17.118 = 239.7 mg/L as CaCO₃ - ÷ 17.848 = **13.4 dGH**.

25.4 Why ppm GH Is Not Conceptually Identical to ppm TDS

[EST] **This deserves explicit treatment because the units look identical and are not.**

Both are reported in mg/L. They measure different things:

	ppm GH	ppm TDS
What is counted	Divalent cations only (essentially Ca ²⁺ + Mg ²⁺)	All dissolved substances
Basis of the number	An <i>equivalent</i> concentration of CaCO ₃ , not the actual mass of anything present	An <i>estimated</i> total mass of dissolved solids, inferred from conductivity
How obtained	Chemical titration (or calculation from ion concentrations)	Conductivity measurement × an assumed conversion factor
Includes sodium?	No	Yes
Includes nitrate?	No	Yes
Includes bicarbonate?	No	Yes
Includes dissolved organics?	No	Partially (only if they conduct)
Is it a direct measurement?	Yes, of hardness	No — it is a proxy

Worked illustration: Water A: 150 ppm GH (Ca/Mg), plus 150 ppm bicarbonate, plus 20 ppm sodium, plus 15 ppm nitrate → **TDS roughly 335 ppm, GH 150 ppm**

(8.4 dGH). Water B: household-softened. GH near 0 ppm; sodium high; bicarbonate unchanged — **TDS roughly 330 ppm, GH ~5 ppm (0.3 dGH).**

Nearly identical TDS. Completely different water. This is the strongest argument for testing GH separately rather than relying on a TDS meter alone — and simultaneously the strongest argument for owning a TDS meter, since it is the fastest way to detect that *something* has changed. The two tools answer different questions and neither replaces the other.

Section 26: TDS and Conductivity

Search terms for this section: TDS, total dissolved solids, conductivity, EC, electrical conductivity, microsiemens, $\mu\text{S}/\text{cm}$, ppm, TDS meter, conversion factor, TDS shock, osmotic shock, shrimp TDS.

26.1 What Is Actually Being Measured

[EST] A “TDS meter” does not measure total dissolved solids. It measures **electrical conductivity** and then multiplies by a conversion factor to *estimate* TDS.

$$\text{TDS (ppm)} \approx \text{conductivity } (\mu\text{S}/\text{cm}) \times k$$

Common values of k : - **0.50** — NaCl scale, the most common default in aquarium meters - **0.55** — KCl scale - **0.64-0.70** — “442” natural water scale - Some meters allow the factor to be set; many do not disclose it.

[INF] Practical consequence: two TDS meters reading the same water can display meaningfully different ppm values simply because they use different conversion factors. **Conductivity in $\mu\text{S}/\text{cm}$ is the raw, unambiguous measurement; ppm TDS is an interpretation of it.** Where precision matters — shrimp keeping, RO consistency — recording $\mu\text{S}/\text{cm}$ avoids an entire class of confusion.

26.2 What Conductivity Responds To

[EST] Conductivity depends on: - **Ion concentration** — more ions, more conductivity. - **Ion charge** — divalent ions (Ca^{2+} , Mg^{2+} , SO_4^{2-}) contribute more per mole than monovalent (Na^+ , Cl^- , K^+). - **Ion mobility** — H^+ and OH^- are exceptionally mobile and conduct far better than their concentration suggests. This is why very acidic water can read surprisingly high conductivity. - **Temperature** — conductivity rises roughly 2% per $^{\circ}\text{C}$. **Good meters compensate automatically; cheap ones may not.** Comparing a 20 $^{\circ}\text{C}$ sample against a 26 $^{\circ}\text{C}$ tank without compensation introduces a ~12% error.

[EST] What conductivity does not respond to: - **Non-ionic dissolved substances.** Sugars, many dissolved organic molecules, urea, and uncharged species do not conduct. They are “dissolved solids” that a TDS meter cannot see. - **Which ions are present.** Conductivity is blind to composition.

26.3 What TDS Can and Cannot Tell You

TDS/conductivity <i>can</i> tell you	TDS/conductivity <i>cannot</i> tell you
That total ionic content changed	Which ions changed
That your RO membrane is degrading	Whether GH or KH specifically changed
That remineralization was consistent between batches	Whether the Ca:Mg ratio is right
That a tank is accumulating dissolved solids over time (top-off ratchet, overdue water changes)	Whether the accumulation is nitrate, minerals, or sodium
That source water changed seasonally	Whether the change matters
That two water samples are ionically similar	That two water samples are chemically identical

[INF] Best framing: TDS is a change detector, not a diagnostic. It answers “is this the same as last time?” extremely well and cheaply. It never answers “what is in this water?”

26.4 Relationship With GH and KH

[INF] In a typical freshwater aquarium on mineralized tap water, GH and KH together account for a large share of the conductivity — but the share varies enormously with water type.

Rough illustrative relationships (highly water-dependent):

Water type	GH	KH	Typical conductivity	GH+KH share of conductivity
RO/DI	0	0	0-5 $\mu\text{S}/\text{cm}$	n/a
Remineralized RO (GH-only, 6 dGH)	6	0	~180-220 $\mu\text{S}/\text{cm}$	High
Soft tap water	3	2	~120 $\mu\text{S}/\text{cm}$	High
Typical municipal	8	6	~350 $\mu\text{S}/\text{cm}$	Moderate-high
Hard limestone tap	15	12	~600 $\mu\text{S}/\text{cm}$	High
Sodium-softened tap	~0	10	~600 $\mu\text{S}/\text{cm}$	Low — conductivity is mostly sodium
Old, under-maintained tank	14	1	~700 $\mu\text{S}/\text{cm}$	Moderate — much is nitrate and sodium

[INF] The two bolded/notable rows are the important ones. They are precisely the cases where a TDS reading alone would mislead: softened water reads “high TDS” while being calcium-free, and an old tank reads “high TDS” mostly from accumulated nitrate and organics rather than useful minerals.

26.5 TDS in Shrimp Keeping

[HOB] Dwarf shrimp keepers track TDS more closely than any other group in the freshwater hobby, and generally more closely than they track GH or KH.

Commonly cited target ranges **[HOB — hobby convention, provenance not traceable to primary research]:**

Species group	Commonly cited TDS	Commonly cited GH	Commonly cited KH
<i>Neocaridina</i> (cherry etc.)	~150-300 ppm	6-14 dGH	2-8 dKH
<i>Caridina</i> (Crystal/Bee)	~100-150 ppm	4-6 dGH	0-2 dKH
Taiwan Bee	~110-140 ppm	4-5 dGH	0-1 dKH
Sulawesi	~150-200 ppm, distinctive composition	6-8 dGH	2-4 dKH

[CON] Important caveats on these figures: 1. They are hobby conventions that propagate between sources; this research could not trace them to primary literature. 2. Different sources give different numbers (compare the *Neocaridina* GH ranges reported by different specialist sites: 4-10, 6-12, 6-14, 7-15 all appear). 3. **TDS alone is insufficient.** 150 ppm TDS composed of calcium and magnesium sulfate is a completely different environment from 150 ppm composed of sodium

chloride. Shrimp keepers who target TDS without also targeting GH are managing the wrong variable. 4. Consistency between batches probably matters more than the specific value within a reasonable range — which is exactly what TDS is good for measuring.

[INF] The defensible version of shrimp TDS practice: use TDS as a **reproducibility check** on remineralization, and use GH as the **biologically meaningful target**. TDS confirms you made the same water as last time; GH confirms the water contains what the shrimp need.

26.6 “TDS Shock”

[CON] Real phenomenon; commonly over-attributed.

The mechanism is sound [EST]: a rapid change in external osmolality forces immediate osmoregulatory adjustment. Aquatic invertebrates in general, and dwarf shrimp in particular, are more sensitive to this than most fish because they are smaller, have thinner permeable surfaces, and — in the case of shrimp — may be mid-molt with a temporarily permeable new exoskeleton.

What is over-attributed [CON]: - Every unexplained shrimp death after a water change gets labelled TDS shock. Other candidates: copper contamination, ammonia from disturbed substrate, temperature swing, chlorine/chloramine, dissolved oxygen change, and simple coincidence with molting. - The specific magnitude thresholds circulated in the hobby (e.g. “never change TDS by more than 20 ppm at once”) are conventions without traceable evidentiary basis.

[INF] Defensible statement: *rapid, large changes in ionic concentration are a plausible and mechanistically supported stressor for dwarf shrimp, and matching the ionic content of replacement water is sound practice — but “TDS shock” is a hypothesis applied retrospectively far more often than it is demonstrated.*

26.7 Conductivity in Fish Breeding

[HOB/STR] Conductivity is arguably a **better predictor of breeding success than pH** for soft-water species, and is the parameter that professional breeders actually watch.

- **Discus:** commercial practice targets low conductivity for fertilisation and hatch; figures under about 150 µS/cm are commonly cited.
- **Cardinal tetra, many small characins:** very low conductivity commonly reported as necessary.
- **Wild-collected Rio Negro species:** native water is often under 20 µS/cm.
- **Mechanism [INF]:** egg chorion permeability, water hardening after fertilisation, and sperm motility are all osmotically sensitive. In high-ionic-strength water, sperm motility duration is reduced and eggs may not hydrate correctly.

[INF] This is a significant point: the hobby discusses breeding in terms of pH, while the professional practice tracks conductivity. If a single number had to be chosen for soft-water breeding management, conductivity would probably be the better one.

Section 27: Osmoregulation

Search terms for this section: osmoregulation, osmotic gradient, ion regulation, ionoregulation, gill, chloride cell, ionocyte, kidney, sodium uptake, chloride uptake, osmotic shock, hypoosmotic, plasma osmolality.

27.1 The Fundamental Situation

[EST] A freshwater fish is **hyperosmotic to its environment**: its body fluids contain far more dissolved solute than the surrounding water.

- Fish plasma osmolality: roughly **270-320 mOsm/kg** (approximately equivalent to 9-11 ppt salinity).
- Freshwater: typically **1-15 mOsm/kg**; Amazonian blackwater can be under 1.

Two consequences follow directly from physics:

- Water flows into the fish** continuously, down the osmotic gradient, across the gills and any permeable surface.
- Ions leak out** of the fish continuously, down their concentration gradients.

The fish must therefore, constantly and at metabolic cost: - **Excrete the excess water** — freshwater fish produce large volumes of very dilute urine. - **Actively reclaim ions** — via specialised gill cells (ionocytes, historically “chloride cells”) using ATP-driven transport, and via ion reabsorption in the kidney. - **Take up ions from the water** to replace losses — sodium and chloride primarily, plus calcium.

[EST] **Freshwater fish do not drink appreciably.** They have no need to; water is arriving passively. (Marine teleosts do the opposite — they drink constantly and excrete salt.)

27.2 Why Water Chemistry Changes the Cost

[EST/STR]

Lower external ion concentration → steeper gradient → faster passive loss → higher active uptake cost. This is the direct physiological basis for the claim that very soft water is “harder work” for fish.

Environmental calcium reduces the leak. As covered in Section 2.5.1, calcium at gill tight junctions reduces paracellular permeability. More calcium = tighter gills = less passive ion loss = lower metabolic cost.

Low pH increases the leak — in many species. Hydrogen ions displace calcium at the tight junctions, and H⁺ also competes with Na⁺ at uptake transporters. This is the mechanism by which acidification causes ionoregulatory failure in acid-sensitive species.

Dissolved organic carbon reduces the leak. As covered in Section 16.6, Rio Negro DOC provided near-complete protection against acid-induced ion loss in zebrafish, reducing diffusive Na⁺ and Cl⁻ loss and stimulating Na⁺ uptake (Duarte et al., 2016).

27.3 The Critical Qualification — Adaptation Is Real and Species-Specific

[STR/CON] **This section exists to prevent the over-generalisation that Section 27.2 invites.**

The “soft water is harder work” model was developed largely in temperate species — salmonids, cyprinids, goldfish — and it does not transfer cleanly to fish that evolved in ion-poor water.

Documented findings from Amazonian species research: - Characids from ion-poor blackwater show **Na⁺ uptake that is largely unaffected even at pH as low as 3.25**, and rates of diffusive loss that strongly resist stimulation by acid. Disturbances on acid exposure are small, brief, or absent. - Cichlids have lower baseline rates of both ion uptake and diffusive loss than characids. At pH ≤ 4.0 their Na⁺ uptake is fully inhibited, but efflux resists stimulation, so **net salt loss stays small and tolerable**. - **Both groups resist acid-induced ion loss far better than temperate models predict.** - Calcium’s protective effect is inconsistent in these species — present in some studies, absent in others, suggesting either very high tight-junction calcium affinity or reliance on other factors such as DOC.

(These findings come from a substantial body of work by Gonzalez, Wood, Wilson, Val and colleagues from the 1990s to the 2020s; see the bibliography.)

[INF] **What this means for the hobby, stated plainly:**

- Soft-water fish are not “coping” with soft water. They are built for it.** The framing of soft water as inherently stressful is a temperate-species generalisation.
- The reverse question is under-asked:** what is the ionoregulatory cost to a blackwater-adapted characid of living in 15 dGH, pH 8.2 water? This is the fish whose machinery is calibrated for a different environment, and the aquarium literature is largely silent on it.
- “Fish adapt to anything” and “fish must have their native chemistry” are both wrong.** The truth is that adaptation is real, specific, and directional — and that a species adapted to one extreme is not automatically comfortable at the other.

27.4 Acute vs. Chronic Change

[EST]

	Acute change	Chronic condition
Timescale	Minutes to hours	Weeks to years
Primary risk	Osmotic/ionic disturbance faster than compensation	Sustained metabolic cost; cumulative physiological effects
Compensation mechanism	Existing transporters work harder; short-term acid-base adjustment	Transporter expression changes; ionocyte density changes; gill morphology remodels
Time required to compensate	Hours	Days to weeks
Visible signs	Rapid gilling, loss of equilibrium, erratic swimming, clamped fins, sudden mortality	Reduced growth, poor colour, reproductive failure, disease susceptibility, shortened lifespan — frequently attributed to other causes
How aquarists usually detect it	Immediately, dramatically	Often not at all

[INF] **The asymmetry in detection is important.** Acute events are visible and get attributed (often to pH, sometimes correctly). Chronic costs are invisible and get attributed to bad luck, poor stock quality, or disease. **The hobby’s evidence base is therefore heavily biased toward acute effects**, which may be part of why “stability” dominates the discourse over “appropriateness.”

27.5 Why Hardness Changes Matter Independently of pH

[INF] **Consolidating the argument made throughout this document:**

1. **Calcium acts directly at the gill**, independently of pH, to reduce ion permeability.
2. **Total ionic strength sets the osmotic gradient**, independently of pH.
3. **Calcium is required for skeletal and — critically — invertebrate exoskeletal mineralisation**, entirely independently of pH.
4. **Egg water-hardening and sperm motility respond to ionic strength**, not to pH per se.
5. **Nothing in points 1-4 is measured by a pH test.**

[INF] Therefore: an aquarist who tests only pH is not measuring any of the variables that osmoregulation actually depends on. **GH and conductivity are the physiologically relevant measurements; pH is a third, separate axis.** This is arguably the strongest single argument in this document for testing GH and conductivity rather than treating pH as the master parameter.

Section 28: Shrimp and Snails

Search terms for this section: Neocaridina, Caridina, cherry shrimp, crystal shrimp, Taiwan bee, Amano shrimp, molting, molt, exoskeleton, white ring of death, calcium, magnesium, snail shell, shell erosion, nerite, mystery snail, ramshorn, bladder snail, biofilm.

28.1 Why Invertebrates Are a Different Problem From Fish

[EST] Fish and calcifying invertebrates have fundamentally different relationships with water hardness:

	Fish	Shrimp and snails
Primary use of environmental calcium	Gill permeability regulation; some skeletal uptake	Structural — exoskeleton and shell are built from it
Dietary alternative available?	Substantially, yes	Partially, but water-column calcium matters directly
Consequence of chronic deficiency	Metabolic cost; subtle	Acute and visible: failed molt, eroded shell, death
Timescale to visible failure	Months to years	Weeks
Recovery after correction	Generally good	Existing shell damage does not repair; new growth can be normal

[INF] **Practical implication:** GH is a **soft constraint** for most fish and a **hard requirement** for calcifying invertebrates. An aquarist can debate whether 3 dGH stresses a tetra. There is much less to debate about whether 3 dGH lets a nerite maintain its shell.

28.2 The Molting Process

[EST] Crustaceans grow by **ecdysis** — periodically shedding the old exoskeleton and expanding a new one before it hardens.

Simplified sequence: 1. **Pre-molt:** the animal reabsorbs some minerals from the old exoskeleton and stores them (in decapods, partly in gastroliths); a new soft cuticle forms underneath. 2. **Ecdysis:** the old shell splits — in dwarf shrimp, between the carapace and abdomen — and the animal backs out. 3. **Post-molt:** the animal takes up water to expand, then **rapidly hardens the new exoskeleton by depositing calcium carbonate** drawn from the water and from internal stores. This is the period of maximum vulnerability and maximum calcium demand. 4. **Inter-molt:** normal hardened state.

[EST] **Where water chemistry intervenes:** - **Insufficient calcium/magnesium (low GH):** the new exoskeleton cannot harden properly. The animal remains soft, vulnerable to injury and infection. - **Excessive hardness:** commonly reported to cause the old shell to be too thick or the split to fail. [CON] — this is widely repeated in the hobby; the mechanism is plausible but the evidence is husbandry observation rather than controlled study. - **Rapid parameter change:** a shrimp mid-molt has a temporarily permeable surface and is at maximum osmotic vulnerability. This is the mechanistic basis for the observation that shrimp deaths cluster after water changes.

[HOB] **“White ring of death”:** a visible white gap at the carapace/abdomen junction, indicating the molt failed to separate cleanly. Almost universally fatal. Commonly attributed to GH problems or to rapid parameter change. [CON] The attribution is plausible and consistent with the mechanism, but “white ring” is a hobby-observed sign rather than a diagnosed condition, and other causes (bacterial infection, injury, general debilitation) can produce similar appearances.

28.3 Neocaridina vs. Caridina — Different Requirements

[HOB] This is the clearest case in the freshwater hobby where two superficially similar animals need genuinely different water.

	<i>Neocaridina davidi</i> (cherry, blue dream, etc.)	<i>Caridina</i> (crystal, bee, Taiwan bee)	<i>Caridina multidentata</i> (Amano)
Origin	East Asian streams, moderately mineralised	Southeast Asian soft, acidic streams	Japanese/Taiwanese streams; amphidromous
Commonly cited GH	6–14 dGH (sources vary: 4–10, 6–12, 7–15 all appear)	4–6 dGH	Wide tolerance, ~4–15 dGH
Commonly cited KH	2–8 dKH	0–2 dKH	Wide
Commonly cited pH	6.8–7.8	5.8–6.8	6.5–7.8
Commonly cited TDS	~150–300 ppm	~100–150 ppm	Wide
Typical setup	Tap water often adequate	RO + GH-only remineralizer + active substrate	Tap water usually fine
Difficulty	Beginner	Intermediate–advanced	Beginner

[CON] **Note the disagreement between sources on *Neocaridina* GH.** Specialist shrimp sites variously give 4–10, 6–12, 6–14 and 7–15 dGH. This spread is itself informative: it suggests the species is genuinely tolerant across a broad range and that no source has a precise empirical basis. **The honest statement is that *Neocaridina* do well across roughly 5–15 dGH, and that consistency probably matters more than the specific value.**

[INF] **Why the *Caridina* system is built the way it is** — this is a good example of a coherent, internally consistent design: 1. Target pH is acidic (5.8–6.5). 2. Acidic pH cannot be held against meaningful KH (Section 14). 3. Therefore KH must be near zero. 4. But the shrimp still need calcium and magnesium for molting. 5. Therefore **GH must be supplied without KH** — hence GH-only remineralizers (calcium sulfate + magnesium sulfate based). 6. And the pH must be actively held down — hence active substrate. 7. And the source water must not reintroduce KH — hence RO.

Every component is required by the others. Substituting tap water, or using a GH/KH remineralizer, or adding crushed coral, breaks the system. This explains why *Caridina* keeping appears fussy: it is not fussiness, it is a chain of dependencies.

28.4 The Magnesium Problem

[INF] **Restating from Section 11.3 because this is where it bites hardest.**

A GH test gives a single number covering calcium and magnesium together. Shrimp need both. Several common mineral sources supply **calcium only**:

- Crushed coral — CaCO₃, no magnesium
- Aragonite — CaCO₃, no magnesium
- Cuttlebone — CaCO₃, no magnesium
- Limestone — CaCO₃, no magnesium (dolomite is the exception)
- Calcium chloride, calcium sulfate — calcium only

[INF] **Failure mode:** an aquarist raises GH from 3 to 8 using crushed coral, sees a “correct” GH reading, and continues to lose shrimp to failed molts — because the

entire 5-degree increase was calcium and the magnesium is still at whatever the source water provided. This is a plausible and frequently proposed explanation for otherwise-unexplained molting failure at apparently adequate GH. **Classification: [INF] — mechanistically sound, widely reported, not formally demonstrated in dwarf shrimp.**

[INF] Diagnostic approach: derive Ca and Mg separately from a municipal water report (Section 2.3), or supplement magnesium (magnesium sulfate, Section 11) and observe over two to three molt cycles.

28.5 Snails

[EST] Gastropod shells are calcium carbonate — aragonite, calcite, or both depending on taxon — secreted by the mantle and laid down over an organic matrix. The outermost layer, the **periostracum**, is proteinaceous and provides a degree of protection to the mineral layers beneath.

[EST] Shell erosion mechanism: in water undersaturated with respect to calcium carbonate — i.e. low calcium, low alkalinity, and especially low pH — the shell **dissolves**. This is the same saturation-state chemistry described for crushed coral in Section 17.2, running in the unwanted direction. Once the periostracum is breached or worn, the mineral layers beneath are directly exposed.

Progression of shell damage: 1. Loss of gloss; slight roughening. 2. White chalky patches, usually first at the apex (the oldest part of the shell, longest exposed). 3. Pitting and thinning. 4. Holes; visible erosion into the whorls. 5. In severe cases, structural failure and death.

[EST] Critical fact: shell erosion does not repair. Correcting the water stops further damage and allows healthy new growth at the aperture, but existing damage is permanent. **Early detection matters, and the apex is where to look.**

Species notes [HOB]:

Snail	Calcium demand	Notes
Nerite (<i>Neritina</i>)	High	Thick shells; frequently show erosion in soft water. Cannot reproduce in fresh water (larvae need brackish)
Mystery snail (<i>Pomacea bridgesii</i>)	Very high	Large, fast-growing shell. Among the most GH-demanding common aquarium animals. Erosion is very common in soft water
Ramshorn (<i>Planorbella</i>)	Moderate	Adaptable; often survives in softer water than nerites tolerate, with visible shell thinning
Bladder snail (<i>Physella</i>)	Low-moderate	Thin shells; very tolerant; thrives where others fail
Malaysian trumpet snail (<i>Melanoides</i>)	Moderate	Thick shells; often shows apex erosion in soft acidic tanks
Assassin snail (<i>Anentome</i>)	Moderate	

[HOB] Commonly cited minimum for shell maintenance: roughly **6+ dGH** with some alkalinity present, and pH not persistently below about 7.0. **[CON]** These are hobby conventions; the underlying chemistry (carbonate saturation state) is solid, but the specific numeric thresholds are not traceable to controlled work.

[INF] Supplementary calcium sources [HOB]: cuttlebone, crushed coral in the filter, calcium-fortified snail foods, blanched vegetables (which contribute dietary calcium), and commercial calcium supplements. **Dietary calcium is not a full substitute for water-column calcium** where the water is actively undersaturated, because dissolution is happening at the shell surface from the outside regardless of what the animal eats.

28.6 Biofilm — the Non-Chemistry Variable

[EST/INF] Worth including in a water chemistry document because it is the most common *misattribution* in shrimp keeping.

Dwarf shrimp are primarily **biofilm and detritus grazers**. Biofilm — the aufwuchs layer of bacteria, diatoms, algae, protozoa and their extracellular matrix on every submerged surface — is their principal food source, particularly for shrimplets.

[INF] Consequences: - **New tanks fail shrimp colonies** not because of chemistry but because there is no established biofilm to feed shrimplets. This is why “let the tank mature for a few months before adding shrimp” is sound advice that has nothing to do with GH, KH or pH. - **Excessive cleaning** removes the food supply. - **Leaf litter, driftwood and botanicals** are valuable primarily as biofilm substrate, secondarily as DOC sources (Section 16.6), and only marginally as pH modifiers. - **A colony that stops producing surviving shrimplets while adults remain healthy** is more often a food/biofilm problem than a chemistry problem.

[INF] This is worth stating explicitly because chemistry gets blamed for outcomes it did not cause. Before adjusting GH, KH or pH in a struggling shrimp tank, the questions “how old is this tank?” and “what are the shrimplets eating?” should be asked.

28.7 Extremely Soft and Extremely Acidic Conditions

[INF]

Extremely soft (GH < 3): - Molting failure in *Neocaridina*; marginal even for *Caridina*. - Snail shell erosion essentially guaranteed for most species. - Requires remineralization for any calcifying invertebrate.

Extremely acidic (pH < 5.5): - Carbonate saturation state is far below 1; **any calcium carbonate structure actively dissolves**. - Snails are not viable long-term. - Some *Caridina* and blackwater specialist shrimp tolerate this range, but it is at the edge. - Nitrification is slowed (Section 20.3), so ammonia management depends more on plants, water changes and the fact that TAN is nearly all ammonium.

[INF] The combination of very low GH and very low pH is specifically hostile to calcifiers — not because either alone is necessarily lethal, but because both push the carbonate saturation state in the same direction.

Section 29: Aquarium Plants

Search terms for this section: aquatic plants, plant nutrients, calcium deficiency, magnesium deficiency, interveinal chlorosis, carbon source, bicarbonate uptake, biogenic decalcification, micronutrient availability, iron availability, chelator, planted tank water chemistry.

29.1 Calcium and Magnesium as Plant Nutrients

[EST]

Calcium: a macronutrient. Structural role in cell walls (calcium pectate in the middle lamella) and in membrane function. **Largely immobile within the plant** — it is not readily relocated from old tissue to new. Deficiency therefore appears in **new growth**: stunted, distorted, twisted or crumpled new leaves, and dieback of growing tips.

Magnesium: a macronutrient and the **central atom of the chlorophyll molecule**. **Mobile within the plant** — it is relocated from old leaves to new when supply is short. Deficiency therefore appears in **older leaves** as **interveinal chlorosis**: yellowing between the veins while the veins themselves stay green.

[INF] The mobility difference is a genuinely useful diagnostic:

Symptom location	Suggests
New growth distorted, stunted, tips dying	Calcium (immobile), or boron
Old leaves yellowing between green veins	Magnesium (mobile)
Old leaves uniformly yellowing	Nitrogen (mobile)
New leaves yellow with green veins	Iron (immobile in plant)

[CON] **Caveat:** these classic deficiency patterns come from terrestrial plant nutrition. Aquatic plants take up nutrients through both roots and leaves, and the substrate can supply what the water column lacks. The patterns are directionally useful and frequently confounded in practice.

29.2 Carbon Availability and KH

[EST] Aquatic plants need inorganic carbon. Two forms are available:

- **CO₂ (dissolved carbon dioxide)** — the preferred and energetically cheapest source for essentially all aquatic plants.
- **Bicarbonate (HCO₃⁻)** — usable by *some* species, at metabolic cost, via mechanisms that actively transport bicarbonate or convert it extracellularly.

[EST] **Biogenic decalcification:** plants that use bicarbonate strip it from the water and release hydroxide, raising local pH dramatically at the leaf surface. Where calcium is present, calcium carbonate precipitates onto the leaves as a visible white crust. This is commonly seen on *Vallisneria*, *Elodea*, *Egeria*, *Hydrilla* and *Potamogeton* in hard water under strong light without CO₂ injection. **It is direct visual evidence that a plant is using bicarbonate.**

[EST] Species differ substantially in bicarbonate use:

Group	Bicarbonate use	Implication
<i>Vallisneria</i> , <i>Egeria</i> , <i>Elodea</i> , <i>Hydrilla</i> , <i>Potamogeton</i>	Strong	Thrive in hard, alkaline water without CO ₂ injection
Many <i>Cryptocoryne</i> , <i>Anubias</i> , <i>Java fern</i> , <i>Bolbitis</i>	Moderate/slow-growing	Tolerant of a wide range; low demand
Most stem plants, <i>Rotala</i> , <i>Ludwigia</i> , carpeting species, <i>Hemianthus</i>	Weak or none	Depend on free CO ₂ ; struggle in hard alkaline water without injection

[INF] **This is a real, mechanistic connection between KH and plant success** — and it is not the connection usually claimed. The claim “plants prefer low KH” is imprecise. The accurate statement is: **in a tank without CO₂ injection, high KH and high pH shift the inorganic carbon balance toward bicarbonate, which only some plants can use.** With CO₂ injection, free CO₂ is supplied directly and the KH becomes far less relevant to carbon availability.

29.3 pH and Micronutrient Availability

[EST/INF] Nutrient availability is pH-dependent, a principle established in soil science and transferring imperfectly to the water column.

Nutrient	Availability trend
Iron (Fe)	Falls sharply as pH rises. Above about pH 7, unchelated Fe ³⁺ precipitates as insoluble hydroxide
Manganese	Similar to iron
Phosphate	Precipitates with calcium at high pH — calcium phosphate formation reduces availability of both
Zinc, copper	Reduced availability at high pH
Calcium, magnesium	Availability is a function of concentration, not pH
Nitrogen (as nitrate)	Largely pH-independent
Potassium	Largely pH-independent

[INF] **Chelators change this picture substantially.** Commercial fertilisers supply iron bound to chelating agents which keep it in solution at higher pH than free iron would tolerate:

Chelator	Effective pH range (approximate)
EDTA	Up to about pH 6.5 for iron; loses effectiveness above this
DTPA	Up to about pH 7.5
EDDHA	Up to about pH 9; distinctive red colour
Gluconate / citrate	Short-lived; rapid uptake required

[INF] **Practical consequence:** an aquarist in hard, alkaline water using an EDTA-only iron fertiliser may see iron deficiency symptoms despite dosing adequately, because the iron precipitates before the plants can use it. **Switching to a DTPA or EDDHA-chelated product is the appropriate response, not changing the water’s pH.** This is a good example of solving a chemistry problem without touching the water chemistry.

29.4 Extremely Hard and Extremely Soft Water for Plants

Extremely hard water (GH > 20, KH > 15, pH > 8): - Calcium and magnesium abundant; no deficiency risk for those. - **Free CO₂ is very low** — most inorganic carbon is bicarbonate. - Iron and phosphate availability reduced. - Bicarbonate-using species thrive; CO₂-dependent species struggle. - Biogenic decalcification crusts appear on leaves. - **[INF]** The realistic approaches are species selection, CO₂ injection, or strongly chelated micronutrients — in that order of effort.

Extremely soft water (GH < 2, KH < 1): - **Calcium and magnesium may be genuinely deficient** for demanding species. - Free CO₂ is proportionally higher at a given pH. - Micronutrient availability is good. - **Very low carbon reserve:** in a heavily planted, brightly lit, non-injected tank, photosynthesis can strip CO₂ and drive pH sharply up during the photoperiod — because there is neither CO₂ nor bicarbonate reserve. - **[INF]** Standard remedy is GH supplementation (Section 11), often with a product like Seachem Equilibrium that supplies calcium, magnesium and potassium without adding KH.

29.5 Why “Plants Prefer pH X” Is the Wrong Claim

[INF] Consolidated: pH itself is not a plant nutrient and plants do not have a pH preference in any direct sense. What pH is correlated with:

1. **The inorganic carbon species available** (CO₂ vs. bicarbonate) — which is really about KH and CO₂, with pH as the indicator.
2. **Micronutrient solubility** — which is manageable with chelators.
3. **Calcium and magnesium availability** — which is really about GH.

[INF] Therefore the accurate framing is: plants respond to **carbon availability, macronutrient concentration, and micronutrient solubility**. pH is a useful *indicator* of all three and a *cause* of none of them. Adjusting pH to please plants is treating the readout rather than the system — a direct parallel to the argument in Section 7.

[HOB] The commonly cited “planted tank sweet spot” of roughly pH 6.5-7.2 with KH 3-6 exists because that range happens to combine adequate free CO₂, workable micronutrient solubility, adequate buffering, and compatibility with most commonly kept fish. It is a convergence of practical constraints, not a botanical requirement.

Section 30: Care Sheets and Parameter Ranges

Search terms for this section: care sheet, fish profile, parameter range, recommended pH, species profile, citation circularity, aquarium literature, Fishbase, published range.

30.1 Where Do Published Ranges Actually Come From?

[INF] Investigating the provenance of aquarium parameter ranges is difficult precisely because most sources do not cite anything. The identifiable origins:

Source of the range	How reliable	How to recognise it
Field measurements of the natural habitat	Good for wild-caught stock; often irrelevant for farmed stock	Cites a locality, a collection date, or a survey; ranges are often extreme (pH 4.2–5.1)
Databases compiling published literature (e.g. FishBase)	Reasonable, but often reports very wide tolerance ranges drawn from disparate sources	Cites primary literature
Commercial breeding experience	Practically valuable; usually undocumented	Often appears on breeder or importer sites; may reflect only that facility’s water
Aquarium book lineage	Variable; frequently decades old	Round numbers; ranges that appear identically across many species
Copied from another website	Poor	Identical ranges across sites, sometimes with identical phrasing; no citation
Inferred from the genus	Poor	The same range given for every species in a genus that spans different habitats
Written to be safe/uncontroversial	Poor	“pH 6.5–7.5” appearing for nearly every species regardless of origin
AI-generated with fabricated citations	Very poor — actively harmful	Precise-sounding statistics attributed to named institutions that cannot be located. See “About This Document”

[INF] **The “pH 6.5–7.5” signature.** When a care sheet gives this range for a cardinal tetra, a mystery snail, a Malawi cichlid and a molly, it is not reporting data. It is reporting the middle of the range of typical tap water. This is not useless — it does indicate “most tap water is fine” — but it should not be read as a species-specific finding.

30.2 Why Sources Disagree About the Same Fish

[INF] Consolidated causes:

- They are answering different questions** (habitat / survival / thriving / breeding / farm origin). See Section 8.1.
- They assume different stock provenance** — wild-caught vs. F1 vs. long-domesticated line.
- They reflect the author’s own water**, which worked for them.
- Different eras.** A 1975 book describes wild-caught imports; a 2024 site describes farmed stock. Both may be accurate for their subject.
- Different geography.** European and British aquarium literature has historically been more soft-water-oriented; much North American literature reflects harder municipal supplies.
- Copying errors compound.** A transcription slip in one influential source propagates.
- Some sources report degrees and some report ppm** without labelling clearly (Section 24.5).
- Commercial motivation.** A source selling buffers, RO systems or remineralizers has an incentive to present narrow ranges.

30.3 Should an Aquarist Target the Middle of a Published Range?

[INF] **Generally no**, for several reasons:

- The range’s midpoint has no special status. If the range came from habitat data, the midpoint is an arithmetic artefact of the sampling. If it came from copying, it means nothing at all.
- Reaching the midpoint often requires intervention**, and intervention carries risks (Section 7) that being slightly off-centre does not.
- The value of a stable, unmanipulated environment usually exceeds the value of hitting a specific number within a broad tolerance band.
- Being inside the range at a stable value is the goal.** Being at the centre of the range via weekly chemical dosing is worse than sitting near the edge without intervention.

[INF] **Better decision procedure:**

- Test the source water** (Section 9.1) — this is the fixed constraint.
- Determine stock provenance** — wild-caught or farmed? Ask the retailer; it genuinely matters.
- Determine the goal** — keeping, or breeding?
- Ask whether the source water falls within the envelope where this stock is routinely kept successfully**, given 2 and 3.
- If yes: change nothing.** Select stock to match the water.
- If no: change the source water** (dilution, remineralization, mineral substrate), not the tank chemistry (Section 7.5).

30.4 How Reliable Are Published Ranges Overall?

[INF] **Honest assessment:**

Aspect	Reliability
Direction (this species is a soft-water fish / this one is a hard-water fish)	Good. The broad classification is generally correct and useful
Order of magnitude (soft vs. hard, acidic vs. alkaline)	Good
Specific boundaries (pH 6.4 vs. 6.6)	Poor. No basis for that precision
Breeding requirements	Moderate , where the source is a breeder
Maintenance requirements	Poor to moderate ; usually conservative
Applicability to farmed stock	Poor , if the range is habitat-derived
Invertebrate GH requirements	Moderate — the underlying chemistry is sound even where the numbers are conventions

[INF] Practical summary: care sheets are good at telling you what kind of water a species comes from and poor at telling you what number to hit. Read them for direction, not for precision.

Section 31: Acclimation

Search terms for this section: acclimation, drip acclimation, float acclimation, plop and drop, shipping bag, transfer, quarantine, ammonia in bag, TDS shock, osmotic shock, temperature acclimation.

31.1 What Acclimation Is Actually For

[INF] Acclimation manages a transition across **several simultaneous parameter differences**:

Parameter	Typical bag-to-tank difference	Rate of harm	Priority
Temperature	Can be large	Fast	High
pH	Often 0.5–1.5 units	Moderate	Moderate
Ammonia / ammonium	Potentially very high in the bag	Fast, and worsened by pH rise	Critical — see 31.3
Conductivity / TDS	Variable, sometimes large	Moderate	High for invertebrates
GH / KH	Variable	Moderate	Moderate
Dissolved oxygen	Bag may be depleted	Fast	High
CO ₂	Bag typically CO ₂ -elevated	Fast	Moderate

[INF] The hobby focuses acclimation discussion almost entirely on pH and temperature. The physiology suggests **osmotic/ionic difference and ammonia** deserve at least equal attention.

31.2 Methods

31.2.1 Float acclimation (temperature only)

Float the sealed bag in the tank 15–30 minutes, then net the animal into the tank, **discarding the bag water**.

- **Addresses:** temperature only.
- **Does not address:** chemistry difference.
- **Advantage:** fast; **does not introduce bag water** (which may contain ammonia, pathogens, medications, or dye).
- **Appropriate for:** short transport, small chemistry differences, robust fish.

31.2.2 Drip acclimation

Siphon tank water into a container holding the animal and its bag water at a slow drip until the volume has increased two- to threefold, then net the animal across.

- **Addresses:** temperature, pH, GH, KH, TDS — all gradually.
- **Advantage:** the gentlest chemistry transition available.
- **Disadvantages:** slow; the container is unheated and can drift cold; **and see 31.3, the critical caveat**.
- **Appropriate for:** invertebrates, wild-caught fish, sensitive species, large chemistry differences.

31.2.3 “Plop and drop”

Net the animal directly into the tank after brief temperature equalisation.

- **[CON] Controversial but with a real rationale.** The argument: it minimises time in ammonia-loaded bag water, minimises handling duration, and avoids the ammonia hazard of 31.3. Proponents include some commercial operations and experienced keepers.
- **Appropriate for:** short local transport, robust farmed fish, small chemistry differences.
- **Inappropriate for:** long shipping, invertebrates, wild-caught fish, large chemistry differences.

31.2.4 Batch/incremental acclimation

Add tank water to the container in measured increments (e.g. 20% of container volume every 10–15 minutes). Functionally similar to drip acclimation, easier to control, faster to complete.

[INF] Often the best practical compromise: gradual chemistry transition without the extended duration of a slow drip.

31.3 The Critical Caveat — When Long Acclimation Becomes Counterproductive

[EST] This is the most important content in this section and it is frequently absent from acclimation advice.

The mechanism: 1. During transport, fish excrete ammonia into the sealed bag. 2. Respiration produces CO₂, which **lowers the bag water’s pH**. 3. At the resulting low pH, **the ammonia is predominantly ammonium (NH₄⁺) — the relatively non-toxic form**. The fish tolerates a TAN level that would be dangerous at higher pH. 4. **When the bag is opened, CO₂ outgasses and the pH rises**. 5. **The stored ammonium rapidly converts to toxic free ammonia (NH₃)**. 6. Dripping higher-pH tank water in accelerates the pH rise further. 7. **The longer the acclimation, the longer the animal sits in water whose ammonia toxicity is actively increasing**.

[INF] **Consequences for practice:** - **The longer acclimation is not automatically the gentler one**. A three-hour drip on a bag from a 48-hour shipment can be actively harmful. - **For long-shipped fish, faster is often safer** — or the acclimation should be done with an ammonia-binding conditioner in the container. - **Never pour bag water into the tank**. This is standard advice for pathogen reasons; the ammonia reason is at least as good. - **The trade-off is real and situation-dependent**: short transport with a big chemistry difference favours slow acclimation; long transport favours speed or ammonia binding.

[INF] **A practical resolution:** for long-shipped livestock, open the bag, add a dose of ammonia-binding water conditioner to the container immediately, and then acclimate at whatever pace the chemistry difference warrants. This decouples the two competing risks.

31.4 Wild-Caught vs. Commercially Raised

[INF]

	Wild-caught	Commercially raised
Chemistry tolerance	Narrower — no generational selection, no developmental acclimation to other water	Wider
Transport duration	Usually longer (collection → exporter → importer → wholesaler → retailer)	Often shorter
Condition on arrival	Frequently depleted; may not have fed for days	Generally better
Acclimation approach	Slow chemistry transition warranted, but see 31.3 — the long transport creates the opposing risk	Faster acceptable
Quarantine	Strongly indicated	Recommended
Realistic strategy	Match the exporter's water if known; transition over days-to-weeks in quarantine rather than hours in a bucket	Standard acclimation

[INF] The best answer for wild-caught, long-shipped fish is usually to move the chemistry transition out of the acclimation bucket and into the quarantine tank — acclimate briskly into quarantine water that approximates the shipping water, then adjust the quarantine tank's chemistry gradually over days via water changes. This gives a genuinely slow transition without holding the animal in deteriorating bag water.

31.5 Invertebrate Acclimation

[INF] Dwarf shrimp warrant slower acclimation than fish, for reasons that are mechanistically sound: - Higher surface-area-to-volume ratio. - Thin permeable cuticle; a shrimp mid-molt is at maximum permeability. - Documented sensitivity to rapid osmotic change (Section 26.6). - Small body mass, so less physiological buffering.

[HOB] **Common practice:** drip acclimation over 1–2 hours, targeting a gradual TDS transition; net the shrimp across rather than pouring; discard the acclimation water.

[INF] **But 31.3 still applies.** Shrimp shipped internationally sit in the same ammonia-accumulating bag. Adding an ammonia binder at the start of acclimation is prudent.

31.6 Separating Tradition From Evidence

Practice	Classification
Temperature equalisation before transfer	[EST] — sound
Not pouring bag water into the tank	[EST] — sound (pathogens, ammonia, medications)
Gradual chemistry transition where the difference is large	[STR] — sound physiological basis
Drip acclimation for invertebrates	[INF] — well-reasoned
Multi-hour acclimation for long-shipped fish	[CON] — may be actively harmful . See 31.3
“Always drip acclimate everything”	[Oversimplified] — ignores the ammonia trade-off
Turning tank lights off during transfer	[HOB] — plausible stress reduction, minimal evidence, no downside
Specific drip rates (“2 drops per second”)	[HOB] — convention, not evidence
Quarantining new arrivals	[STR] — strongly supported for disease reasons; also gives a slow-chemistry-transition venue

Section 32: Myths and Controversies — Evidence Evaluation

Search terms for this section: aquarium myths, water chemistry myths, pH myths, hardness myths, misconceptions, folklore, debunking, evidence.

Classification scale used: Strongly supported / Generally supported but context-dependent / Plausible but oversimplified / Primarily anecdotal / Misleading / Unsupported (myth).

Claim 1: “A stable pH is more important than the correct pH.”

Classification: Generally supported but context-dependent. Supported for maintenance of commercially bred generalist species. Not supported for breeding, wild-caught specialists, or invertebrate mineral requirements. The best evidence for it is the ammonia-speciation mechanism (Section 6.5), which is almost never the reason given. The trade-off framing is rhetorical — most aquarists can have both. **See Section 6 in full.**

Claim 2: “Never chase pH.”

Classification: Strongly supported as method guidance; misleading as a universal prohibition. The chemistry is sound (Section 7.2): acid dosed against alkalinity produces oscillation then collapse. But the advice is routinely misapplied to dismiss legitimate goals (breeding, invertebrate needs, correcting genuine alkalinity depletion). Correct form: *do not repeatedly dose chemicals to move tank pH against the water’s buffering*. **See Section 7.**

Claim 3: “Fish adapt to any pH.”

Classification: Misleading. Contains a true core (tolerance is much wider than care sheets suggest; Merck cites lethal limits near pH 4 and 10 for many species) wrapped in a false generalisation. Individual acclimation is real and documented. Generational adaptation is real but slow, incomplete, and not established for most ornamental strains. Breeding requirements frequently do not adapt at all. And “survives” is not “unaffected.”

Claim 4: “Captive-bred fish don’t care about hardness.”

Classification: Plausible but oversimplified. Farmed stock is genuinely more tolerant than wild stock. But: “captive-bred” does not specify a water chemistry — farms use everything from soft Southeast Asian water to hard Florida well water. Breeding requirements transfer poorly regardless of provenance. And for calcifying invertebrates the claim is simply false — hardness is a structural requirement, not a preference. **See Section 6.6.**

Claim 5: “Hard water is healthier because it is more stable.”

Classification: Plausible but oversimplified. Conflates hardness (GH) with buffering (KH). GH does not buffer; KH does, and they can diverge completely (Section 4.3). The Old Tank Syndrome profile is *high GH with zero KH and low pH* — hard water that is anything but stable. What is true: waters with high GH usually also have high KH, and high KH is genuinely more forgiving. What is false: that hardness itself confers stability, or that stability makes hard water healthier for species not suited to it.

Claim 6: “Soft water causes pH crashes.”

Classification: Plausible but oversimplified. Low KH permits rapid pH change; low GH does not. The two are correlated, so the claim points in a real direction while naming the wrong parameter. Also, low KH does not *cause* a crash — it removes the resistance to one. Whether pH moves depends on the acid production rate. A lightly stocked, well-maintained 0 dKH tank is entirely stable. **See Sections 3.7 and 5.4.**

Claim 7: “KH controls pH.”

Classification: Misleading. KH controls **resistance to pH change**. Where pH actually sits is set by the balance of alkalinity against dissolved CO₂ and other acids. At a fixed KH of 4, pH can range from about 6.3 to 7.6 depending on CO₂ alone (Section 3.8). The correct statement: **KH controls how far pH moves for a given acid input; CO₂ and other acids control where it sits.**

Claim 8: “GH and KH are basically the same thing.”

Classification: Unsupported (myth). GH counts divalent cations; KH counts acid-neutralizing capacity, mostly from bicarbonate anions. They correlate in limestone-derived water because CaCO₃ dissolution releases both in equal equivalents — but they diverge routinely: gypsum water (GH >> KH), softened or sodium-bicarbonate water (KH >> GH), aging tanks (GH stable, KH falling). **See Section 3.6.**

Claim 9: “High pH means hard water.”

Classification: Unsupported (myth). A statistical tendency mistaken for a rule. Sodium-bicarbonate water and household-softened water are both alkaline with near-zero GH. Lake Tanganyika is pH 9 with only moderate GH. **See Section 4.2.**

Claim 10: “Low pH means soft water.”

Classification: Unsupported (myth). The Old Tank Syndrome profile — pH below 6.0 with total hardness in the hundreds of mg/L — is the direct counterexample, and it is documented in veterinary literature. Gypsum-derived water and CO₂-injected hard water are others.

Claim 11: “Driftwood lowers pH.”

Classification: Misleading as stated; supported with a KH qualification. Driftwood releases weak organic acids that are consumed by alkalinity. In water above roughly 5–6 dKH the effect on pH is negligible; the water browns and nothing else happens. Below about 1–2 dKH the effect can be substantial. **Both the people who say it works and the people who say it doesn’t are reporting accurately about different water. See Section 16.2–16.3.**

Claim 12: “Indian almond leaves lower pH.”

Classification: Same as Claim 11. Higher leachate per unit mass than driftwood, so a larger effect at the same KH — but the same total dependence on alkalinity. Additionally: the antimicrobial claims have some *in vitro* support for *Terminalia catappa* extracts but the extrapolation to aquarium concentrations is unestablished. **And the best-evidenced benefit — DOC protection of ionoregulation in soft, acidic water — is not a pH effect at all (Section 16.6).**

Claim 13: “Crushed coral stabilises pH.”

Classification: Generally supported but context-dependent. It establishes a self-regulating alkalinity **floor** via negative feedback: as pH falls, undersaturation increases, dissolution accelerates. Genuine and useful. But it does not hold a specific value, does not act quickly enough to counter a fast acid input, does not prevent upward movement, and **does nothing at all in water already at carbonate saturation** — which is why so many aquarists report “I added crushed coral and nothing happened.” **See Section 17.**

Claim 14: “A pH swing will kill fish.”

Classification: Plausible but oversimplified. Depends entirely on magnitude, rate, direction and species. Fish in CO₂-injected planted tanks routinely experience 1.0-unit daily swings without harm. Fish moved abruptly from pH 6.0 ammonia-loaded water to pH 7.8 water can die within hours — but from ammonia, not from pH.

Direction matters more than magnitude because of the ammonia interaction (Section 6.5). The claim conflates a mechanism (rapid change is stressful) with an outcome (death) without the intervening conditions.

Claim 15: “Never change pH by more than 0.2 per day.”

Classification: Primarily anecdotal — origin untraceable. This research could not trace this figure, or the competing 0.3/day version, to any controlled study. The underlying principle (slower is safer) is sound. The specific number appears to be hobby convention. It is also inconsistent with observed practice: CO₂-injected tanks swing far more than this daily without incident, and drip acclimation routinely moves fish faster than this. **The figure is conservative and harmless to follow, but it should not be presented as an established threshold.**

Claim 16: “A one-point pH change is automatically deadly.”

Classification: Unsupported (myth). Directly contradicted by routine practice: CO₂ injection produces roughly 1.0-unit daily swings as a design target. What can be dangerous about a 1.0-unit change: the ammonia conversion if it is upward and ammonia is present; the accompanying osmotic change if the two waters differ ionically; and the CO₂ concentration if the change was caused by CO₂ at high KH. **The pH number alone does not determine the risk.**

Claim 17: “RO water is too pure for fish.”

Classification: The conclusion is broadly right; the usual explanation is wrong. The popular mechanism (“osmosis will make their cells burst”) is not how fish physiology works — fish are osmoregulators with functional gills and kidneys. The real issues are ion loss, the metabolic cost of active uptake, absence of calcium at the gill, and zero buffering. **And there is a crucial qualification:** natural blackwater is equally ion-poor and supports enormous biodiversity — the difference being dissolved organic carbon, which measurably protects ionoregulation (Section 10.3, 16.6). **Pure RO is not aquarium water; but “too pure” is the wrong diagnosis, and it is not the ion-poverty alone that is the problem.**

Claim 18: “TDS shock kills fish/shrimp.”

Classification: Plausible but oversimplified; over-attributed. The mechanism (rapid osmolality change forcing immediate osmoregulatory adjustment) is sound, and invertebrates are genuinely more vulnerable. But the label is applied retrospectively to a wide range of unexplained losses whose actual causes may be copper, ammonia, temperature, chlorine, or molt-cycle coincidence. Specific thresholds circulating in the hobby have no traceable basis. **See Section 26.6.**

Claim 19: “You should match the exact pH listed on a fish care sheet.”

Classification: Unsupported (myth). Care-sheet ranges have unknown provenance, conflate five different questions, and rarely distinguish wild from farmed stock. Achieving a specific number usually requires exactly the intervention that Claim 2 correctly warns against. **See Sections 8 and 30.**

Additional Claims Identified During Research

Claim 20: “Nitrate causes the pH to drop in old tanks.”

Classification: Misleading. Nitrate is the **conjugate base of a strong acid** and does not acidify water. The acid is the H⁺ produced during nitrification. Nitrate accumulation is a **marker** of the process that consumed the alkalinity, not the cause. **Correlation is real; causation is misattributed.**

Claim 21: “Boiling water softens it.”

Classification: Plausible but impractical. Boiling precipitates temporary (bicarbonate-associated) hardness only; permanent (sulfate/chloride-associated) hardness is unaffected. Evaporation during boiling partly re-concentrates what remains. Real chemistry, useless at aquarium scale. **See Section 12.2.2.**

Claim 22: “Household softened water is good for soft-water fish.”

Classification: Misleading — potentially harmful. Sodium-exchange softeners remove GH, leave KH unchanged, leave TDS unchanged, and add sodium. The water is soft in the plumbing sense only; the osmotic load is unchanged, the buffering is intact, the pH will not fall, and calcifying invertebrates will fail. **See Section 9.5.**

Claim 23: “Aeration adds CO₂ to a planted tank.”

Classification: Unsupported (myth) as applied to injected tanks. Gas exchange moves water toward atmospheric equilibrium. Injected tanks are supersaturated with CO₂, so agitation **removes** it and **raises** pH. Only water that is undersaturated (below ~2–3 ppm) gains CO₂ from agitation. **See Sections 1.5 and 19.3.**

Claim 24: “You can estimate your CO₂ from pH and KH.”

Classification: Misleading in most real aquariums. Valid only where carbonic acid is the sole acid and carbonate the sole alkalinity contributor. Driftwood, botanicals, aquasoil, nitric acid and phosphate buffers all break the assumption, and the error runs consistently in the direction of **overestimating** CO₂. **The pH-drop-from-degassed-baseline method cancels this interference and is the better hobby technique. See Section 19.4–19.5.**

Claim 25: “Adding CO₂ lowers KH.”

Classification: Unsupported (myth). CO₂ produces one H⁺ and one HCO₃[−] together; net acid-neutralizing capacity is unchanged. pH falls, KH does not. This is a genuinely useful diagnostic: **falling pH with stable KH suggests CO₂; falling pH with falling KH suggests nitrification, organics or active substrate. See Sections 3.4 and 19.1.**

Claim 26: “A big water change is always risky.”

Classification: Misleading. Risk scales with (parameter difference) × (fraction changed), not with fraction alone. An 80% change with matched water is essentially harmless and is standard practice in discus and breeding operations. A 20% change into a neglected tank with mismatched water can be lethal. **The percentage rule is a proxy for the mismatch nobody measured. See Section 22.2.**

Claim 27: “Old tanks are seasoned and therefore better.”

Classification: Conflates two different states. Mature biofilm, established microbial communities and settled biology are genuine advantages. Accumulated nitrate, depleted alkalinity, concentrated minerals and accumulated organics are not. **“Mature” and “neglected” are different conditions that share the word “old.”**

Claim 28: “You need a specific pH for plants.”

Classification: Misleading. Plants respond to carbon availability, macronutrient concentration and micronutrient solubility. pH indicates all three and causes none of them. Iron availability at high pH is solved by chelator selection, not by pH adjustment. **See Section 29.5.**

Claim 29: “Shrimp need TDS X.”

Classification: Primarily anecdotal. Widely circulated targets are hobby conventions with no traceable primary basis, and different specialist sources give materially different numbers. TDS is composition-blind: 150 ppm of calcium magnesium sulfate and 150 ppm of sodium chloride are not interchangeable. **TDS is the right tool for reproducibility and the wrong tool for a biological target; GH is the biologically meaningful measurement. See Section 26.5.**

Claim 30: “The API/hobby test kit range is the full range.”

Classification: Misleading — a genuine and common measurement trap. Standard low-range liquid pH kits typically bottom out near 6.0. A soft-water or aquasoil tank reading “6.0” may actually be 5.4 or 4.9. Many aquarists have never seen their tank’s true pH. **A reading at the end of a kit’s scale is a floor or ceiling, not a measurement. See Section 1.4.1.**

Claim 31: “Crushed coral will raise my pH to 8.2 and cook my fish.”

Classification: Unsupported (myth). Calcium carbonate dissolution is self-limiting at the carbonate saturation point, which in freshwater is typically reached around pH 7.6–8.0. It cannot drive pH indefinitely upward. **This is the opposite failure mode to Claim 13, and both come from the same misunderstanding of saturation-limited dissolution. See Section 17.2.**

Claim 32: “Peat removes calcium from the water.”

Classification: Contested. Peat demonstrably lowers a measured GH reading. Whether it *removes* calcium or *complexes* it (leaving it present but invisible to the EDTA titration) is disputed; a hobby-technical account reports that atomic absorption spectrophotometry found calcium unchanged after peat treatment despite a large drop in titrated hardness. This could not be traced to peer-reviewed work. If true, “peat softens water” and “peat removes calcium” are different claims and only the first is reliable. **See Section 12.2.3.**

Claim 33: “Nitrifying bacteria die below pH 6.5.”

Classification: Misleading — a wastewater figure misapplied to aquariums. The figure comes from activated sludge literature built around *Nitrosomonas* and *Nitrobacter*. Aquarium biofilters are dominated by *Nitrospira*, including comammox organisms adapted to low-nutrient conditions, and acid-tolerant archaeal ammonia oxidisers exist. Acidic aquariums demonstrably process ammonia. **Nitrification at low pH is slower and performed by a different community — not absent. See Section 20.3.**

Claim 34: “GH of 8 means the shrimp have enough minerals.”

Classification: Plausible but incomplete. A GH reading does not resolve the calcium:magnesium split. Crushed coral, aragonite, cuttlebone and limestone all supply calcium with no magnesium. A tank whose GH was raised entirely by a calcium-only source can show an adequate GH with a magnesium deficit. **See Sections 11.3 and 28.4.**

Section 33: Realistic Scenario Library

Search terms for this section: troubleshooting, aquarium problem, diagnosis, why is my pH, why is my KH dropping, water chemistry problem, parameter troubleshooting.

Format used for each scenario: Observation → Most plausible explanations → Additional possibilities → Questions to ask → Tests that would help → Actions that may be appropriate → Actions that may create unnecessary instability.

Scenario 1: “My tap water is pH 7.6 but my aquarium is pH 6.5.”

Most plausible explanations 1. **Acid production in the tank exceeding alkalinity replacement.** Nitrification plus organic acids have drawn KH down; pH followed. Most likely if tap KH is low-to-moderate and water changes are small or infrequent. 2. **CO₂ — injected or biological.** If CO₂ is injected, this is the design working. If not, high bioload and low surface agitation can hold CO₂ above atmospheric equilibrium. 3. **Active substrate** doing exactly what it is designed to do. 4. **Botanicals, driftwood, peat** in low-KH water. 5. **Tap water was measured as-drawn.** If the tap sample was CO₂-supersaturated, its *equilibrated* pH may be well above 7.6 — making the gap even larger than it appears.

Additional possibilities: phosphate buffer added previously; a partially exhausted aquasoil; a large dead organism decomposing; over-filtration of a very lightly stocked tank producing an unusually clean but unbuffered environment.

Questions to ask: What is the tank’s KH, and what is the tap’s KH? Is this a change or has it always been this way? What is the substrate? Is CO₂ injected? Botanicals present? How often and how large are water changes? How long has the tank been running?

Tests that would help: KH of tank and tap (this is the decisive test). Tap pH after 24h aeration. Nitrate. Ammonia. A pH profile across the day. KH again in two weeks to establish a trend.

Actions that may be appropriate: If KH is stable and healthy and the fish are fine, **nothing**. If KH is declining toward zero, increase water change volume/frequency, or add a slow carbonate source. If it’s CO₂, verify the drop checker and confirm safe concentration.

Actions that may create instability: Dosing pH Up. Adding a phosphate buffer. A sudden large water change to “reset” it — especially without testing ammonia first (Section 21).

Scenario 2: “My GH is extremely high but my KH is low.”

Most plausible explanations 1. **Non-carbonate (permanent) hardness in the source water** — calcium and magnesium paired with sulfate or chloride rather than bicarbonate. Gypsum-influenced geology. Check the source water: if tap shows the same profile, this is simply what your water is. 2. **KH consumed in the tank while GH stayed** — nitrification and organic acids remove alkalinity but do not remove calcium and magnesium. **This is the Old Tank Syndrome trajectory.** Check whether tap KH is much higher than tank KH. 3. **GH-only remineralizer used on RO water.** If this is a *Caridina* setup, it is intentional. 4. **Evaporative concentration** raising GH while nitrification lowers KH — the two processes running in opposite directions on the same tank.

Questions to ask: What is the source water’s GH and KH? Is the tank’s KH lower than the source’s? Is the GH higher than the source’s? How is evaporation topped off? What is the nitrate?

Tests that would help: Source water GH and KH (decisive). Nitrate. TDS of tank vs. source. KH trend over four weeks.

Actions that may be appropriate: If source water is naturally like this — nothing, unless pH stability is a problem, in which case a small carbonate source raises KH. If it is tank-derived depletion — increase water changes, switch to RO/distilled for top-off, address bioload.

Actions that may create instability: Adding a large bicarbonate dose without first checking ammonia (Section 6.5). Assuming high GH means the water is fine.

Scenario 3: “My fish’s care sheet says pH 6.5–7.0 but my tap water is 7.6.”

Questions the aquarist should ask before altering anything 1. **Is this fish wild-caught or commercially bred?** This is the most important question and it changes the answer entirely. 2. **Am I keeping it or breeding it?** Maintenance tolerance is broad; breeding requirements are narrow. 3. **Where did the care sheet’s range come from?** Habitat data, farm practice, or copying? (Section 30.) 4. **What is my tap water’s GH, KH and conductivity?** pH is the least informative of the four. 5. **Is the fish actually showing a problem?** Feeding, colour, activity, behaviour. 6. **What water is the retailer keeping it in right now?** If it’s thriving at the store in 7.6 water, that is direct evidence. 7. **What would achieving 6.8 actually require, and could I sustain it indefinitely?**

Most likely correct answer: for farmed stock kept (not bred), **pH 7.6 with a stable KH is very likely fine**, and intervention carries more risk than the 0.6-unit gap does.

When intervention is warranted: wild-caught specimens; a breeding project; or an observed, persistent problem that survives elimination of ammonia, nitrite, temperature, oxygen, disease and aggression.

If intervening: change the source water (RO blending), not the tank chemistry.

Scenario 4: “My KH keeps dropping.”

Most plausible explanations 1. **Nitrification consuming alkalinity faster than water changes replace it.** Run the mass balance in Section 20.2 — this is arithmetic, not mystery. 2. **Low source water KH.** If tap is 2 dKH and 30% weekly changes add 0.6 dKH against consumption of 1.5 dKH/week, the deficit is structural. 3. **Active substrate** consuming alkalinity via cation exchange. 4. **Botanicals, driftwood, peat.** 5. **Topping off evaporation instead of changing water** — no alkalinity is being added at all. 6. **Source water changed seasonally** — utility switched sources, or well water shifted.

Additional possibilities: a phosphate buffer previously in use that has now washed out (KH readings were inflated and are now reading true); a previously-added carbonate source that has become coated and stopped dissolving.

Questions to ask: Source KH? Water change volume and frequency? Substrate? Bioload and feeding? Botanicals? When did this start? Has the source water been retested recently?

Tests that would help: Source KH (retest — do not rely on a figure from a year ago). Nitrate. Ammonia. KH logged weekly for a month to establish rate of decline.

Actions that may be appropriate: Increase water change volume or frequency. Add crushed coral or aragonite in the filter for a self-limiting floor. Reduce bioload or feeding. Dose bicarbonate for a precise, immediate correction — **after testing ammonia**. Top off with RO/distilled instead of tap.

Actions that may create instability: A large bicarbonate dose all at once, especially with ammonia present. Ignoring it because pH still looks fine — **pH is the lagging indicator** (Section 3.7).

Scenario 5: “I added crushed coral and nothing happened.”

Most plausible explanations 1. **The water is already at or above carbonate saturation.** $\Omega \geq 1$, so nothing dissolves. **This is the correct behaviour, not a failure** — and it means the water did not need it. 2. **Not enough time.** Dissolution is slow; changes take one to three weeks to become measurable. 3. **Insufficient quantity or surface area.** A small handful in a large tank. 4. **Insufficient flow.** Coral sitting in a low-flow corner develops a locally saturated boundary layer and stops dissolving. 5. **Coated surfaces.** Older material coated with detritus, biofilm or phosphate compounds. 6. **Something is consuming the alkalinity as fast as it dissolves** — active substrate, heavy bioload, botanicals. The coral is working; the effect is being cancelled.

Questions to ask: How much, what grain size, where placed, for how long? What was the starting KH and pH? Is there aquasoil? What is the flow through the media bag?

Tests that would help: KH and pH now vs. before. KH of the water immediately downstream of the media bag vs. the far end of the tank (a flow/saturation check).

Actions that may be appropriate: Wait longer. Increase quantity. Move to higher flow. Rinse the material in tank water. **If the water is already hard and alkaline, remove it — the water needs nothing.**

Actions that may create instability: Piling in large quantities out of impatience (though note the self-limiting property means overshoot in chemistry is unlikely). Switching to bicarbonate dosing without first understanding why coral did nothing.

Scenario 6: “I added driftwood but my pH didn’t change.”

Most plausible explanation: KH is high enough to absorb the organic acids completely. Above roughly 5–6 dKH this is the expected, normal outcome (Section 16.2).

Additional possibilities: the wood was pre-boiled or pre-cured and has little leachate left; the piece is small relative to the volume; activated carbon or Purigen-type resin in the filter is removing the tannins as fast as they leach; not enough time has passed.

Questions to ask: What is the KH? How large is the piece relative to tank volume? Was it boiled or soaked first? Is carbon or a chemical resin in use? Is the water visibly tinting?

Tests that would help: KH. Whether the water is browning (if yes, tannins are entering and the alkalinity is absorbing the acidity — confirming the diagnosis).

Actions that may be appropriate: Accept it — wood is valuable for structure, biofilm and DOC regardless of pH. If low pH is genuinely the goal, **reduce KH first** (RO dilution) and the same wood will then work. Remove chemical filtration media if tannins are wanted.

Actions that may create instability: Adding more and more wood expecting a pH effect that the alkalinity will continue to absorb. Adding pH Down on top of it.

Scenario 7: “My pH drops when the CO₂ turns on.”

Is that expected? Yes. That is precisely what CO₂ injection does. A drop of roughly 0.8–1.2 units from the degassed baseline over 1–2 hours after injection starts, returning overnight, is the normal and intended pattern.

What to verify rather than “fix”: 1. **The magnitude corresponds to a safe CO₂ concentration.** The same pH drop means very different CO₂ at different KH — about 35 ppm at 3 dKH, about 140 ppm at 12 dKH (Section 19.6). **Check KH.** 2. **Drop checker with 4 dKH reference solution reads green,** not yellow. 3. **Fish behaviour.** Surface gasping overrides every number. 4. **KH is stable over time.** If KH declines, the same injection rate produces a lower pH — and a **pH controller will over-inject** to compensate.

Actions that may be appropriate: Confirm KH; confirm drop checker; ensure CO₂ is off overnight; add night aeration; establish the degassed baseline pH once so the drop can be measured properly.

Actions that may create instability: Adding buffer to “stop the pH swing” — this raises the CO₂ level required to reach the same pH, which is the opposite of safer. Trying to hold pH constant with a controller while KH is drifting.

Scenario 8: “My tap water tests differently after sitting overnight.”

Explanation [EST]: the water was **supersaturated with CO₂** under mains or well pressure. Once exposed to air, CO₂ outgasses, carbonic acid is consumed, and **pH rises**. Very common with well water and with deep municipal groundwater; a rise of 0.5–1.0+ units is unremarkable.

Key confirming observation: KH does not change during this process. Only pH moves. If KH also changed, something else is happening (sample contamination, test error, or precipitation).

Which number to use: the equilibrated (aerated) value predicts what the tank will settle toward. The as-drawn value predicts the transient pH the fish see at the moment of a water change. Both matter; they answer different questions.

Actions that may be appropriate: Aerate replacement water for an hour before use (also equalises temperature). Record both values in the log. Recognise that a tank whose pH is *higher* than the as-drawn tap pH is normal, not anomalous.

Actions that may create instability: Chasing the difference with chemicals. Concluding the tap water is “unstable.”

Scenario 9: “I use RO water. What should I add back?”

The answer depends entirely on what is being kept.

Goal	Remineralization approach	Target
General community fish	GH/KH remineralizer, or blend RO with tap	~6–8 dGH, ~3–5 dKH
<i>Neocaridina</i> shrimp	GH/KH remineralizer	~6–10 dGH, ~2–4 dKH
<i>Caridina</i> shrimp	GH-only remineralizer + active substrate	~4–6 dGH, 0–1 dKH
Soft-water fish, no breeding	GH-only or light GH/KH	~3–5 dGH, ~1–2 dKH
Soft-water breeding	GH-only, minimal	~2–3 dGH, ~0–1 dKH; track conductivity
Rift Lake cichlids	GH/KH remineralizer or Rift Lake salt mix	~8–10 dGH, ~10–14 dKH
Planted, CO ₂ injected	GH/KH, modest	~5–7 dGH, ~3–4 dKH

Questions to ask: What livestock? What substrate (active or inert)? Is breeding a goal? What target pH?

Method [INF]: dose **by mass**, into a **marked volume** in a **mixing container** (never the tank), and **verify with a TDS meter** before use. Consistency between batches matters at least as much as the specific target.

Actions that may create instability: Dosing into the tank. Scoop-based measurement. Using a GH/KH product in a *Caridina* tank. Failing to check RO output TDS as the membrane ages. Using unrem mineralized RO for water changes.

Scenario 10: “My aquarium has very hard water. Do I need RO?”

Most likely answer: no.

Questions to establish that: 1. What is actually being kept? Livebearers, most community fish, Rift Lake cichlids, goldfish, snails and *Neocaridina* are all fine or better in hard water. 2. Is anything actually wrong? Feeding, colour, behaviour, breeding, invertebrate shell condition. 3. Is the water hard in GH, in KH, or both? (These have different implications.) 4. Is there a breeding goal? 5. Is the stock wild-caught?

RO becomes justified if: wild-caught soft-water specialists are the goal; *Caridina* keeping; a soft-water breeding project; the source water is genuinely extreme (>20 dGH, pH >8.5, or unusual contaminants); or a persistent problem has survived elimination of every other cause.

Otherwise the better answer is species selection. Hard, alkaline, well-buffered water is one of the more forgiving environments to keep an aquarium in — it resists the alkalinity depletion that causes most long-term parameter failures.

Actions that may create instability: Adopting RO without a consistent remineralization discipline. Inconsistent RO is worse than consistent hard tap water.

Scenario 11: “My snails’ shells are eroding but my GH is 8.”

Most plausible explanations 1. **pH is low enough that carbonate saturation is below 1**, so the shell dissolves regardless of GH. Check pH and KH — with KH near zero and pH under ~6.8, shells erode even at adequate GH. 2. **The GH is calcium-poor / magnesium-heavy**, or the reverse — the reading does not resolve composition. 3. **The damage is historical.** Erosion does not repair. If the water was corrected recently, the old damage remains; check whether **new growth at the aperture** is healthy — that is the diagnostic. 4. **The species is unusually demanding** — mystery snails and nerites both are. 5. **CO₂ injection** driving daytime pH low enough to shift saturation state.

Tests that would help: KH (likely the missing variable). pH, including overnight/pre-dawn low. Source water GH and KH. Ca and Mg separately from the water report.

Actions that may be appropriate: Raise KH modestly (crushed coral is well suited — it raises KH, GH and pH together and is self-limiting). Add cuttlebone. Verify magnesium adequacy.

Actions that may create instability: Raising GH further when the actual limiting factor is alkalinity and pH.

Scenario 12: “Everything was stable for two years and now my pH is rising and my shrimp are struggling.”

Most plausible explanation: the active substrate has exhausted. Cation exchange capacity is finite. When it is spent, the tank reverts to behaving like inert substrate and drifts toward the source water’s chemistry. **[INF] This is a scheduled event, not a fault.**

Additional possibilities: source water changed seasonally or the utility switched sources; a previously-dissolving carbonate object was added; the keeper changed remineralizer product or batch; RO membrane degradation raising output TDS.

Tests that would help: pH and KH trend over the past months (if logged). Source/change water KH and TDS. RO output TDS. Compare against records from a year ago.

Actions that may be appropriate: Plan a substrate refresh. Transition gradually. Reduce KH in the change water if that is the driver.

Actions that may create instability: Dosing acid to hold the old pH. Sudden substrate replacement in a stocked shrimp tank.

Scenario 13: “My pH is 6.0 on the test kit and won’t go lower.”

Most plausible explanation: the kit has bottomed out. Standard low-range liquid pH kits typically end at 6.0. The actual pH may be 5.6, 5.2, or lower. **A reading at the end of a scale is a floor, not a measurement.**

Tests that would help: a calibrated digital pH meter, or a wide-range/low-range test kit. KH — if KH is 0, a pH well below 6.0 is entirely plausible.

Actions that may be appropriate: Get an accurate reading before making any decisions. Assess whether the actual value is appropriate for the stock.

Actions that may create instability: Making management decisions based on a pegged reading — in either direction.

Scenario 14: “I did a big water change and my fish died.”

Most plausible explanations, in order 1. **Ammonia unmasking (Section 21.2).** A low-pH tank with accumulated ammonium; the water change raised pH; the ammonium converted to free ammonia. **Most likely if the tank was neglected.** 2. **Dechlorinator failure** — not dosed, under-dosed, expired, or chloramine treated as if it were chlorine. 3. **Temperature shock.** 4. **Large osmotic/ionic change** between tank and source water. 5. **Source water changed** — utility flushing mains, source switch, seasonal change. 6. **Substrate disturbance** releasing hydrogen sulfide or trapped organics from a deep bed. 7. **Contamination** — soap residue, copper from unflushed plumbing, a cleaned bucket.

Tests that would help: Ammonia in the tank now. Source water: chlorine/chloramine, pH, KH, GH, TDS, nitrate. Tank: KH, pH, nitrate. **If tank nitrate was very high before the change, that supports explanation 1.**

What this scenario is not evidence for: that large water changes are inherently dangerous (Section 22.2, Claim 26). It is evidence that **large changes into mismatched water** are dangerous, which is a different and correctable statement.

Scenario 15: “My KH is 8 but my pH is 6.6.”

Most plausible explanations 1. **Dissolved CO₂.** At 8 dKH, pH 6.6 corresponds to roughly 24 ppm CO₂ (Section 19.4). If CO₂ is injected, this is normal. If not, the tank may be CO₂-accumulating from heavy bioload plus low surface agitation. 2. A **phosphate buffer** inflating the KH reading — the measured “8 dKH” may not be

carbonate at all. 3. **Test error** — reagent age, wrong sample volume, or a pH kit misread. 4. **Substantial non-carbonate alkalinity** from organic conjugate bases in a heavily botanical tank.

Tests that would help: pH after 24 hours of vigorous aeration (degassed baseline). If pH rises to ~7.8, the answer is CO₂. If it stays at 6.6, the KH reading is not carbonate and something else is going on. **This single test resolves the scenario.**

Actions that may be appropriate: Increase surface agitation if CO₂ accumulation is unintended. Discontinue phosphate buffers. Re-test with fresh reagents.

Scenario 16: “My GH went up and I didn’t add anything.”

Most plausible explanation: evaporative concentration from topping off with mineralized water (Section 23.2). Minerals enter with every top-off and leave with nothing.

Additional possibilities: a carbonate rock, shell, or gravel in the tank slowly dissolving; source water hardened seasonally; a previously-added mineral product still dissolving; new decor.

Tests that would help: Source water GH (retest). Tank TDS trend. Tank GH vs. source GH — **if the tank exceeds the source, top-off concentration is nearly certain.** Vinegar test on any suspect rock (fizzing indicates carbonate).

Actions that may be appropriate: Switch top-off water to RO/distilled. Increase water change volume to export the accumulated minerals.

Scenario 17: “My cycle stalled and my pH is 6.2.”

Most plausible explanations 1. **Alkalinity exhausted.** Nitrification consumed the KH; pH fell; the rate of nitrification declined. Common in soft-water tanks cycling with a heavy ammonia dose. 2. **Ammonia is present but mostly as ammonium** — the biofilter has less free NH₃ substrate available. 3. **The community is establishing slowly** at low pH — genuinely slower, not absent (Section 20.3).

Tests that would help: KH (likely at or near zero). Ammonia, nitrite, nitrate.

Actions that may be appropriate: Restore KH gradually with bicarbonate — **but note the ammonia interaction.** In a fishless cycle there is no livestock at risk, so raising KH and pH is straightforward and appropriate. **In a stocked tank, this is the dangerous case** and requires small incremental changes.

Actions that may create instability: In a *stocked* tank, a large buffer dose while ammonia is present. In a fishless cycle, this concern does not apply — a useful distinction that is often blurred.

Scenario 18: “Two aquarists with the same tap water report completely different tank parameters.”

Most plausible explanations, in order of likely magnitude 1. **Substrate type** — active aquasoil vs. inert gravel produces different GH, KH and pH from identical source water (Section 18.5). 2. **CO₂ injection** — a 1.0-unit pH difference from this alone. 3. **Water change volume and frequency** — determines whether alkalinity is replenished faster than it is consumed. 4. **Bioload and feeding** — sets the acid production rate. 5. **Botanicals and driftwood.** 6. **Surface agitation** — sets the equilibrium CO₂ and therefore the baseline pH. 7. **Tank age** — the accumulated position on the alkalinity-depletion curve. 8. **Time of day of testing.** 9. **Top-off practice.** 10. **Test method and kit calibration.**

[INF] The general lesson: source water is an *input*, not a *result*. Two tanks on identical tap water are running different processes on it. **When comparing parameters between aquarists, the substrate, CO₂ status and maintenance schedule are more informative than the tap water report.**

Section 34: Decision-Making Reference

Search terms for this section: when to change water chemistry, when to leave water alone, decision framework, intervention, aquarium decision making, troubleshooting framework.

[INF] These are structured considerations, not universal prescriptions. Every item is conditional on species, system, and goals.

34.1 When Should an Aquarist Leave Their Water Alone?

Considerations pointing toward **no intervention**:

- Livestock is feeding, coloured up, behaving normally, and — where applicable — breeding.
- Parameters are **stable over time**, whatever their absolute values.
- KH is stable and not trending toward zero.
- Source water falls within the envelope where the stock is routinely kept successfully.
- Stock is commercially bred and the goal is maintenance rather than breeding.
- The only reason for concern is a number on a care sheet.
- The proposed intervention would require ongoing dosing to maintain.
- The aquarist cannot commit to the consistency the intervention would require.

[INF] The default should be non-intervention. The chemistry of intervention (Sections 7, 14, 15) means a poorly executed change is usually worse than no change.

34.2 When Should an Aquarist Investigate Further?

Signals warranting investigation (not yet action):

- **KH declining month over month** — the single most useful early-warning signal.
- Tank pH substantially below source water's equilibrated pH.
- Tank GH substantially above source GH (top-off concentration).
- TDS rising steadily over months.
- Nitrate persistently high despite water changes.
- Invertebrate shell erosion or molting failure.
- Recurring unexplained losses after ammonia, nitrite, temperature, oxygen, disease and aggression have been excluded.
- Parameters changed without the aquarist changing anything — **retest the source water first**.
- Plants showing deficiency patterns.
- A tank that has run for years without a substrate refresh, if it uses active substrate.

34.3 When Might Modifying GH Be Justified?

- **Calcifying invertebrates present and GH is low** (< ~5 dGH for *Neocaridina* and snails). The clearest case in the hobby.
- Molting failures or shell erosion despite adequate KH and pH.
- Plants showing calcium or magnesium deficiency not correctable by fertilisation.
- Building water from RO — GH must be supplied by definition.
- Livebearers (especially mollies) in very soft water.
- Source water is extreme in either direction and the stock is unsuited to it.
- GH is adequate but the **Ca:Mg ratio** may be distorted by a calcium-only mineral source.

Not justified by: a care sheet number, or a desire to “improve” water that is already working.

34.4 When Might Modifying KH Be Justified?

- **KH is measurably declining toward zero in a stocked tank.** Correcting drift, not chasing a target.
- pH declining in parallel with KH.
- Source water KH is too low to replace what nitrification consumes at the current bioload and change schedule.
- Biofilter performance appears impaired (ammonia or nitrite in an established tank) at low pH.
- The stock requires alkaline, buffered water (Rift Lake cichlids).
- The keeper's realistic maintenance cadence does not match the system's acid load — buffer substitutes for maintenance frequency.
- **Downward:** an acidic system is the goal and the source water's alkalinity prevents it. **Reduce KH by dilution before attempting acidification.**

Caution in all cases: test ammonia before raising KH/pH in an established tank (Section 6.5).

34.5 When Might Deliberately Modifying pH Be Justified?

- **Breeding a species with documented chemistry-dependent reproduction.** The strongest case.
- Wild-caught specialists of soft-water species.
- Blackwater or biotope display where the environment is the point.
- *Caridina* systems.
- Source water at a genuine extreme (pH > 8.5 or < 5.5) with unsuited stock.
- Persistent, otherwise-unexplained problems in a tank far from the species' native chemistry.

In all cases: modify the **source water and the system**, not the tank chemistry. See Section 7.5.

34.6 When Might RO Water Be Justified?

- Species genuinely require soft water (wild-caught blackwater fish, *Caridina*, soft-water breeding).
- Source water is extreme, contaminated, or seasonally erratic.
- **Independent control of GH and KH is required** — the one thing blending cannot deliver.
- Reproducibility is the goal (sensitive invertebrates, breeding programmes).

Preconditions before adopting RO: the keeper must dose by mass, into a marked volume, in a mixing container, verified by TDS meter, every time. **Without that discipline, RO is a downgrade.**

34.7 When Might Crushed Coral Be Useful?

- Soft source water with a declining KH trend in a stocked tank.
- Preventing alkalinity depletion / Old Tank Syndrome.
- Rift Lake cichlids on soft-to-moderate tap water.
- Snail and *Neocaridina* tanks needing a passive calcium and buffer source.
- Where a self-limiting safety floor is more valuable than a precise target.

Not useful when: water is already hard and alkaline (it will do nothing); an acidic system is the goal (it directly opposes it); a specific KH value is needed (use bicarbonate); or it is the sole mineral source for calcifying invertebrates (no magnesium).

34.8 When Might Active Substrate Be Influencing the Result?

Always ask what the substrate is before interpreting any water chemistry observation.

Signs active substrate is the driver: - pH substantially below source water pH with no CO₂ injection. - KH and GH both below source water values. - Sawtooth pH pattern around water changes. - Ammonia in a new tank without a corresponding bioload. - Parameters drifting **upward** after one to three years — the exhaustion signature. - CO₂ estimated from the pH/KH table appearing implausibly high.

34.9 A General Decision Sequence

[INF] For any water chemistry concern:

1. **Test the source water.** Both as-drawn and equilibrated pH; plus KH, GH, TDS, nitrate.
2. **Test the tank.** The same panel.
3. **Compare.** The difference between them, and its direction, identifies the process at work (Section 4.7 / 9.1).
4. **Establish the trend.** One reading is a data point; three over six weeks is information.
5. **Rule out non-chemistry causes** — ammonia, nitrite, temperature, oxygen, disease, aggression, stock quality, food supply.
6. **Determine whether the observed state is actually causing a problem.**
7. **If intervention is warranted, change the inputs** (source water, substrate, mineral source, maintenance schedule) **rather than dosing the tank.**
8. **Change one variable at a time**, slowly, and re-test.

Section 35: Additional Questions Discovered During Research

Questions that recur in hobbyist discussion and were not explicitly listed in the original research brief.

Q: Why does my pH read differently in the morning than at night? Confusion source: aquarists assume pH should be a constant. **Answer [EST]:** diurnal CO₂ cycling. Photosynthesis consumes CO₂ during the photoperiod (pH rises); respiration produces it overnight (pH falls). The swing's magnitude is inversely related to KH. This is normal. **Test at a consistent time of day or the readings are not comparable.** See Section 1.6.

Q: Can I use my pH meter to measure my RO water? Confusion source: the meter gives an unstable, implausible reading. **Answer [EST]:** measuring the pH of pure water is close to meaningless. With no buffer, pH is set by trace dissolved CO₂ and by any contamination, and it drifts continuously. Meters also struggle in very low-conductivity solutions. **Measure the pH after remineralization, not before.** See Section 10.2.

Q: Does adding salt (NaCl) affect GH, KH or pH? Answer [EST]: No to all three. Sodium and chloride are monovalent and do not contribute to GH; sodium chloride is a neutral salt and contributes no alkalinity; it does not shift pH. **It does raise TDS and conductivity substantially.** This is a clean demonstration that TDS and GH are not the same measurement.

Q: Why does my nitrate test read high right out of the tap? Answer [INF]: agricultural runoff into groundwater and surface supplies. Well water in farming regions can have substantial nitrate; municipal supplies are limited to 10 mg/L nitrate-nitrogen in the US (≈45 mg/L as nitrate). If source nitrate approaches tank nitrate, water changes stop reducing it — which is confusing until the source is tested. RO removes most nitrate.

Q: Do plants raise pH? Answer [EST]: During the photoperiod, yes — by consuming CO₂. In hard water with strong light and no CO₂ injection, bicarbonate-using species can drive local pH very high (biogenic decalcification, visible as white leaf crust). Overnight, plant respiration lowers pH. **Net over 24 hours, plants do not raise pH; they cycle it.**

Q: Does an air stone raise pH? Answer [EST]: in CO₂-supersaturated water, yes — by removing CO₂. In water already at atmospheric equilibrium, negligibly. In a CO₂-injected tank, an air stone will substantially raise pH and reduce dissolved CO₂. See Section 19.3.

Q: Why did my pH drop after I cleaned my filter? Answer [INF]: several possibilities — disturbed detritus releasing organic acids and CO₂; a temporary reduction in biofilter capacity; or (more often) a coincidental water change done at the same time. If cleaning coincided with a large water change, the water change is the more likely explanation.

Q: My KH is 0 but my pH is 7.5. How? Answer [INF]: possible explanations: (a) the KH test bottomed out — 0.4 dKH and 0 dKH read the same on a drop kit; (b) there is very little dissolved CO₂, so even a tiny alkalinity holds pH up; (c) test error. Note that **pH 7.5 at 0 dKH is a precarious state** — there is no reserve, and any acid input will move it. Use the double-volume titration trick (Section 24.2) to get 0.5-degree resolution.

Q: Do water conditioners/dechlorinators change GH, KH or pH? Answer [INF]: at label doses, negligibly for GH and KH. Some conditioners are slightly acidic or basic but the effect is small relative to any buffering present. **Some products bind ammonia into a less toxic form**, which changes an ammonia test result without changing the water's hardness or alkalinity — a distinct effect worth not confusing with a chemistry change.

Q: Can I mix crushed coral and aquasoil? Answer [INF]: they work directly against each other. The coral supplies alkalinity; the soil consumes it. The result is that **both are consumed faster and neither achieves its purpose.** The soil's finite exchange capacity is spent fighting the coral instead of holding pH down. Choose one.

Q: Why is my GH higher than my KH / KH higher than my GH? Answer [EST]: GH > KH indicates non-carbonate (permanent) hardness — calcium and magnesium paired with sulfate or chloride — **or** that KH has been consumed in the tank. KH > GH indicates alkalinity paired with sodium or potassium rather than calcium: sodium-bicarbonate groundwater, household-softened water, baking soda dosing, or Lake Tanganyika-type water. See Section 3.6.

Q: Does a UV steriliser, protein skimmer, or ozone affect pH? Answer [INF]: UV, no meaningfully. Freshwater protein skimmers are rarely effective and their gas exchange contribution is minor, though any device that agitates water contributes to CO₂ exchange. Ozone can affect pH through oxidation of organics, but is rare in freshwater aquariums.

Q: Why do my parameters differ between the top and bottom of the tank? Answer [INF]: in a well-circulated tank they should not, meaningfully. Differences indicate poor circulation, or measurement near a CO₂ diffuser, a dissolving mineral, or the substrate surface. Persistent stratification in a deep or unfiltered tank is possible. **Sample from mid-water, away from equipment, consistently.**

Q: Should I test my water before or after a water change? Answer [INF]: both, and consistently. Before a change gives the “worst case” — the maximum accumulation the tank reaches. After gives the baseline. Testing at random points relative to the change schedule produces uninterpretable data. Most aquarists find “before the weekly change” the most informative single time.

Q: How long does it take for a parameter change to be “safe” for livestock? Answer [CON]: no validated figure exists. Physiological compensation timescales run from hours (gill acid-base adjustment) to weeks (ionocyte and gill morphology remodeling). Hobby conventions of 1–2 dGH or dKH per week and 0.2 pH per day are conservative and harmless but are not evidence-based thresholds.

Q: Why does my aquarium's pH differ from what the KH/CO₂ chart says it should? Answer [STR]: because the chart assumes carbonic acid is the only acid and

carbonate the only alkalinity contributor. Driftwood, botanicals, aquasoil, nitric acid from nitrification, and phosphate buffers all break the assumption. See Section 19.4.

Q: Do I need to match my quarantine tank's parameters to my display tank? Answer [INF]: yes, and this is under-emphasised. Moving a fish from quarantine to display is another chemistry transition. Running quarantine on the same source water and maintenance regime as the display makes the eventual transfer a non-event — and quarantine is also the ideal venue for a *gradual* chemistry transition for wild-caught stock (Section 31.4).

Q: Is rainwater safe to use? Answer [CON]: chemically it is essentially distilled — zero GH, zero KH, pH around 5.0-5.6. The risk is entirely in collection: roofing material leachate, atmospheric pollution, bird droppings. Cleanly collected from a purpose-built surface into a covered vessel, it is usable and free. Off an aged composite roof, it is a gamble.

Q: Why did my shrimp die right after a water change even though the parameters matched? Answer [INF]: candidate explanations beyond chemistry — a shrimp mid-molt at maximum permeability; copper or metal contamination from plumbing or a container; temperature difference; dechlorinator under-dosed; disturbed substrate releasing ammonia; or coincidence with a molt cycle. **“Parameters matched” usually means GH, KH and pH matched — it rarely means TDS, temperature and trace metals were verified.**

Section 36: Additional Topics Discovered During Research

Concepts that emerged as important during research but were not in the original brief.

36.1 Ammonia Speciation as the Hidden Variable

[EST] Treated at length in Section 6.5. Flagged here because it is arguably **the most under-explained concept in hobby pH discussion**. The fraction of total ammonia present as toxic un-ionised NH_3 rises roughly tenfold per pH unit. This single fact explains: - Why raising pH in a neglected tank kills fish. - Why low-pH tanks tolerate ammonia readings that would be lethal elsewhere. - Why Old Tank Syndrome kills at the moment of correction rather than during the decline. - Why acclimating long-shipped fish slowly can be counterproductive. - Why the direction of a pH change matters more than its magnitude.

It is the strongest mechanistic argument for pH stability, and it is almost never the argument given.

36.2 Dissolved Organic Carbon as a Protective Factor

[STR] Treated in Sections 10.3, 16.6 and 27.2. Flagged here because it **reframes what botanicals are for**. The hobby discusses tannins as a pH tool; the physiological literature indicates DOC directly protects ionoregulation in soft, acidic water — reducing ion loss and stimulating ion uptake — independently of pH. The extrapolation from Rio Negro DOC to aquarium leaf litter is not established, but the mechanism is sound and the hypothesis is well grounded.

36.3 Comammox and the Modern Understanding of Aquarium Nitrification

[STR] The organisms named in aquarium literature (*Nitrosomonas*, *Nitrobacter*) largely come from wastewater engineering and do not reflect what dominates aquarium biofilters. *Nitrospira*, including **complete ammonia-oxidising (comammox) *Nitrospira*** described in 2015, are adapted to the low-substrate conditions of aquariums and drinking-water systems. Archaeal ammonia oxidisers function at lower pH and lower ammonia concentrations than bacterial ones. **This directly undermines the widely repeated “nitrification stops below pH 6.5” claim** and explains why acidic aquariums process ammonia successfully. See Section 20.3.

36.4 The Carbonate Saturation State (Ω) as a Unifying Concept

[EST] Introduced in Section 17.2 for crushed coral, but it explains far more: - Why crushed coral works in some tanks and not others. - Why crushed coral cannot overshoot. - Why snail shells erode in soft acidic water. - Why calcium carbonate precipitates on plant leaves in hard water (biogenic decalcification). - Why aragonite dissolves more readily than calcite. - Why cuttlebone works quickly and limestone rock slowly.

A single concept that unifies six separate hobby observations. It deserves to be understood as a general principle rather than as a crushed-coral footnote.

36.5 Non-Carbonate Alkalinity

[EST] The KH test measures **total alkalinity**, not carbonate specifically. Phosphate buffers, borate, silicate and the conjugate bases of organic acids all contribute. Consequences: inflated KH readings; invalidated CO_2 estimation; and apparent contradictions between KH and pH. See Section 3.1.

36.6 Test Kit Range Ceilings and Floors

[INF] A reading at the end of a kit's scale is not a measurement. Standard low-range liquid pH kits bottom out near 6.0, so many soft-water and aquasoil keepers have never seen their tank's actual pH. **See Section 1.4.1 and Claim 30.**

36.7 Sodium and Potassium Accumulation From Repeated Dosing

[INF] Neither sodium nor potassium is consumed by tank processes, and neither is removed by anything except water changes. Repeated buffer dosing accumulates them indefinitely. Each dKH of baking soda adds roughly 8.2 mg/L sodium; a year of weekly dosing without adequate water changes is a substantial cumulative load that will not appear on a GH, KH or pH test but will appear on a TDS meter.

36.8 The Ca:Mg Ratio

[INF] A GH test does not resolve composition. Calcium-only mineral sources (crushed coral, aragonite, cuttlebone, limestone, calcium salts) can produce an adequate GH with a magnesium deficit. See Sections 11.3 and 28.4.

36.9 Conductivity as the Better Breeding Parameter

[HOB/INF] Professional soft-water breeding practice tracks conductivity, not pH. Egg chorion permeability, water hardening and sperm motility duration are osmotically sensitive. **The hobby discusses breeding in pH terms while practitioners manage it in $\mu\text{S}/\text{cm}$.** See Section 26.7.

36.10 Buffer Intensity vs. Buffer Capacity

[EST] The carbonate system's buffer *intensity* peaks near pK_{a1} (≈ 6.35) and pK_{a2} (≈ 10.33), not in the middle of the aquarium range. What protects a typical aquarium is the *reservoir* of bicarbonate — which is what KH measures. Distinguishing capacity from intensity resolves an apparent paradox in how buffering is described. See Section 5.2.

36.11 CO_2 's Direct Physiological Effect, Independent of pH

[EST] Elevated dissolved CO_2 causes respiratory acidosis in fish blood and reduces haemoglobin oxygen affinity and carrying capacity (Bohr and Root effects). **This — not pH — is how CO_2 overdose kills fish.** It means the same pH drop is safe at 3 dKH and lethal at 12 dKH, because it corresponds to a fourfold different CO_2 concentration. See Section 19.6.

36.12 Stock Quality as a Confounding Variable

[INF] Chemistry gets blamed for outcomes it did not cause. German blue rams, fancy guppies and some discus lines are widely regarded as fragile; plausible causes include intensive line-breeding, high farm stocking density, hormone treatment, prophylactic antibiotic use and pathogen load. **Attributing losses in these strains primarily to GH, KH or pH may be systematic misdiagnosis.** See Section 8.2.

36.13 Biofilm as a Confounding Variable in Shrimp Systems

[INF] Shrimplet survival depends on established biofilm. A new tank fails shrimp colonies for food reasons that look like chemistry reasons. **Before adjusting**

parameters in a struggling shrimp tank, ask how old the tank is. See Section 28.6.

36.14 Fabricated Citations in Aquarium Content

[INF] During this research, multiple recently published aquarium sites were found citing precise-sounding studies with named institutions and specific percentages that could not be located in any indexed database. This appears to be AI-generated content contamination and it is actively degrading the hobby's information environment. **Precise statistics on hobby sites without a traceable primary citation should be treated as unreliable.** See "About This Document."

36.15 Temperature as a Confounding Factor Throughout

[EST] Temperature affects: the neutral point of water (Section 1.3); ammonia speciation (higher temperature = more toxic NH_3); gas solubility (warmer water holds less CO_2 and less O_2); conductivity readings ($\sim 2\%/^\circ\text{C}$); reaction and dissolution rates; and metabolic rate, and therefore the acid production rate. **Temperature is a variable in nearly every relationship in this document and is rarely controlled for in hobby comparisons.**

Glossary

Acclimation — Physiological adjustment by an individual organism to changed environmental conditions, over hours to weeks. Distinct from adaptation.

Acid-neutralizing capacity (ANC) — The formal term for what alkalinity measures: the water's ability to consume added acid without a large pH change.

Adaptation — Genetic change in a population over generations under selection. Not something an individual does.

Alkalinity — The water's capacity to neutralize acid, measured by titration and expressed as mg/L as CaCO₃ or meq/L. **Not the same as "alkaline."** In the aquarium hobby, reported as KH in German degrees.

Alkaline — Having a pH above 7. A description of intensity, not capacity.

Aragonite — A crystalline polymorph of calcium carbonate, more soluble than calcite. The mineral of most mollusc shells and coral skeletons.

Bicarbonate (HCO₃⁻) — The dominant contributor to alkalinity in nearly all freshwater. The conjugate base of carbonic acid.

Biogenic decalcification — Precipitation of calcium carbonate onto plant leaves when a plant strips bicarbonate from hard water and releases hydroxide, raising local pH. Visible as a white crust.

Buffer — A solution containing a weak acid and its conjugate base, resisting pH change when acid or base is added.

Buffering capacity — The total quantity of acid a system can neutralize before the buffer is exhausted. What KH measures.

Calcite — A crystalline polymorph of calcium carbonate, less soluble than aragonite. The mineral of limestone.

Carbonate hardness — Aquarium hobby term for KH. Technically a misnomer, since the test measures total alkalinity rather than carbonate specifically.

Comammox — "Complete ammonia oxidiser." A *Nitrospira* organism performing both ammonia→nitrite and nitrite→nitrate in a single cell. Described in 2015; abundant in aquarium and drinking-water biofilters.

Conductivity — The water's ability to conduct electricity, in µS/cm. Responds to total ionic content weighted by charge and mobility.

dGH / °dH — Degrees of general hardness. 1 dGH = 17.848 mg/L as CaCO₃.

dKH — Degrees of carbonate hardness / alkalinity. 1 dKH = 17.848 mg/L as CaCO₃.

Divalent cation — A positively charged ion with a charge of 2+, such as Ca²⁺ or Mg²⁺. What GH measures.

DOC (dissolved organic carbon) — Dissolved organic material, including humic and fulvic substances. Demonstrated to protect fish ionoregulation in acidic, ion-poor water.

Ecdysis — Molting; shedding of the exoskeleton in arthropods.

Fulvic acid — Lower-molecular-weight, more soluble organic acid from decomposing plant material. Contributes acidity, colour and metal chelation.

GH (general hardness / total hardness) — Combined concentration of dissolved divalent cations, essentially calcium and magnesium, expressed as CaCO₃ equivalent.

Humic acid — Higher-molecular-weight organic acid from decomposing plant material.

Interlamellar cell mass (ILCM) — Tissue between gill lamellae whose height changes in response to environmental conditions including water calcium, altering effective gill surface area.

Ionocyte (chloride cell, mitochondria-rich cell) — Specialised gill cell performing active ion transport.

KH — See Carbonate hardness / Alkalinity.

meq/L (milliequivalents per litre) — Charge-normalised concentration unit. 1 meq/L alkalinity = 50.04 mg/L as CaCO₃ = 2.80 dKH.

Nephrocalcinosis — Calcium deposition in kidney tissue; reported in soft-water species chronically held in hard, alkaline water.

Nitrification — Microbial oxidation of ammonia to nitrite and then nitrate. Produces acid and consumes alkalinity at approximately 7.14 mg CaCO₃ per mg ammonia-nitrogen.

Non-carbonate hardness (permanent hardness) — Hardness associated with sulfate, chloride or nitrate rather than bicarbonate. Not removable by boiling.

Old Tank Syndrome (OTS) — Progressive alkalinity depletion in an under-maintained aquarium, leading to low pH, high ammonia, high hardness, and biofilter failure. Documented in veterinary literature.

Osmoregulation — Physiological maintenance of internal water and solute balance against an environmental gradient.

Periostracum — Proteinaceous outer layer of a mollusc shell, protecting the mineral layers beneath.

pKa — The pH at which a weak acid is half dissociated; the point of maximum buffer intensity for that acid/base pair. Carbonic acid: pKa₁ ≈ 6.35, pKa₂ ≈ 10.33.

Root effect / Bohr effect — Reduction in haemoglobin oxygen affinity (Bohr) and carrying capacity (Root) at low blood pH. The mechanism of CO₂ toxicity in fish.

RO / RO-DI — Reverse osmosis; reverse osmosis with deionization polishing.

Saturation state (Ω, omega) — Ratio describing whether water is undersaturated (Ω < 1, mineral dissolves), saturated (Ω = 1), or supersaturated (Ω > 1, mineral may precipitate) with respect to a mineral such as calcium carbonate.

TAN (total ammonia nitrogen) — Combined NH₃ and NH₄⁺, the quantity an aquarium ammonia test reports.

TDS (total dissolved solids) — Total mass of dissolved substances. In aquarium practice, *estimated* from conductivity via an assumed conversion factor.

Temporary hardness — Hardness associated with bicarbonate; removable by boiling as calcium carbonate precipitate.

Titration — Analytical technique of adding a reagent incrementally until a defined endpoint. The basis of hobby GH and KH tests.

Quick Reference Tables

R1: Parameter — What It Measures / What It Does Not

Parameter	What it measures	What it does NOT measure
pH	Hydrogen ion activity right now	Mineral content; buffering; stability
GH	Dissolved $\text{Ca}^{2+} + \text{Mg}^{2+}$ as CaCO_3 equivalent	Buffering; pH; Ca:Mg ratio; total dissolved solids
KH	Total acid-neutralizing capacity	Where pH sits; calcium content; whether the alkalinity is carbonate
TDS	Conductivity, converted to an estimated mass	Which ions; GH; KH; non-ionic dissolved substances
Conductivity	Ionic content weighted by charge and mobility	Composition

R2: Unit Conversions

From	To	×
dGH or dKH	mg/L as CaCO_3	17.848
mg/L as CaCO_3	dGH or dKH	0.05603
meq/L	mg/L as CaCO_3	50.04
meq/L	dKH	2.804
gpg	mg/L as CaCO_3	17.118
mg/L Ca	mg/L as CaCO_3	2.497
mg/L Mg	mg/L as CaCO_3	4.118
$\mu\text{S}/\text{cm}$	ppm TDS (0.5 factor)	0.5

R3: Method — Raises GH? Raises KH? Changes pH?

Method	GH	KH	pH	Speed	Control
Calcium sulfate (gypsum)	↑	—	—	Slow to dissolve	High
Magnesium sulfate (Epsom)	↑	—	—	Fast	High
Calcium chloride	↑	—	—	Fast	High
Sodium bicarbonate	—	↑	↑	Fast	High
Potassium bicarbonate	—	↑	↑	Fast	High
Sodium carbonate	—	↑ ↑	↑ ↑	Fast	Hazardous
Crushed coral / aragonite	↑ (Ca only)	↑	↑	Slow	Self-limiting
Cuttlebone	↑ (Ca only)	↑	↑	Slow	Self-limiting
GH-only remineralizer	↑	—	—	Fast	High
GH/KH remineralizer	↑	↑	↑	Fast	High
RO dilution	↓	↓	↓ (usually)	Immediate	High
Aquasoil	↓	↓	↓	Days-weeks	Finite lifespan
Peat	↓	↓	↓	Slow	Low
Botanicals / driftwood	—	↓ slightly	↓ if KH low	Slow	Low
CO_2 injection	—	—	↓	Minutes	High, reversible
Aeration	—	—	↑ if CO_2 -rich	Minutes	Moderate
Household softener	↓ ↓	—	—	Immediate	Not recommended

R4: Crushed Coral vs. Baking Soda vs. Remineralizer vs. RO

	Crushed coral	Baking soda	GH/KH remineralizer	RO (alone)
Raises GH	Yes (Ca only)	No	Yes	No — removes
Raises KH	Yes	Yes	Yes	No — removes
Raises pH	Yes, to a ceiling	Yes	Yes	No
Supplies magnesium	No	No	Usually yes	No
Speed	Days-weeks	Minutes	Minutes	Immediate
Precision	Low	High	High	High
Can overshoot?	No — self-limiting	Yes	Yes	n/a
Adds sodium	No	Yes	Usually no	No
Ongoing effort	Very low	Per dose	Per batch	Per batch
Best for	Passive buffer floor	Precise correction	Building water from scratch	Removing everything

R5: Problem → Possible Explanation → Test Next

Problem	Possible explanation	Test next
Tank pH < source pH	Acid production; KH depletion	KH of tank and source
Tank pH > source pH	Source CO ₂ degassing; carbonate dissolving	Source pH after aeration
KH falling over months	Nitrification, organics, aquasoil	Nitrate; source KH; KH trend
GH rising, no dosing	Evaporative concentration from top-offs	Source GH; TDS trend
GH high, KH near zero	Non-carbonate hardness, or KH consumed	Source KH vs. tank KH
KH high, GH near zero	Sodium bicarbonate water or softener	Source water report: sodium
pH swings daily	Photosynthesis/respiration with low KH	KH; pH profile across a day
Crushed coral doing nothing	Water already at carbonate saturation	Current KH and pH
Driftwood not lowering pH	KH absorbing the organic acids	KH
Snail shells eroding	Low pH / low KH / carbonate undersaturation	KH; overnight pH low
Shrimp failing to molt	Low GH, or magnesium deficit	GH; Ca and Mg from water report
Cycle stalled at low pH	Alkalinity exhausted	KH ; ammonia
Fish died after water change	Ammonia unmasking; dechlorinator; temperature	Ammonia; source chlorine; nitrate history
Parameters drifting after 2 years	Active substrate exhausted	KH/pH trend vs. records; source values

R6: Fish Type → Typical Natural Water → Captive Considerations

Type	Typical natural water	Captive consideration
Amazonian blackwater (cardinal, discus, wild <i>Betta</i> , <i>Apistogramma</i>)	pH 4-6; <20-80 µS/cm; GH <2; KH ~0; high DOC	Farmed stock tolerant for maintenance; breeding needs low conductivity; DOC may matter more than pH
SE Asian peat swamp (<i>Boraras</i> , <i>Trigonostigma</i> , chocolate gourami)	pH 3.5-6; very soft; high DOC	Similar; some genuinely narrow
Central American hard-water (<i>Xiphophorus</i> , <i>Amatitlania</i>)	Moderate-hard, alkaline	Very adaptable
Coastal livebearer (mollies)	Hard, alkaline, often brackish	Genuinely mineral-demanding ; soft water is a real failure case
Lake Malawi	pH 7.7-8.6; ~210-285 µS/cm; ~6 dKH; ~6-9 dGH	Alkaline but only moderately hard . Target KH/pH, not extreme GH
Lake Tanganyika	pH 8.6-9.2; ~600 µS/cm; ~18 dKH; ~10-12 dGH	KH > GH . Sodium/potassium bicarbonate dominated
East Asian (goldfish, <i>Neocaridina</i>)	Moderate-hard	Forgiving; buffer reserve valuable for goldfish waste load

R7: Testing Method → Strengths → Weaknesses

Method	Strengths	Weaknesses
Liquid titration (GH/KH)	Accurate; cheap; the hobby standard	Endpoint judgement; 1-degree resolution (use double volume for 0.5)
Liquid colorimetric pH	Cheap; no calibration	±0.2-0.4; range-limited ; useless in tinted water
Test strips	Fast; multi-parameter	Least precise; humidity-sensitive; timing-critical
Digital pH meter	0.01 resolution; works in tinted water	Calibration mandatory ; probe care; false confidence
TDS meter	Cheapest useful tool; instant change detection	Composition-blind; conversion factor assumption
Conductivity meter	Raw unambiguous measurement	Composition-blind
Lab analysis	Only source of separate Ca and Mg	Cost; turnaround
Municipal report	Free; comprehensive; gives Ca:Mg	Annual average; plant not tap; source only

R8: Source Water → Potential Issue → Possible Approach

Source	Potential issue	Possible approach
Very hard municipal (>15 dGH)	Unsuited to soft-water specialists	Species selection first; RO blending if genuinely needed
Very soft municipal (<3 dGH)	Alkalinity depletion; invertebrate mineral shortage	Monitor KH; crushed coral; GH supplement
Household-softened	GH ~0, KH unchanged, high sodium	Use an unsoftened bypass tap , or RO
Well water	CO ₂ -supersaturated; iron; nitrate; untested	Aerate before use ; one-time lab panel
Chloraminated municipal	Standard carbon inadequate for RO prefilter	Chloramine-rated carbon block; appropriate conditioner
Seasonally variable	Unexplained parameter shifts	Quarterly source testing; TDS monitoring
High-nitrate agricultural	Water changes stop reducing nitrate	Test source nitrate; RO if substantial
Rainwater	Collection contamination	Clean purpose-built collection; covered storage
RO/DI	Zero buffer, zero minerals	Always remineralize ; dose by mass; verify with TDS

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Sources Deliberately Excluded

Several recently published aquarium websites citing specific studies — including a claimed 2020 study from a named university on shrimp molting mortality below a specified dGH, and a claimed 2021 *Journal of Fish Biology* study on cortisol elevation outside “optimal hardness” — were located during research and **excluded**. The cited studies could not be found in any indexed literature database. These claims are not reproduced anywhere in this document. See “About This Document.”

Document Notes

Compiled: August 2026.

Intended use: ingestion into NotebookLM as a retrieval corpus for podcast and content development. Not intended for direct publication.

Evidence handling: claims are tagged by evidence tier throughout. Where credible sources disagree, the disagreement is documented rather than resolved. Where a commonly repeated figure could not be traced to a primary source, it is labelled as hobby convention.

Known limitations of this document: - The physiological literature cited is weighted toward Amazonian and temperate model species, because that is where the research exists. Southeast Asian, Central American and African species are less well covered by ionoregulation research. - Very little controlled research exists on the “thriving” tier for ornamental species (Section 8.5). Much of what the hobby believes about long-term effects of chronic parameter mismatch is inference. - Manufacturer dosing figures are reproduced as manufacturer claims and were not independently verified. - Specific numeric thresholds widely used in the hobby (rate-of-change limits, shrimp TDS targets, snail GH minimums) are labelled as conventions because their provenance could not be established. This should not be read as evidence they are wrong — only that they are unverified.